MIL-HDBK-141 figure review

Left: crop of the 1962 scan (auto-detected figure region + caption).

Right: the rebuilt figure from build/lens_designs/.

73 rebuilt figures, then 13 flagged, 11 halftone (scan only), 5 3D-coordinate (scan only), 2 tables.

Mark any page where the rebuild does not match the scan.

Rebuilt figures

scan vs rebuild, one per page

Figure 16.3 — DONE

1962 scan

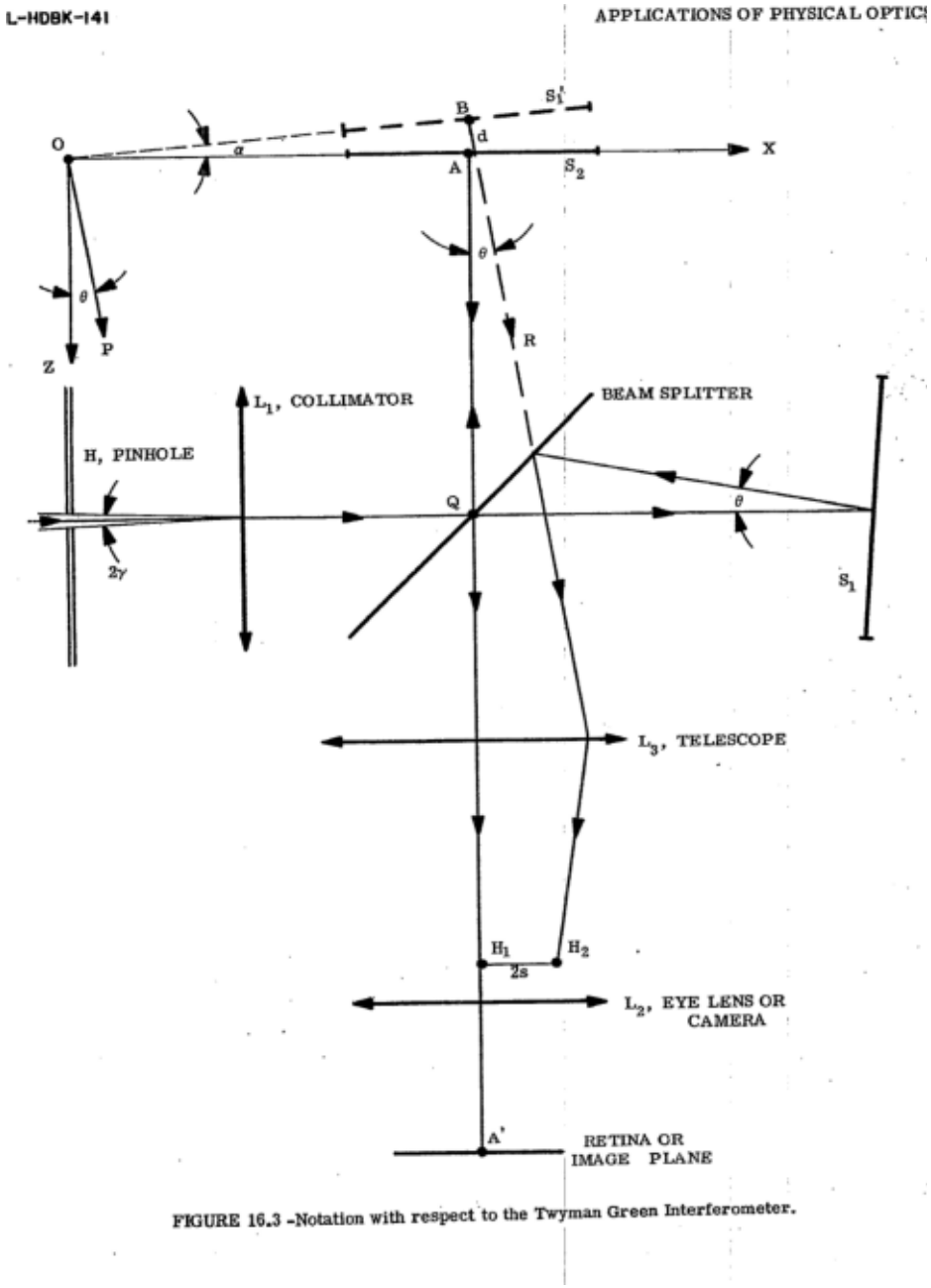


FIGURE 16.3 -Notation with respect to the Twyman Green Interferometer.

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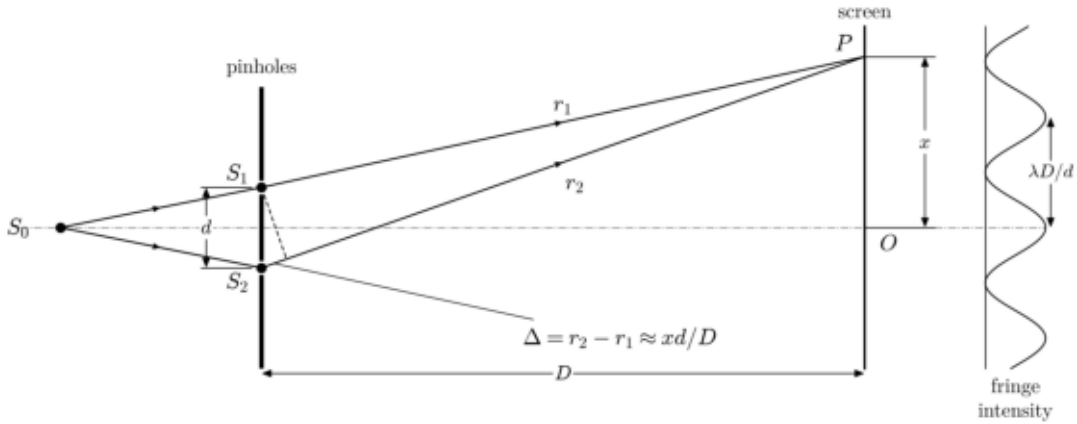


FIGURE 16.3 -Notation with respect to the Twyman Green Interferom

Figure 16.4 — DONE

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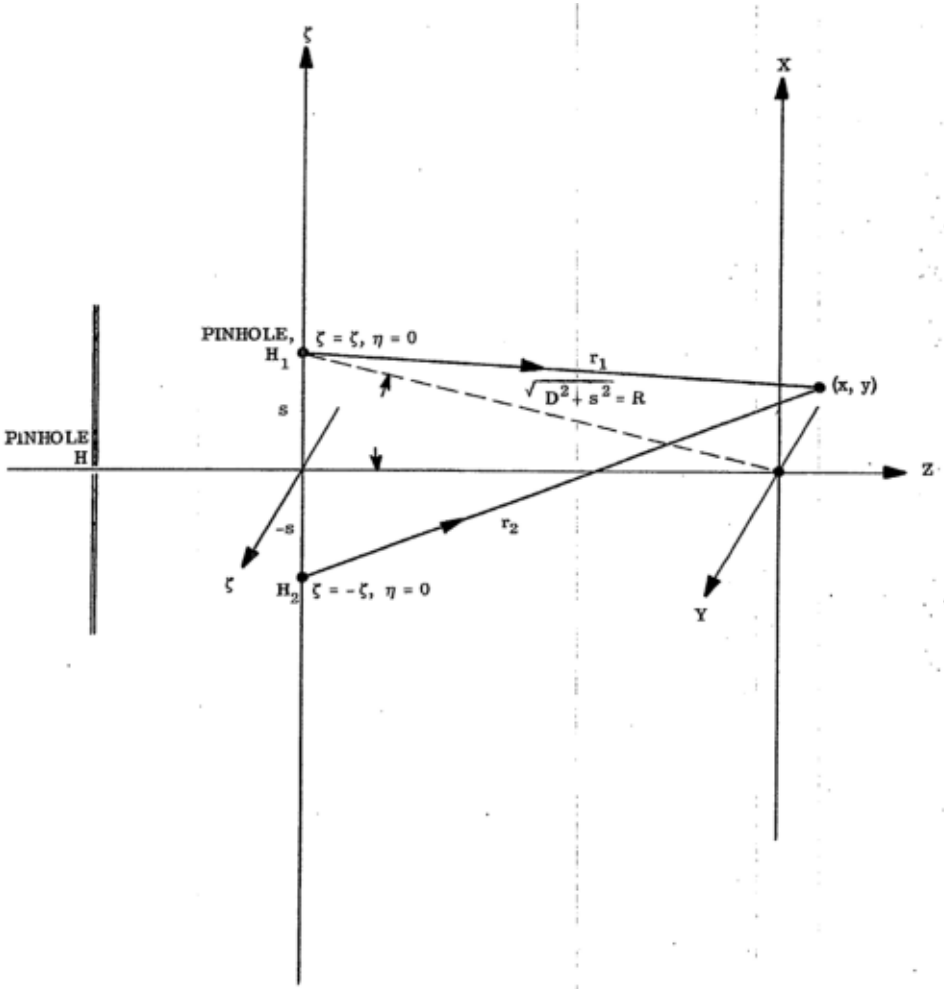


FIGURE 16.4 - Young's Pinhole Interferometer

FIGURE 16.4 - Young's Pinhole Interferometer

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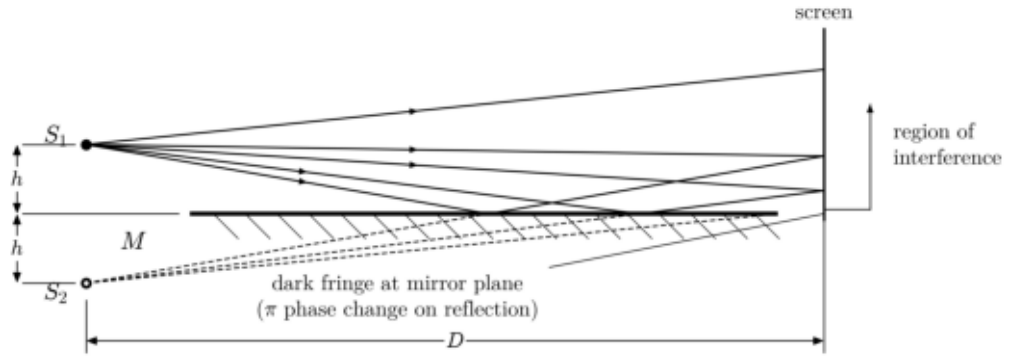


Figure 16.6 — DONE

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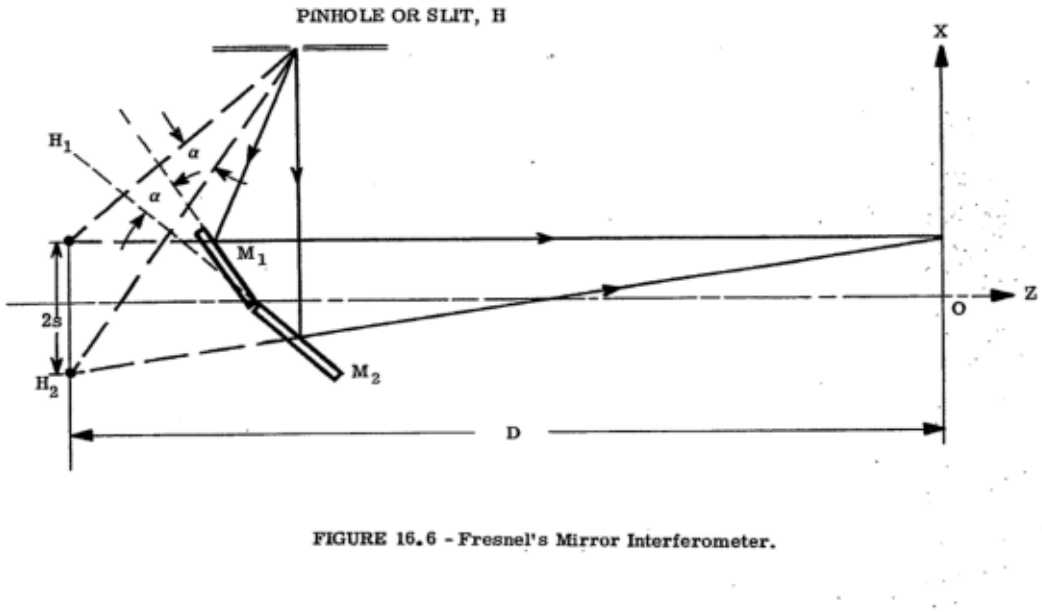


FIGURE 16.6 - Fresnel's Mirror Interferometer.

FIGURE 16.6 -Fresnel's Mirror Interferometer.

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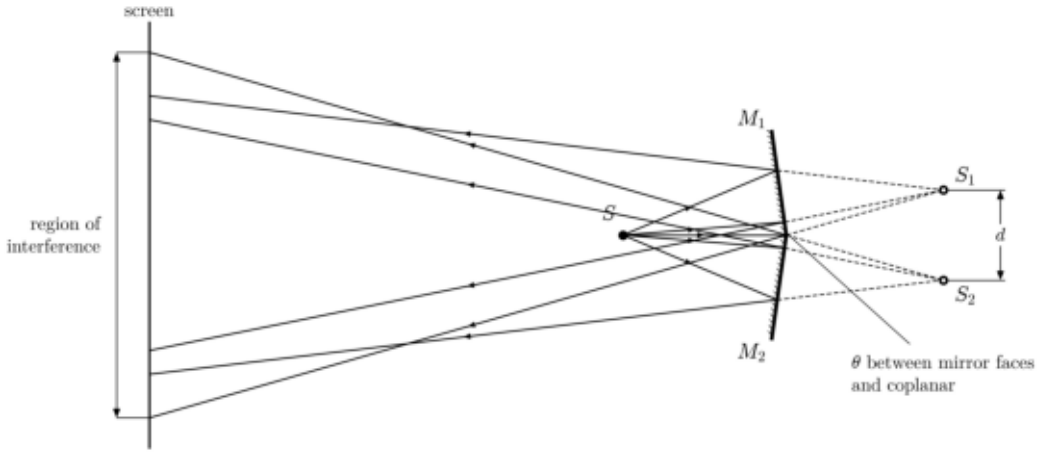


Figure 16.7 — DONE

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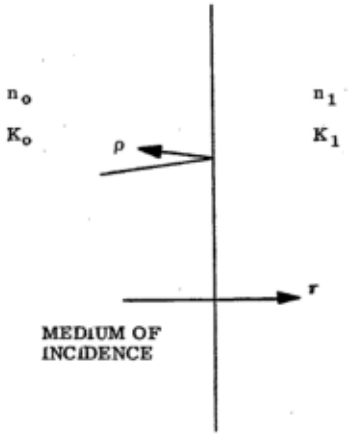


FIGURE 16.7 –Transmittance and reflectance at Translucent Interface

FIGURE 16.7 -Transmittance and reflectance

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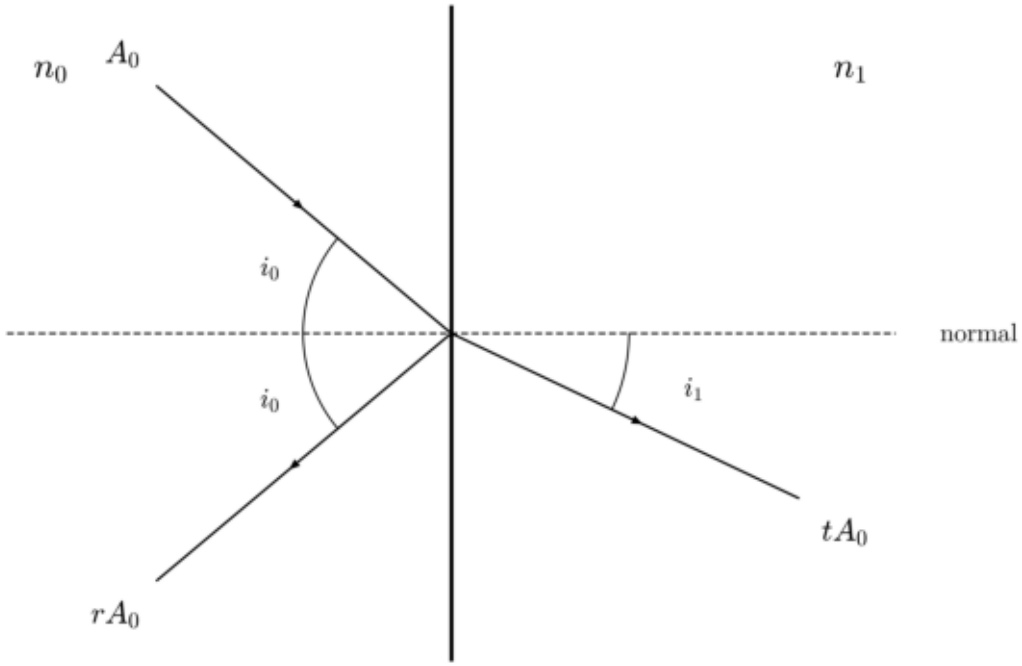


Figure 16.11 — DONE

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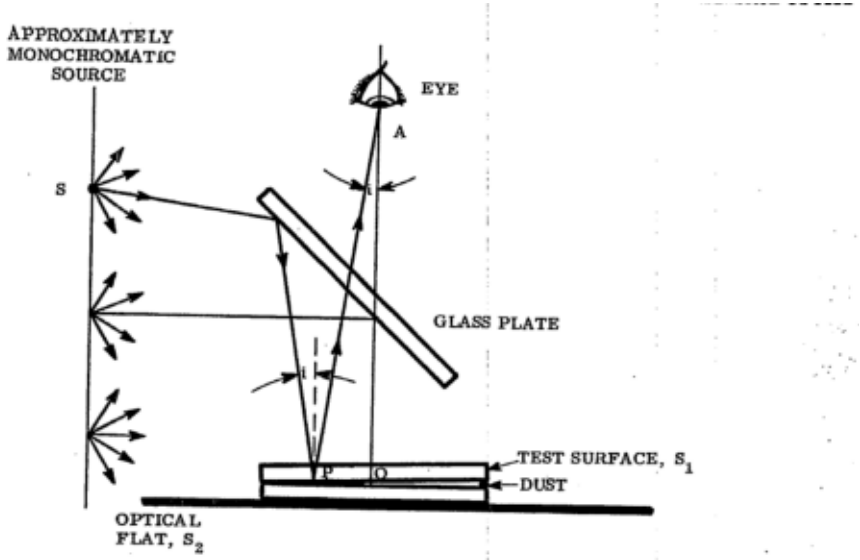


FIGURE 16. 11- Method for obtaining Fizeau Fringes.

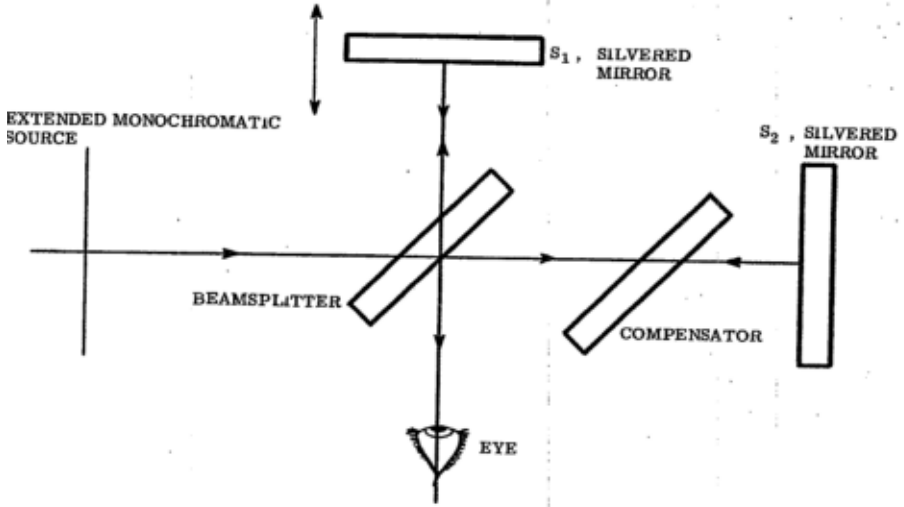


FIGURE 16. 12-Michelson's Interferometer.

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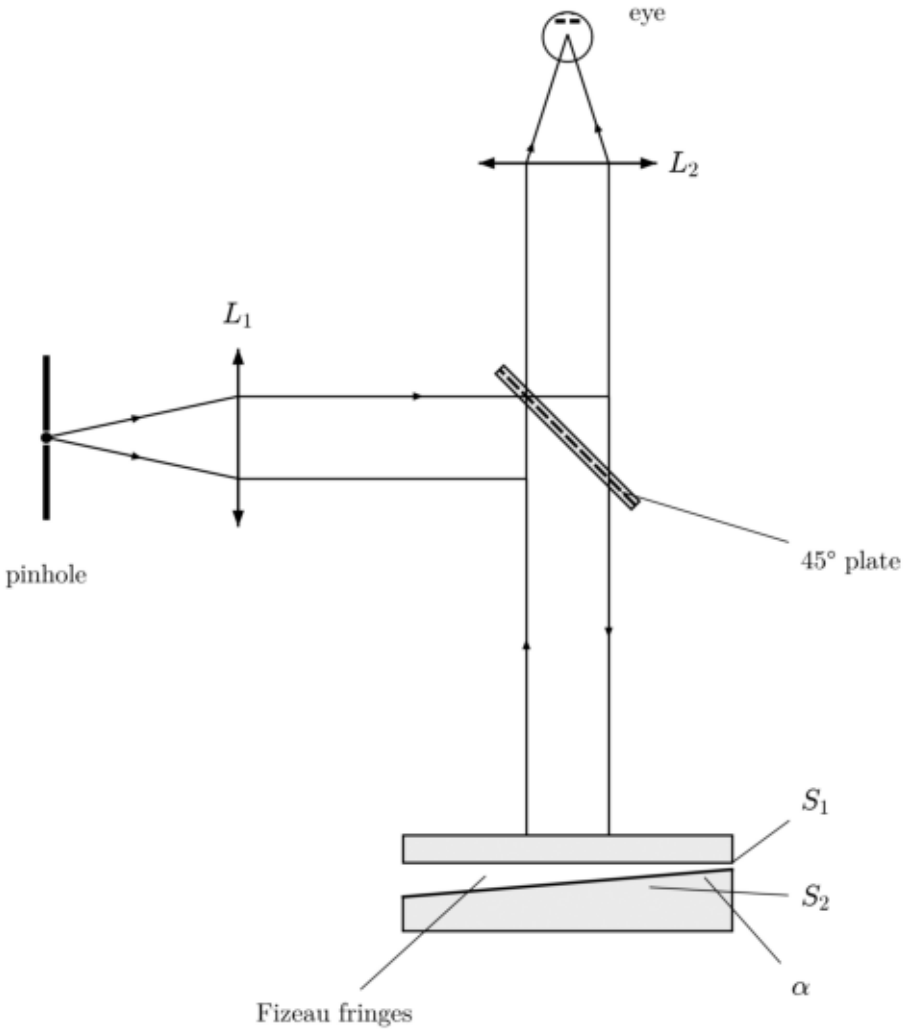


Figure 16.12 — DONE

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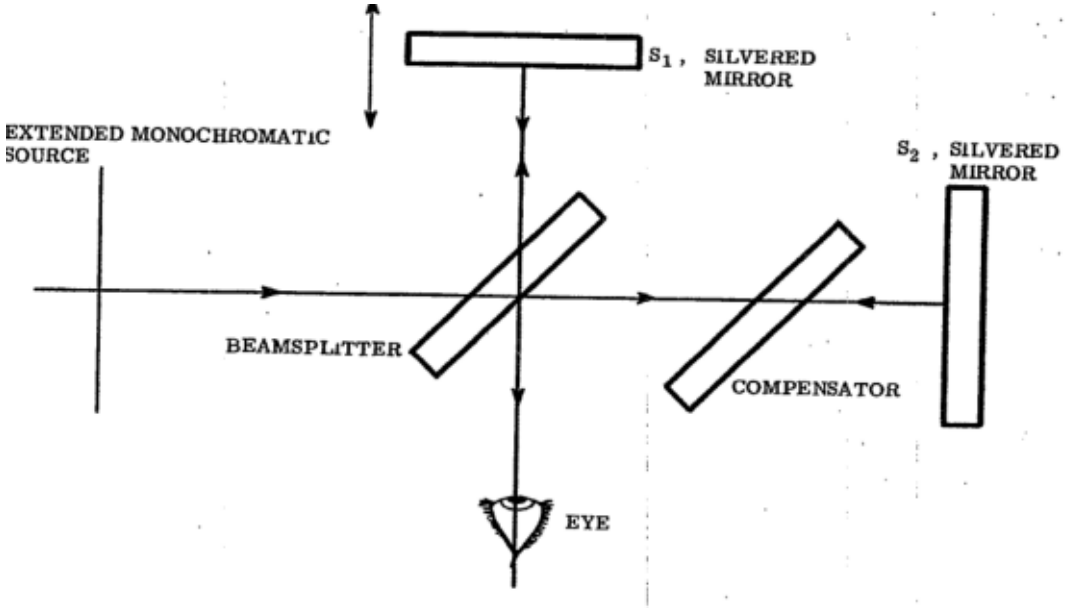


FIGURE 16. 12–Michelson's Interferometer.

FIGURE 16, 12-Michelson's Interferometer.

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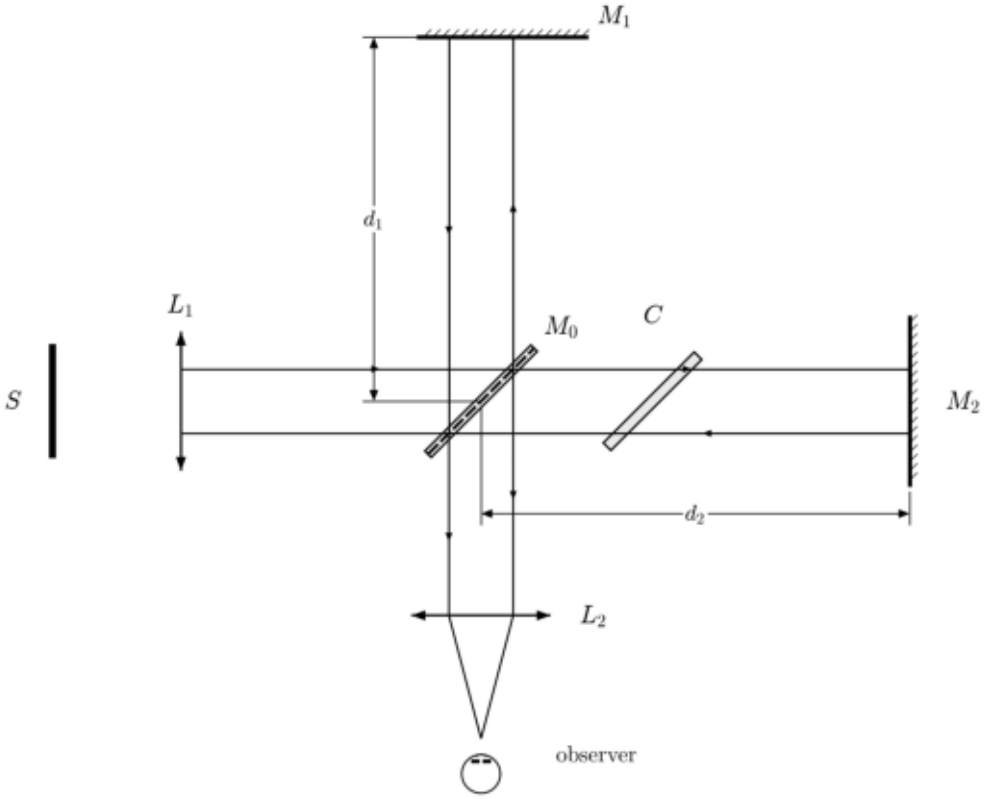


Figure 16.13 — DONE

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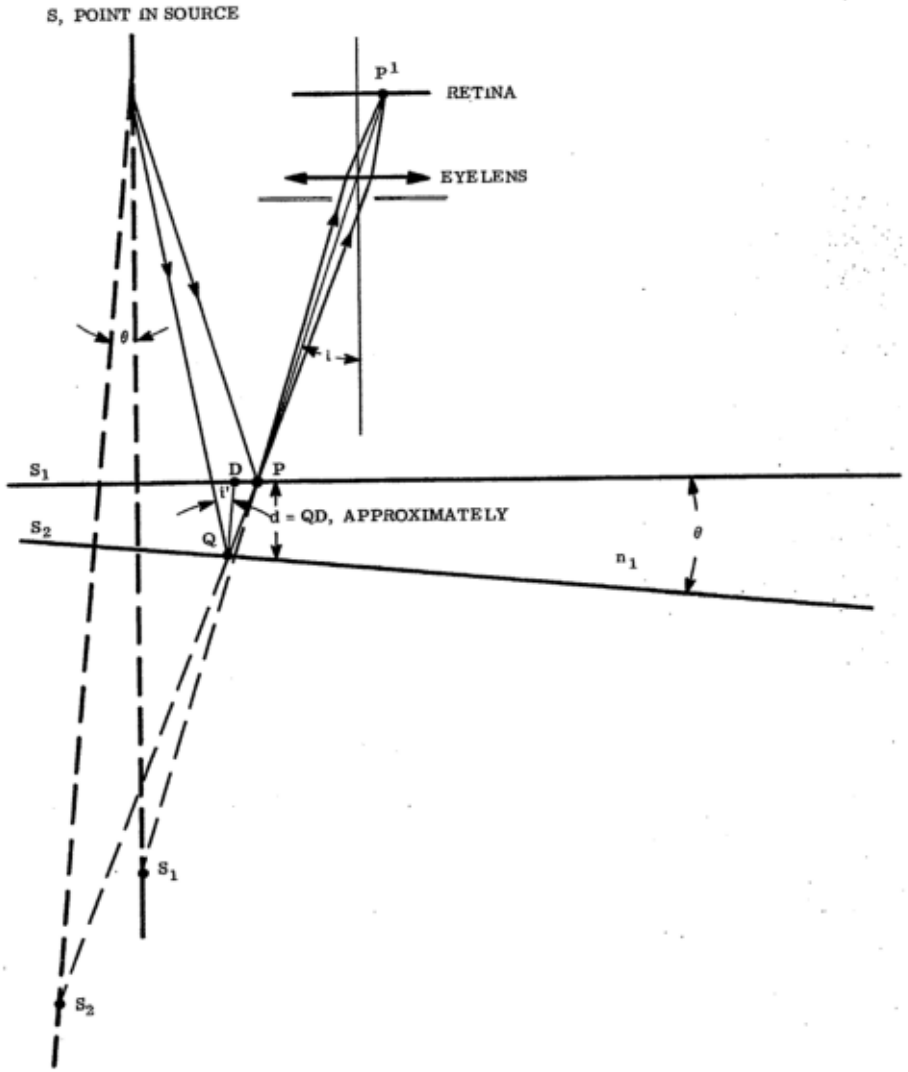


FIGURE 16. 13 - Construction showing how the Fizeau Fringes can appear localized at points, P, near the reflecting surfaces S_1 and S_2

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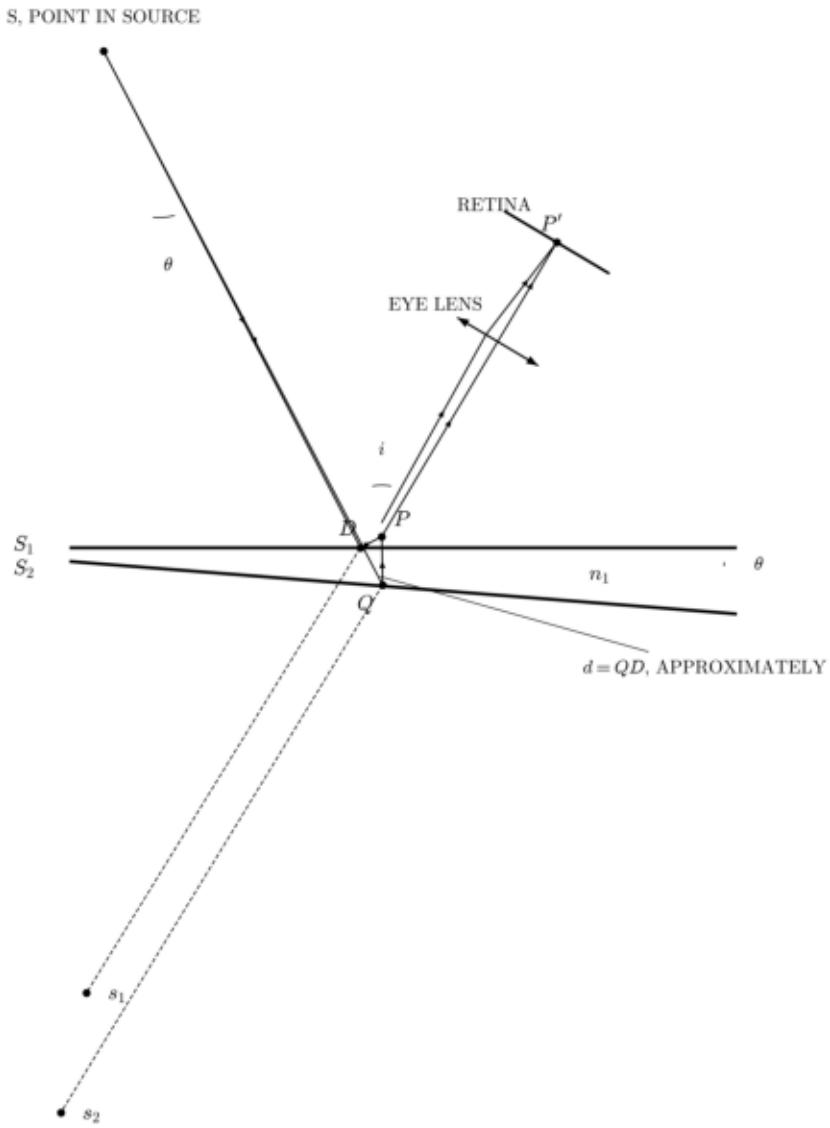


FIGURE 16, 13-Construction showing how the Fizeau Fringes can appear localized

Figure 16.14 — DONE

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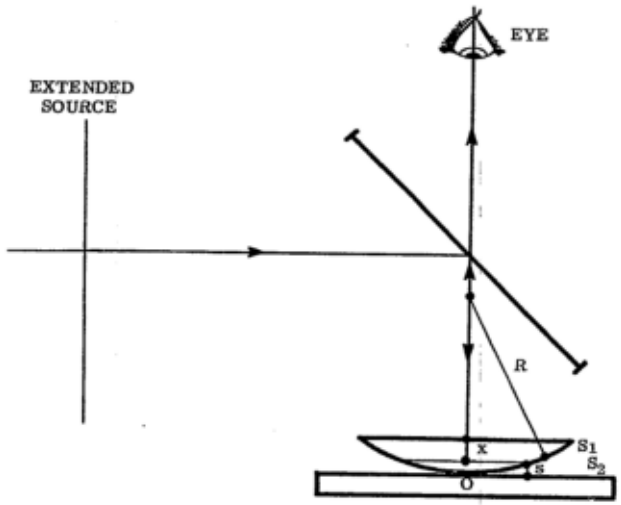


FIGURE 16. 14–Arrangement for obtaining Newton's Rings or Newton's Fringes.

Newton's fringes are Fizeau fringes along which the sagitta s of Figure 16. 14 is constant such that dark fringes

FIGURE 16, 14- Arrangement for obtaining Newton's Rings or Newton's Fringes.

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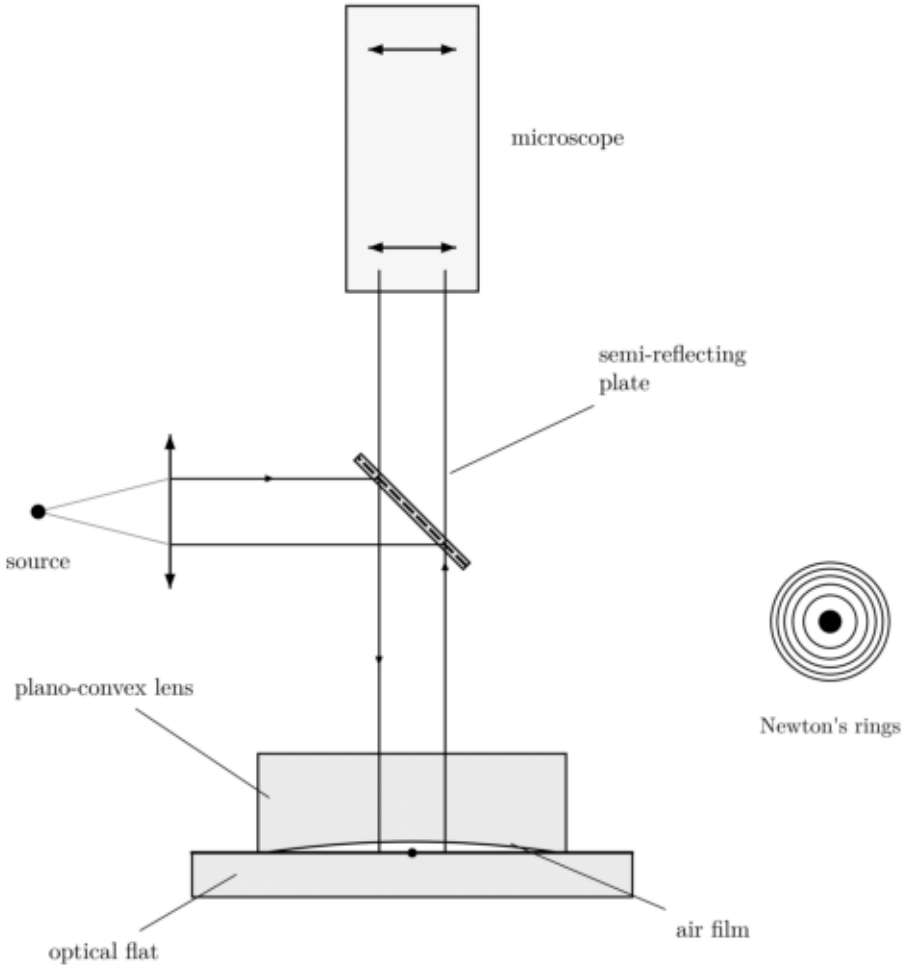


Figure 16.15 — DONE

1962 scan

16.13.1.7 We suppose that the end-mirror S_1 has a large radius R and seek to compute R from the radii of the circular fringes seen about point O when the eye lens and telescope focus the plane z = 0 upon the retina. We may suppose for simplicity that pinhole H and the center C of spherical surface S_1 fall upon the axis of the instrument. We take plane z = 0 through point O as the plane of reference. The plane wave reflected from S_2 appears to return to the observer as a plane wave along the direction OZ . The wave reflected from S_1 appears (apart from spherical aberration produced on reflection) as a spherical wave that expands from point F located at distance R/2 behind point O. We suppose that the distances x are small enough that spherical aberration on reflection can be ignored.

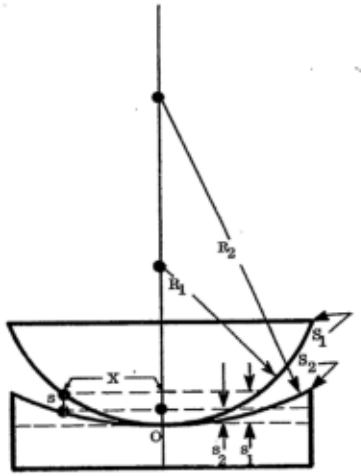


FIGURE 16.15--The Sagitta Method when the Reference Surface S_2 is a sphere.

FIGURE 16,15--The Sagitta Method when the Reference Surface S_ is a sphere.

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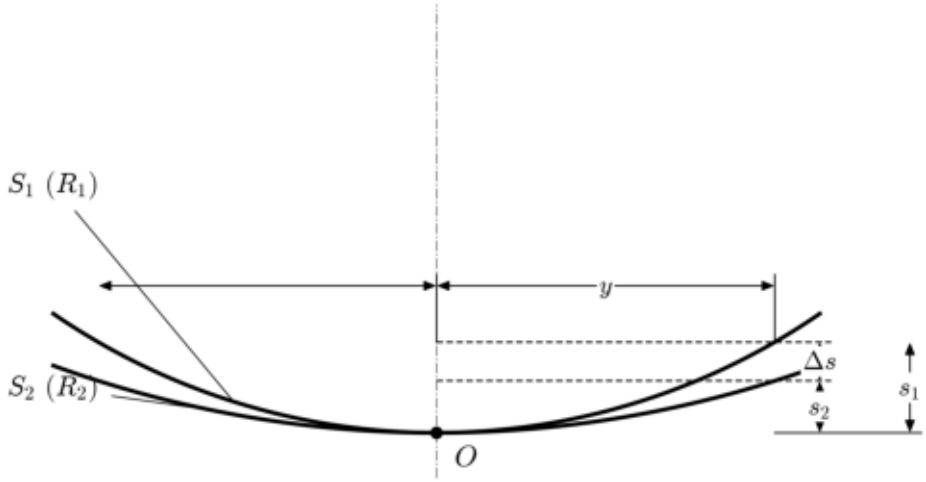
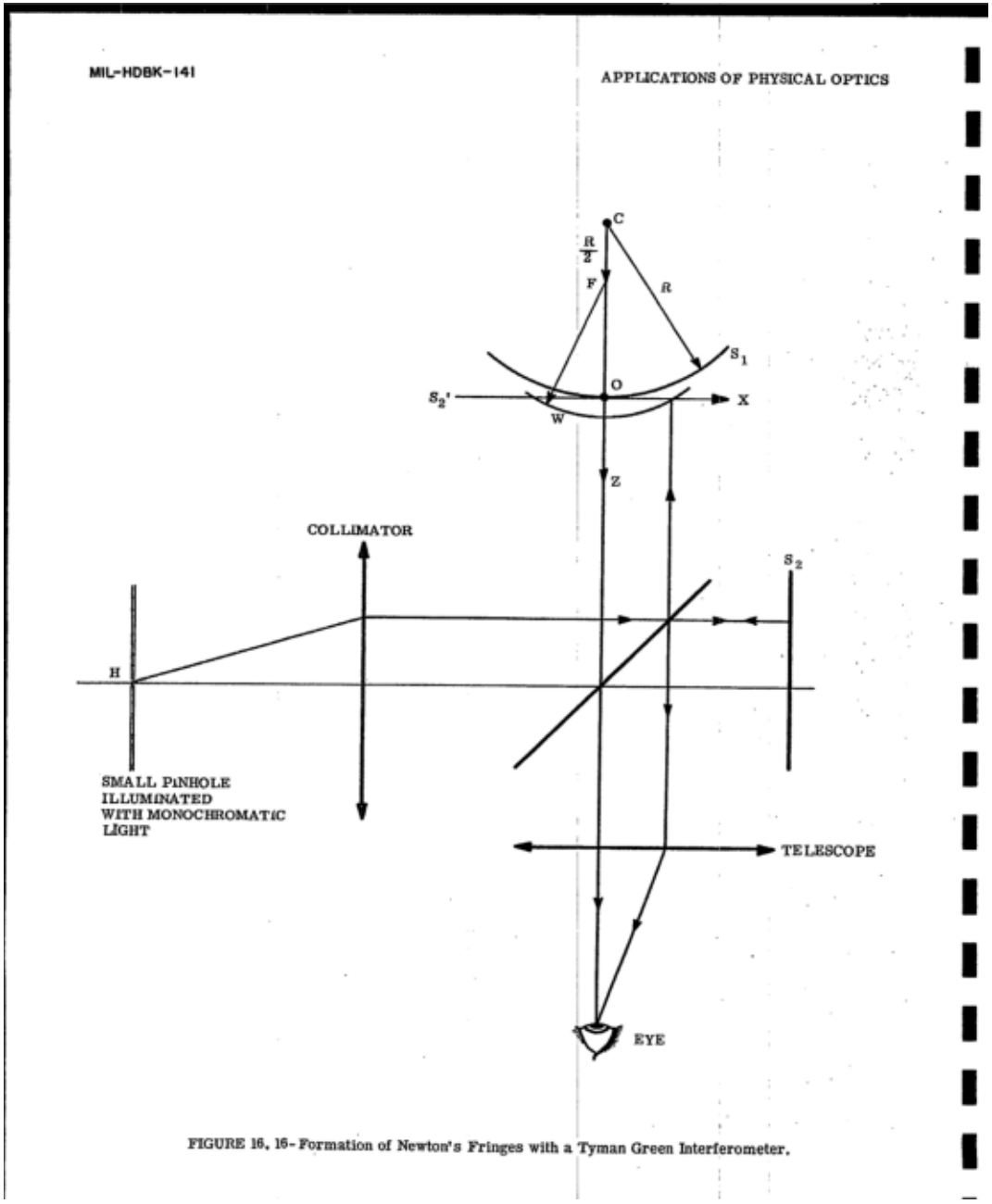


Figure 16.16 — DONE

1962 scan



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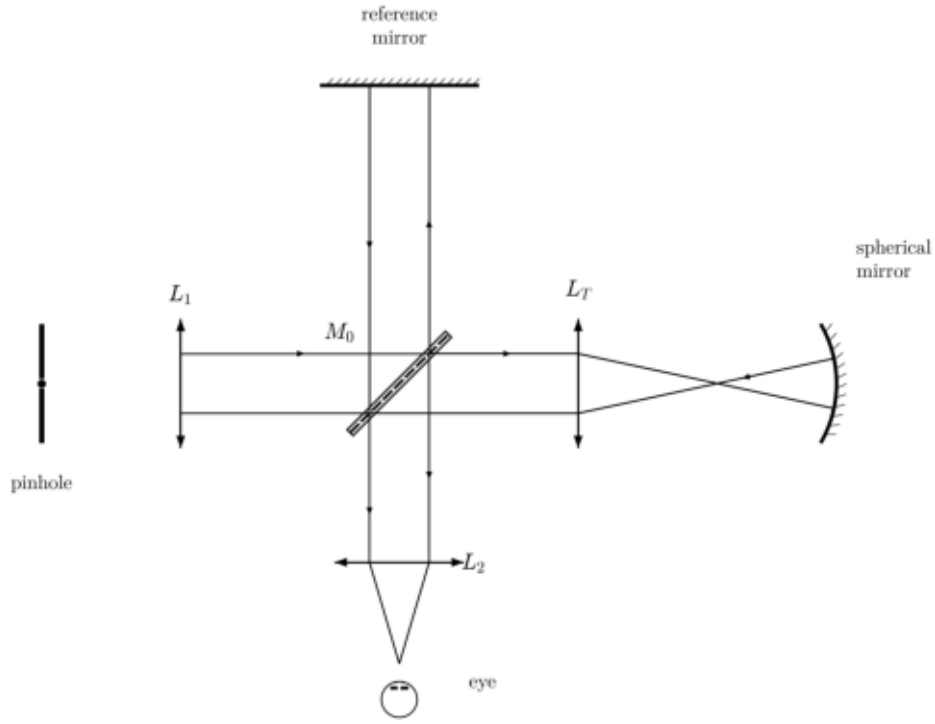
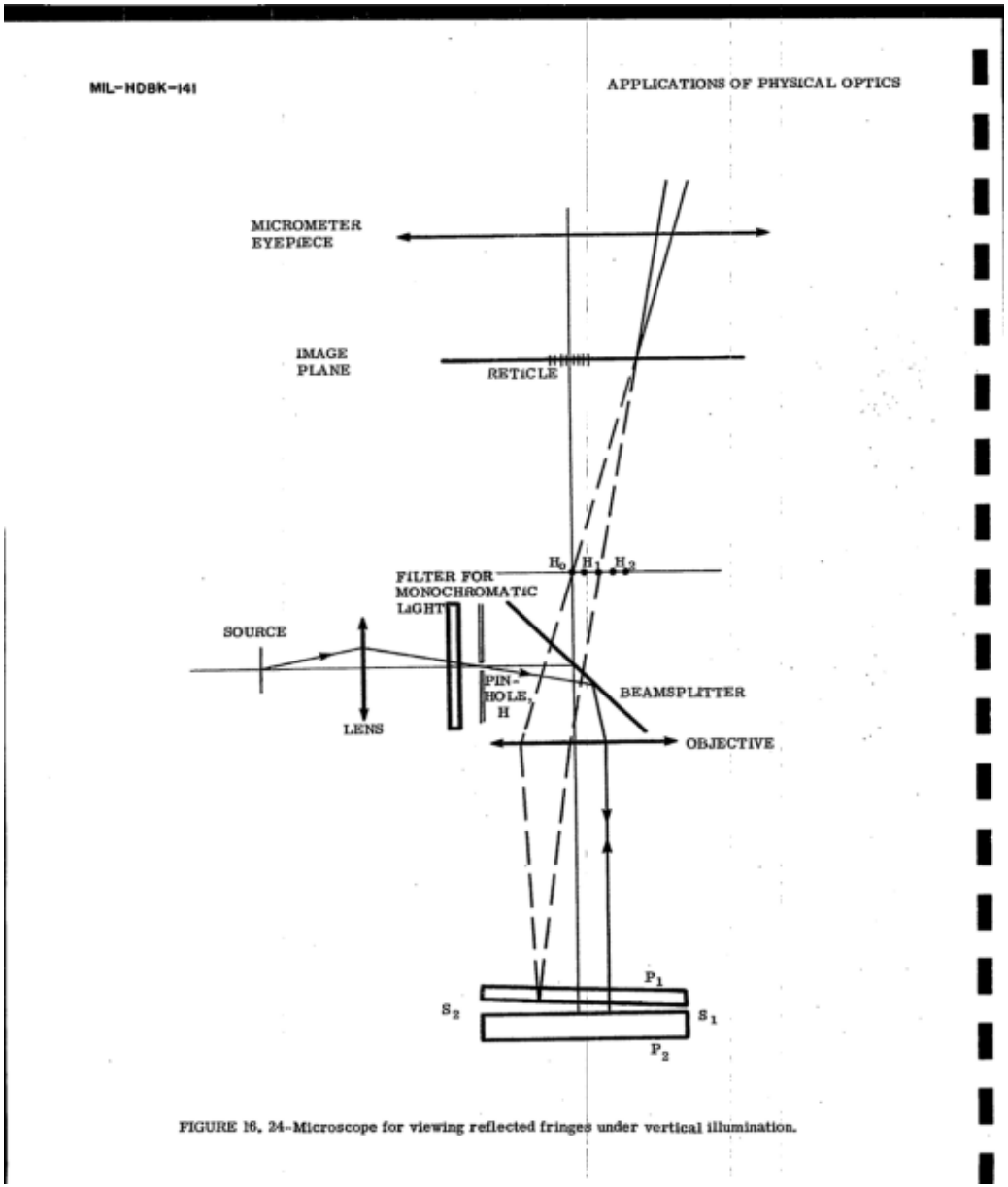


FIGURE 16, 16- Formation of Newton's Fringes with a

Figure 16.24 — DONE

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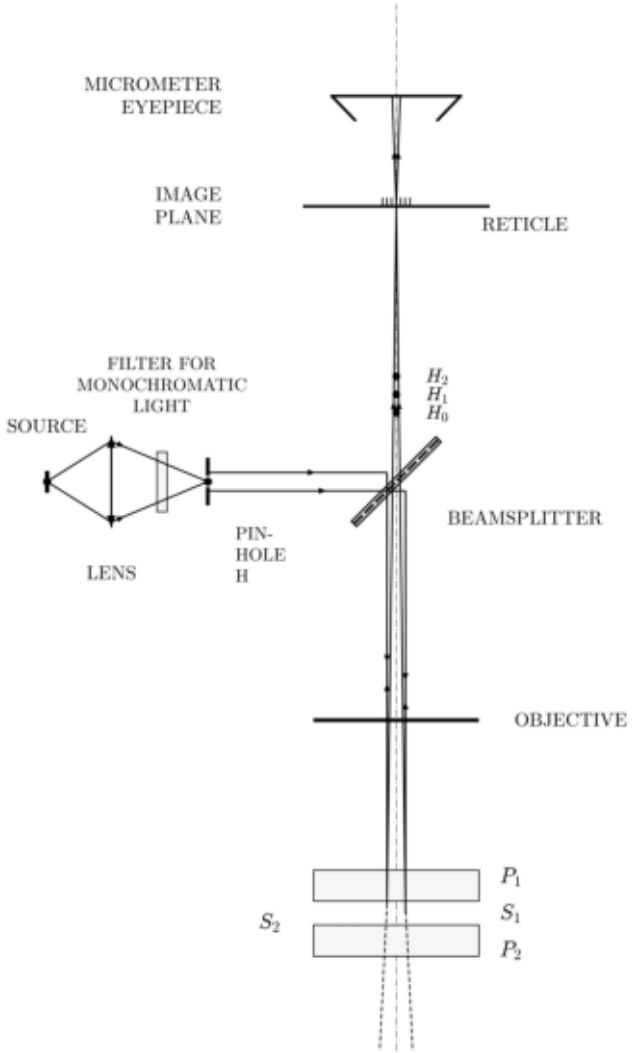


FIGURE 16, 24-Microscope for viewing reflected fringes under vertical illumination.

Figure 16.27 — DONE

1962 scan

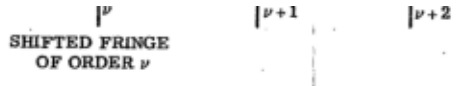


FIGURE 16. 26– Appearance of the narrow fringes when the thickness of the film is a small fraction of a wavelength and the film occupies the portion below the edge AB. If the surfaces S_1 and S_2 are optically flat, the indicated fringe widths, h, will be alike within the expiramental error of measuring h.

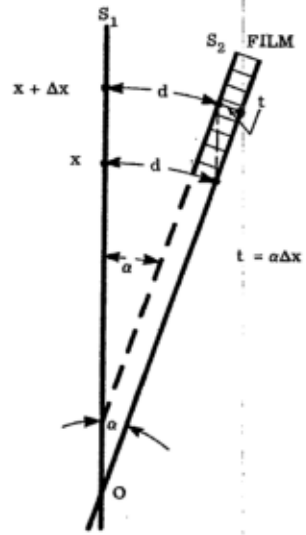


FIGURE 16. 27– Movement of an interference frings, x, to the position, x + Δ x, by the introduction of a film at thickness, t.

FIGURE 16, 27- Movement of an interference frings, x, to! the position, x + A x, by the ,

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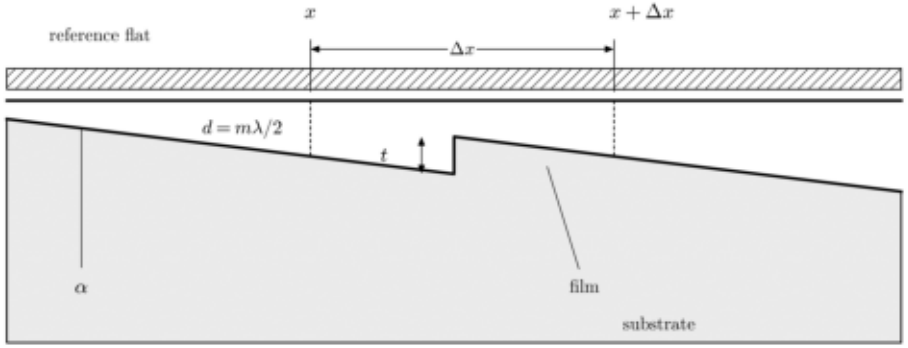
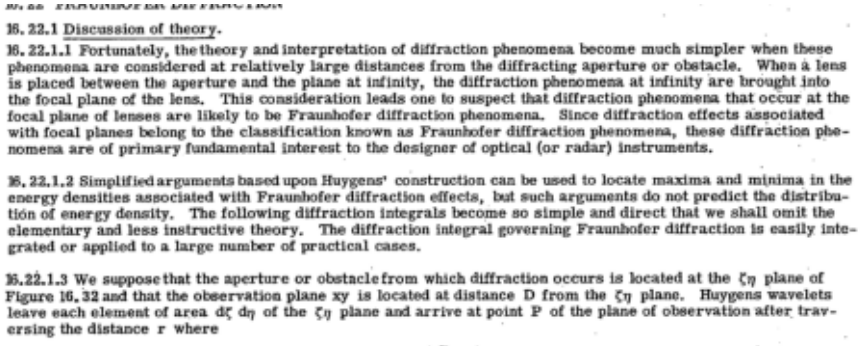


Figure 16.32 — DONE

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16.22.1 Discussion of theory.

16.22.1.1 Fortunately, the theory and interpretation of diffraction phenomena become much simpler when these phenomena are considered at relatively large distances from the diffracting aperture or obstacle. When a lens is placed between the aperture and the plane at infinity, the diffraction phenomena at infinity are brought into the focal plane of the lens. This consideration leads one to suspect that diffraction phenomena that occur at the focal plane of lenses are likely to be Fraunhofer diffraction phenomena. Since diffraction effects associated with focal planes belong to the classification known as Fraunhofer diffraction phenomena, these diffraction phenomena are of primary fundamental interest to the designer of optical (or radar) instruments.

16.22.1.2 Simplified arguments based upon Huygens' construction can be used to locate maxima and minima in the energy densities associated with Fraunhofer diffraction effects, but such arguments do not predict the distribution of energy density. The following diffraction integrals become so simple and direct that we shall omit the elementary and less instructive theory. The diffraction integral governing Fraunhofer diffraction is easily integrated or applied to a large number of practical cases.

16.22.1.3 We suppose that the aperture or obstacle from which diffraction occurs is located at the $\zeta\eta$ plane of Figure 16.32 and that the observation plane xy is located at distance D from the $\zeta\eta$ plane. Huygens wavelets leave each element of area $d\xi\, d\eta$ of the $\zeta\eta$ plane and arrive at point P of the plane of observation after traversing the distance r where

Figure 16.32 and that the observation plane xy is located at distance D from the tq plane. Huygens wavelets

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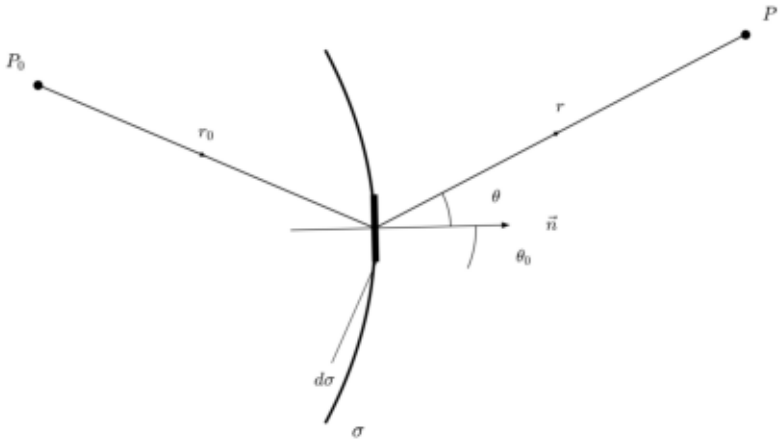
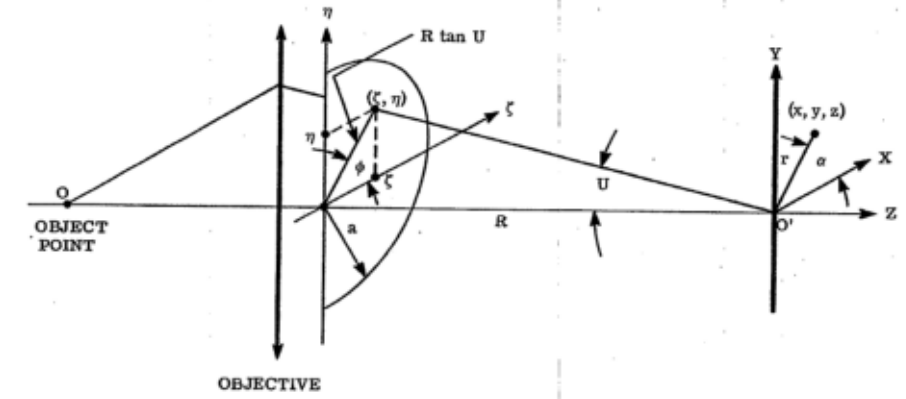


Figure 16.36 — DONE

1962 scan



GURE 16. 36—Notation with respect to objectives that have axial symmetry and circular apertures of radius a.

FIGURE 16. 36~'Notation with respect to objectives that have axial symmetry and circular apertures of

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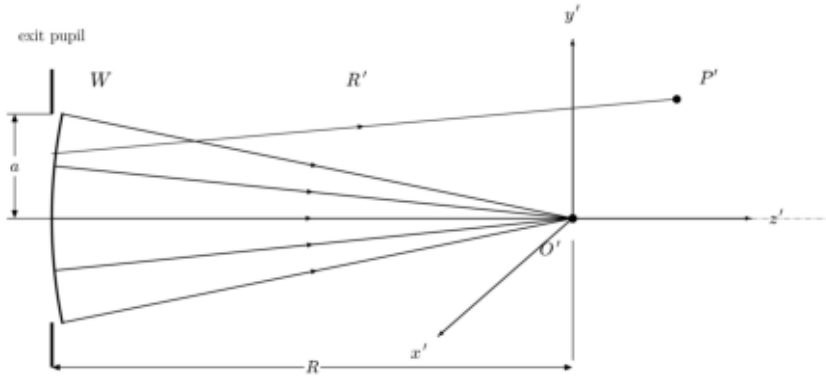


Figure 16.37 — DONE

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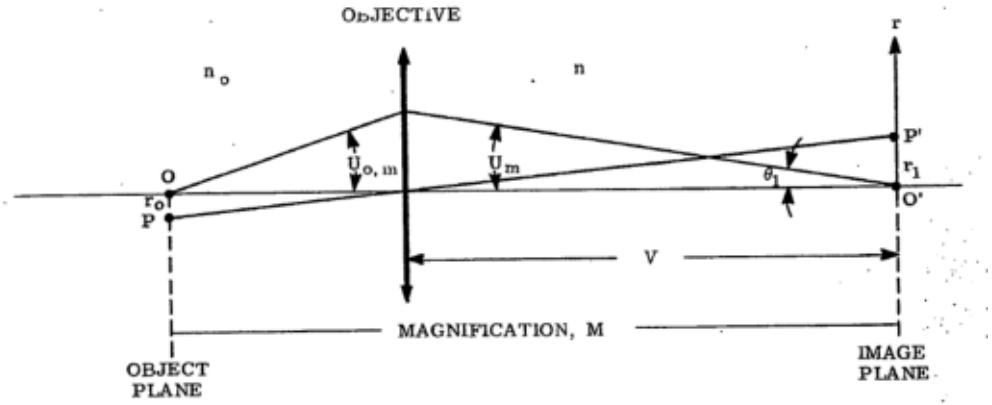


FIGURE 16. 37- Notation with respect to the resolution of two self-luminous object points by objectives having circular apertures.

FIGURE 16. 37-Notation with respect to the resolution of two self-luminous object points

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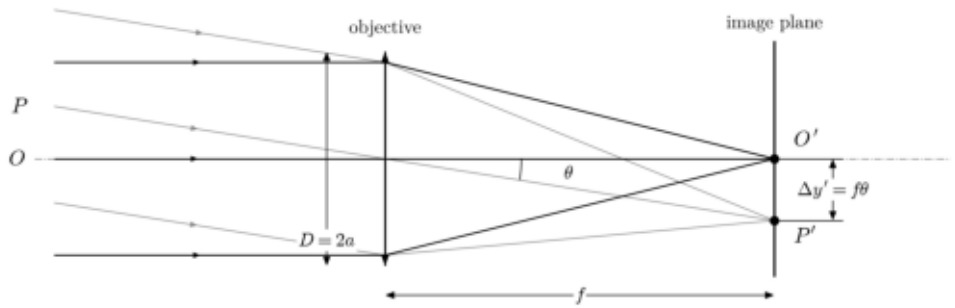


Figure 20.1 — DONE

1962 scan

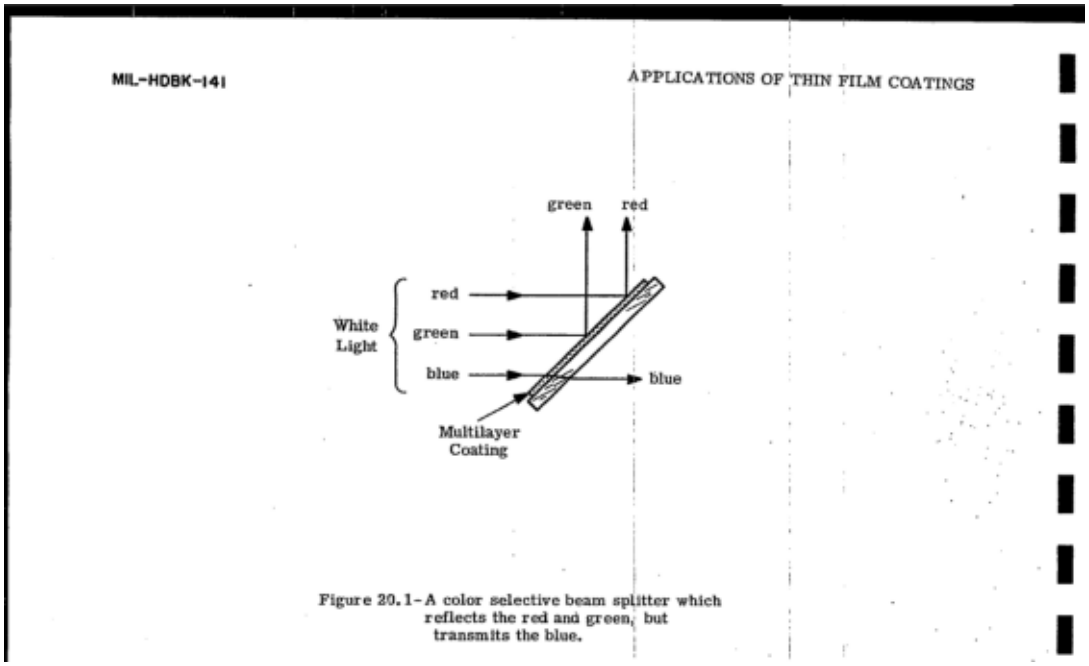


Figure 20,1-A color selective beam splitter which

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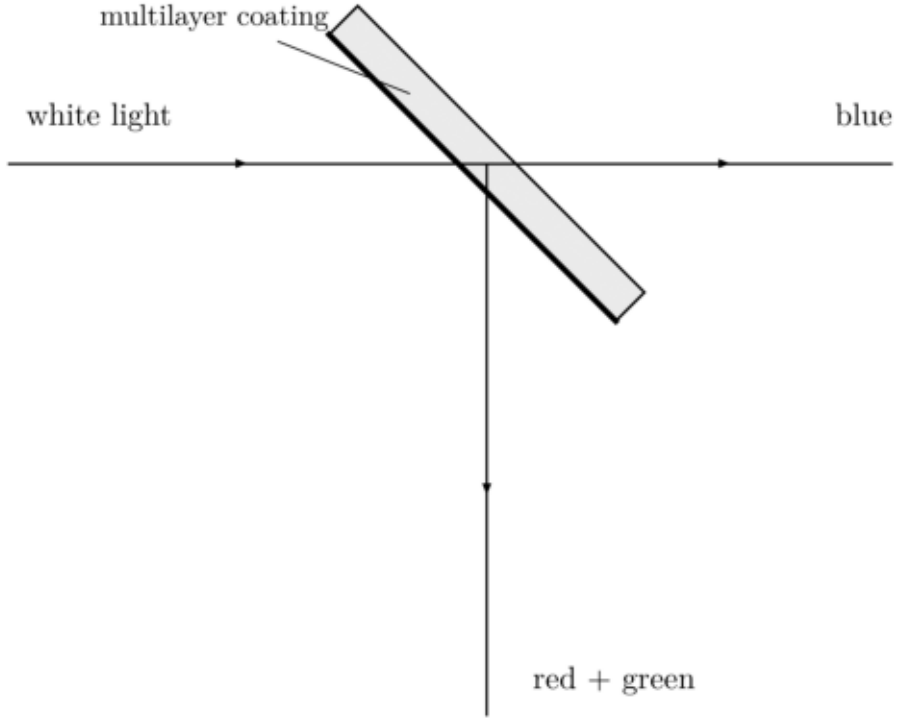


Figure 20.2 — DONE

1962 scan

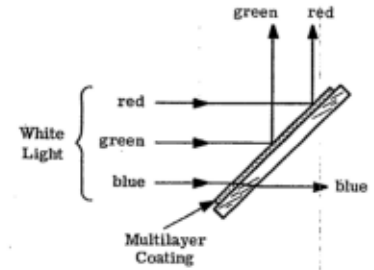


Figure 20.1–A color selective beam splitter which reflects the red and green, but transmits the blue.

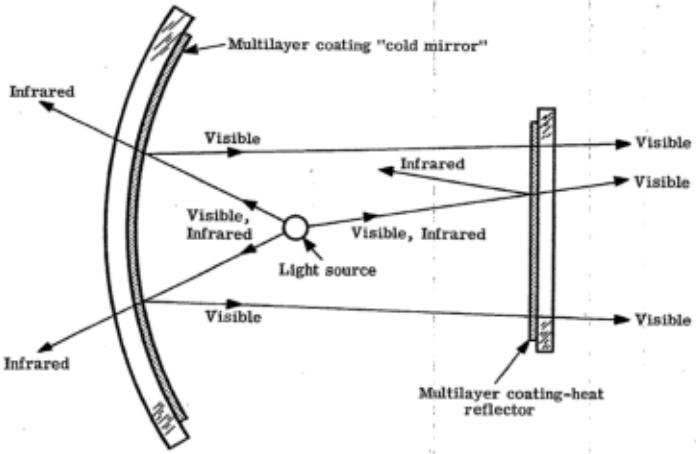


Figure 20.2–A cold mirror and heat reflector reduce the heat directed towards the film in a projection system.

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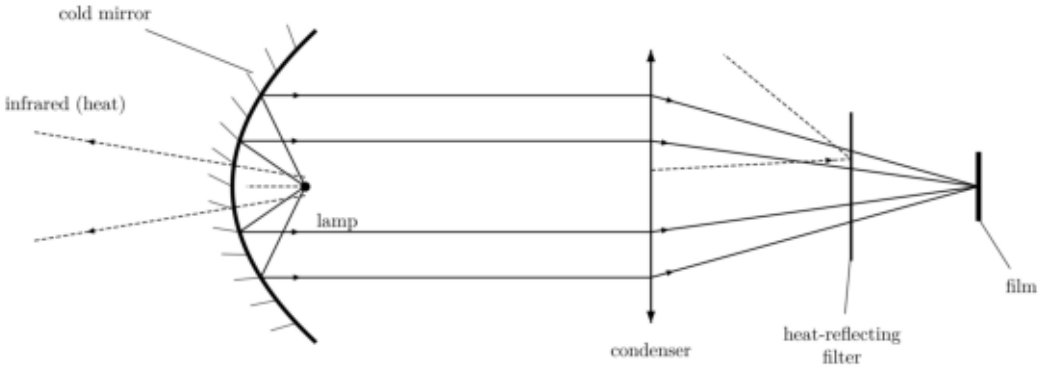


Figure 20.88 — DONE

1962 scan

Kodak Company.

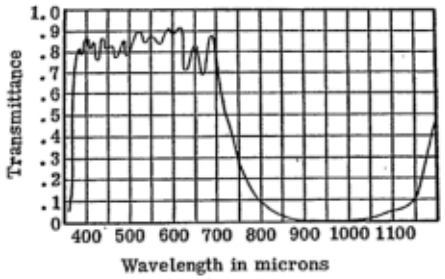


Figure 20.87– Measured spectral transmittance of a multilayer heat reflecting filter. Courtesy of Balzers Aktiengesellschaft, Liechtenstein.

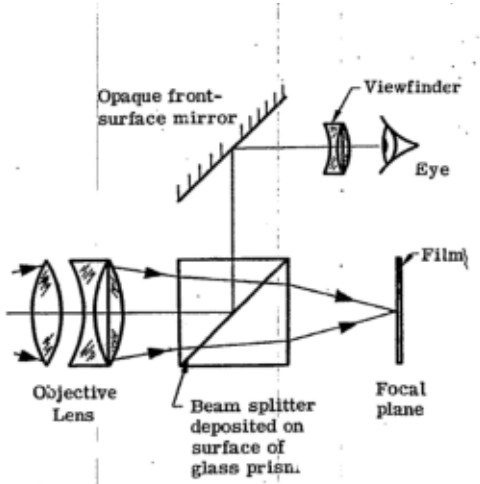


Figure 20.88– The utilization of a beam splitter in a camera to deflect an image into the view finder.

Figure 20, 88-

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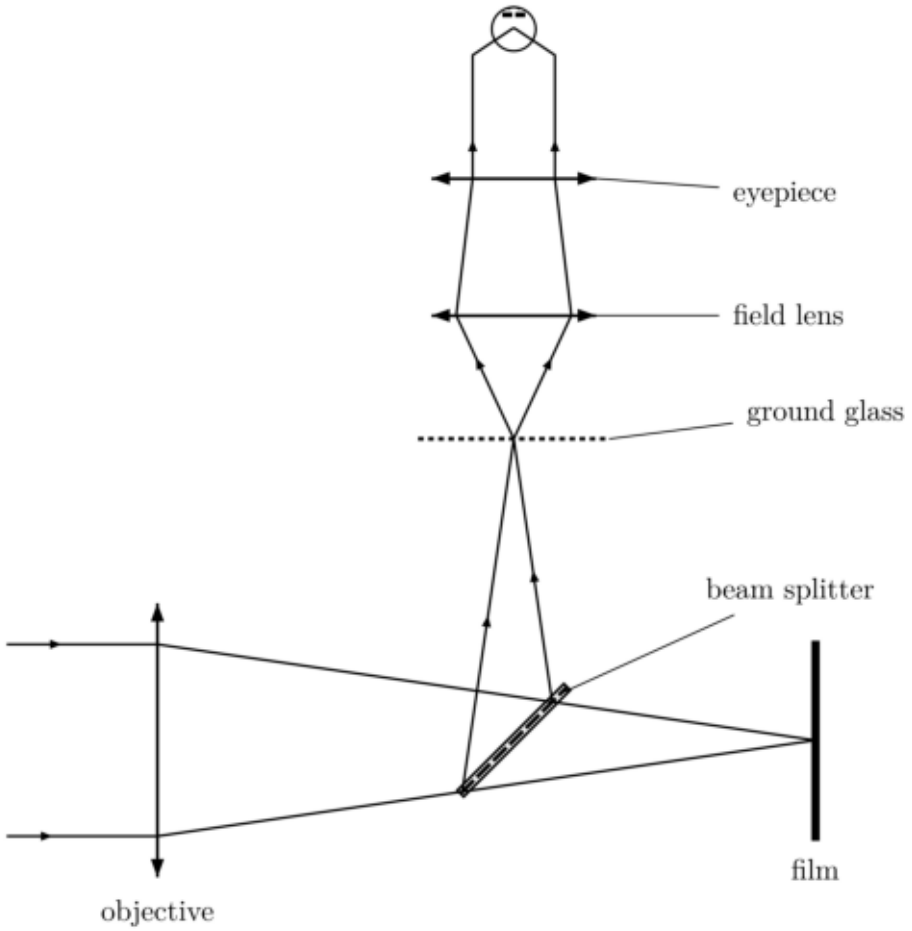


Figure 20.98 — DONE

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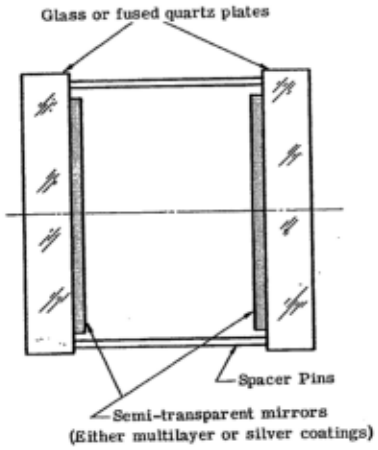


Figure 20.98- The essential parts of a Fabry-Perot interferometer.

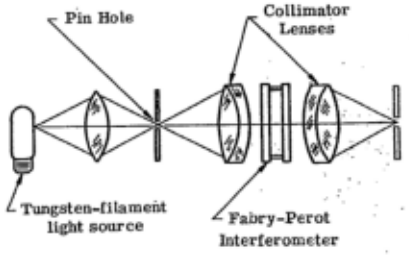


Figure 20.99- A method of illuminating the Fabry-Perot interferometer with collimated light so that it can be used as an optical filter.

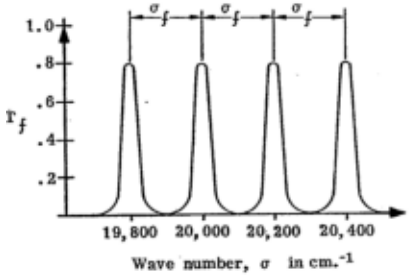


Figure 20.100 – The transmission as a function of wave number of a Fabry-Perot interferometer with t_g = 0.0025 cm .

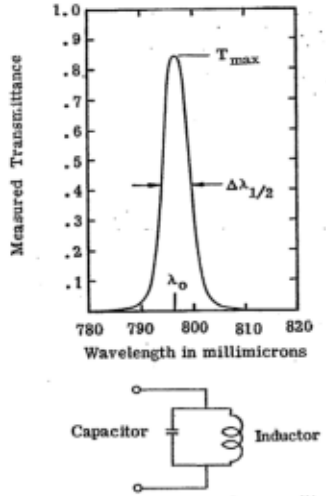
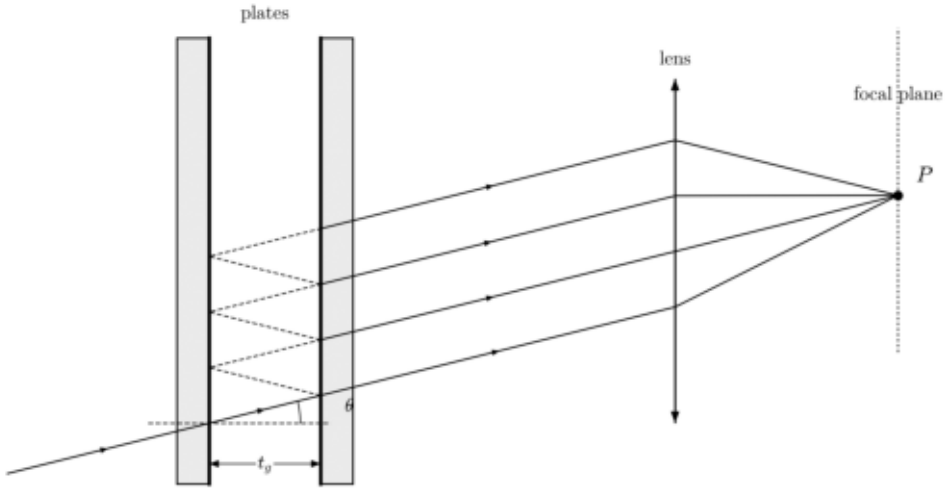


Figure 20.101- Measured spectral transmittance of Fabry-Perot type multilayer filter. Courtesy of Dr. C. Alley.
A parallel resonant circuit.

20-75

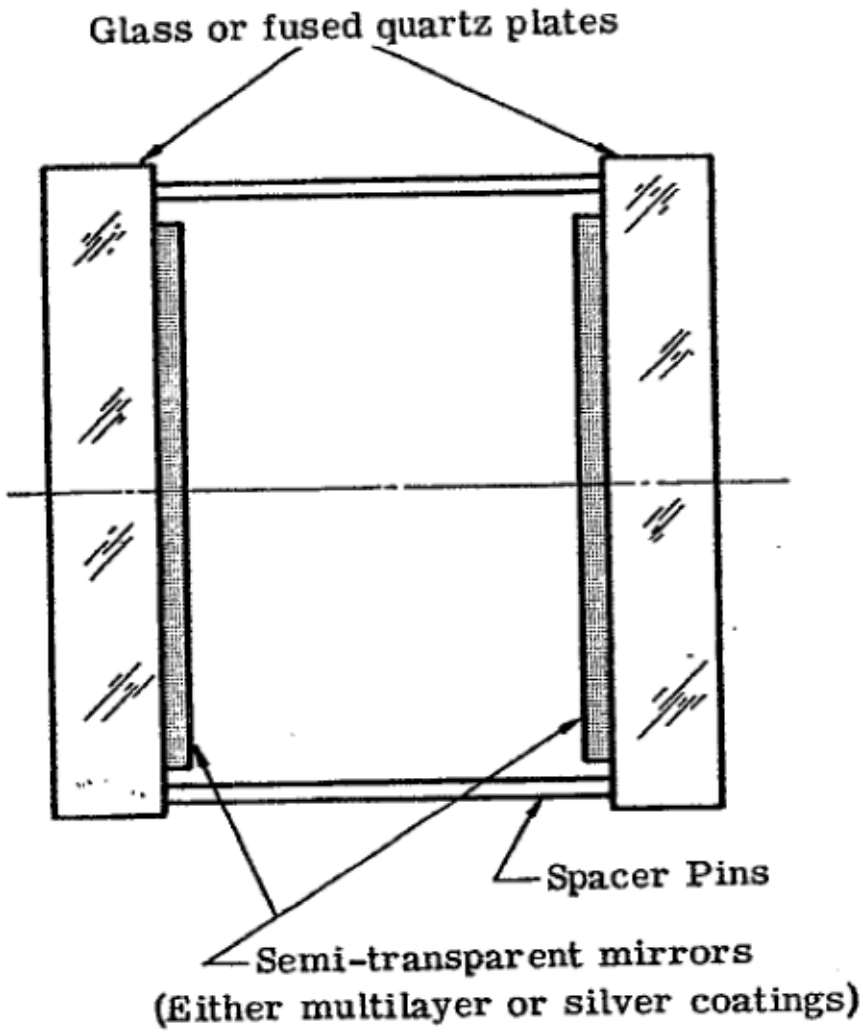
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Figure 20.99 — DONE

1962 scan



'igure 20.98- The essential parts of a Fabry-Perot interferometer.

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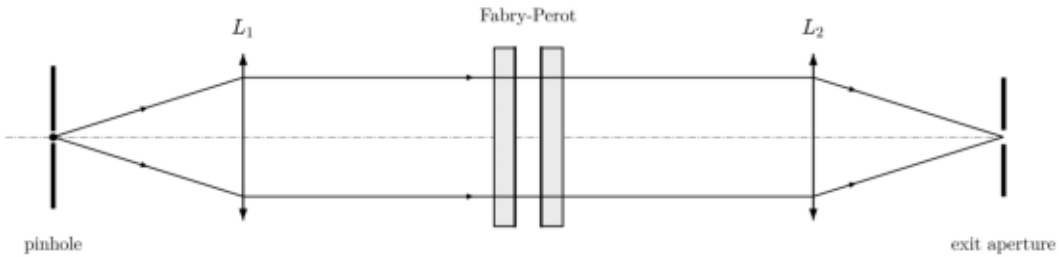


Figure 20.99- A method of illuminating the Fabry-

Figure 21.2 — DONE

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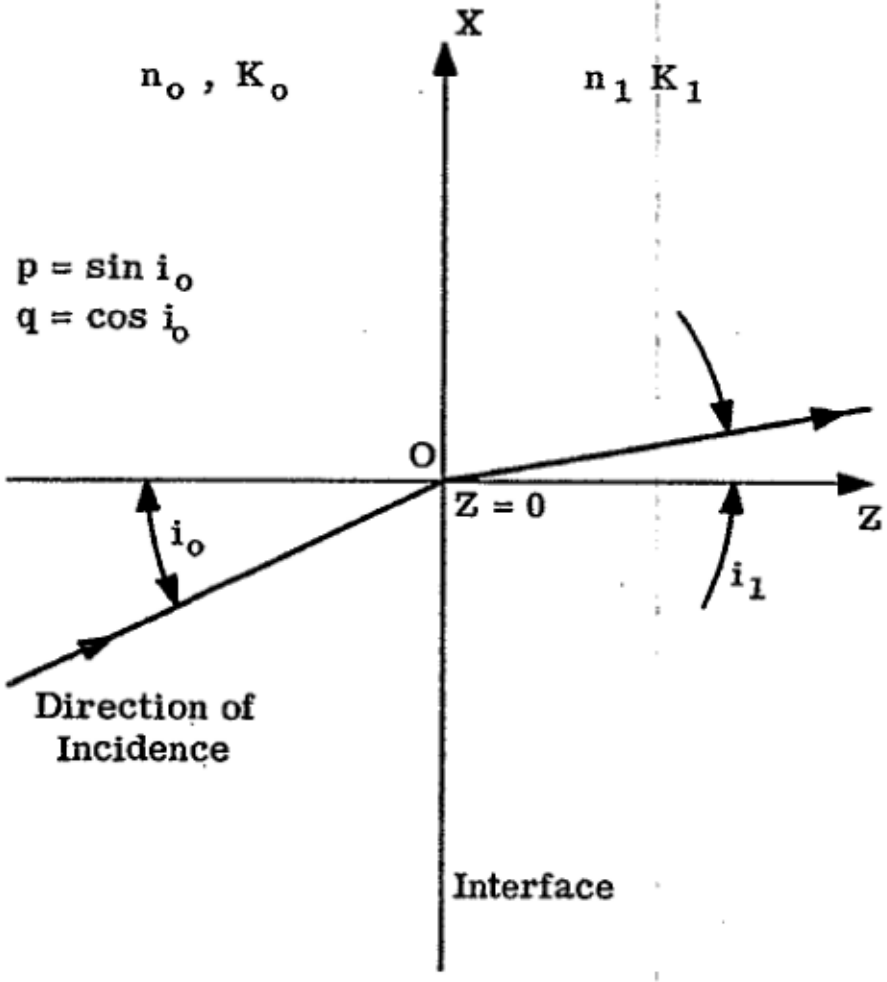


Figure 21. 2- Notation with respect to Fresnel's coefficients for oblique incidence. Plane XZ is chosen as the plane of incidence. i_o is the angle of incidence.

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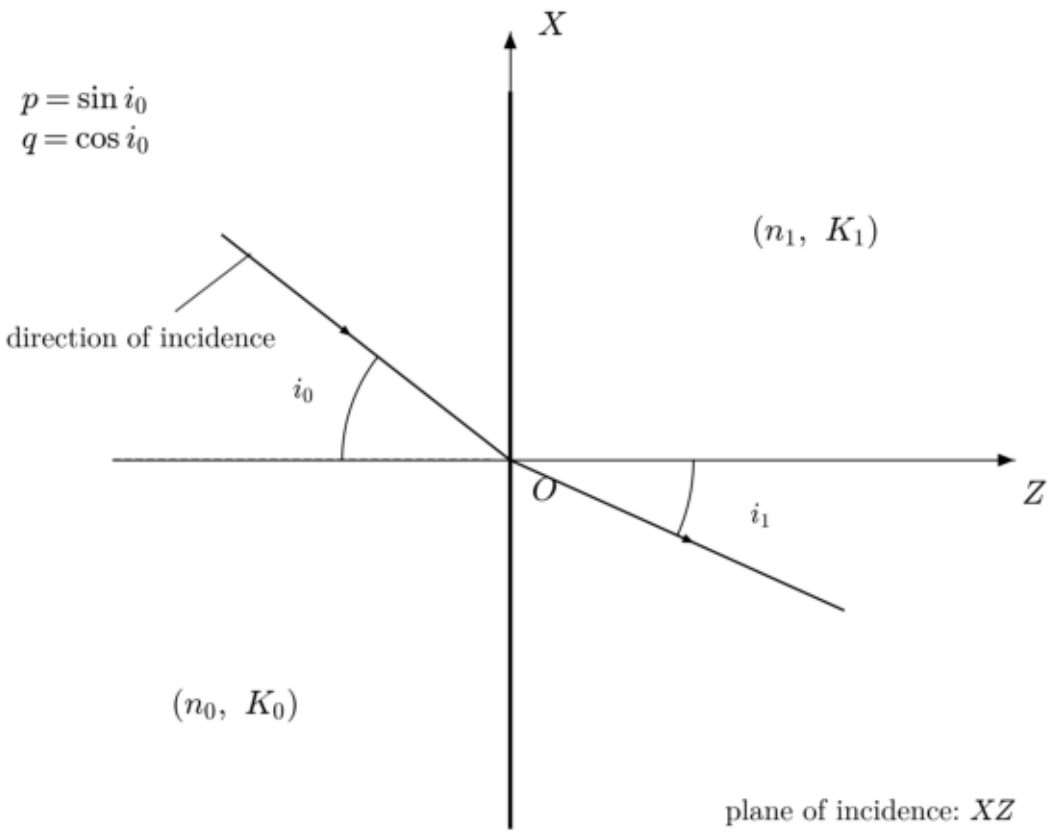


Figure 21.5 — DONE

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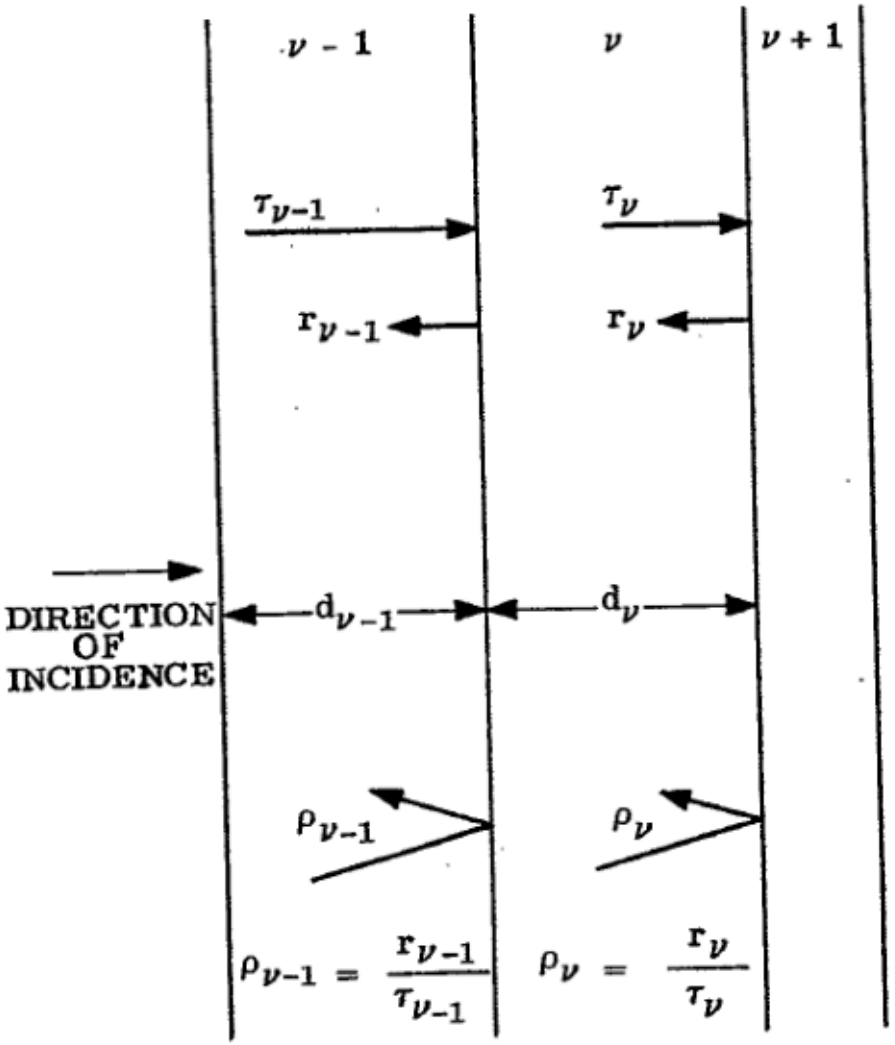


Figure 21. 5- Flow conditions at the $(\nu\text{-}1)^{th}$ and ν^{th} layers.

...has the factor $e^{i\beta\nu}$ because the flow r_{ν} must c...

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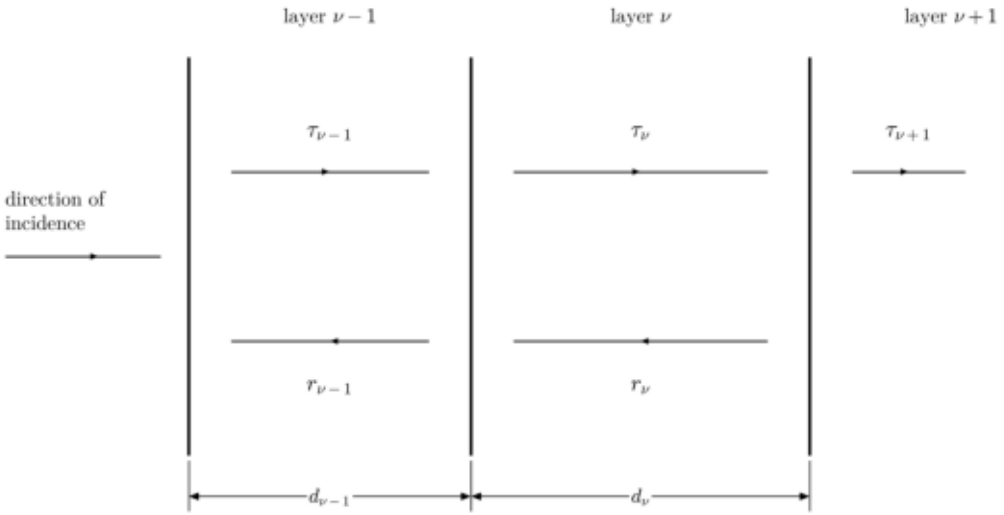


Figure 21, 5- Flow conditions at the (v-1)th and

Figure 22.3 — DONE

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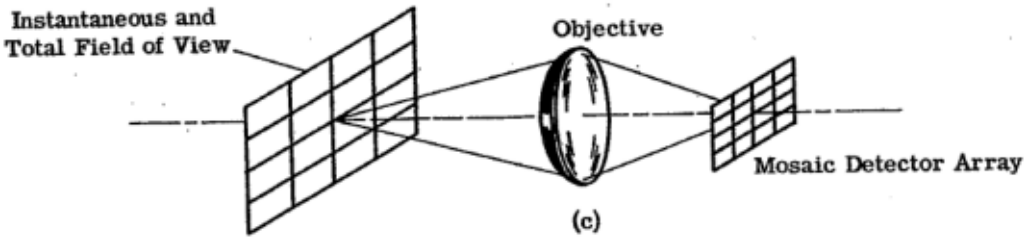


Figure 22.3 – Examples of basic scanning systems.

Figure 22.3 - Examples of basic scanning systems.

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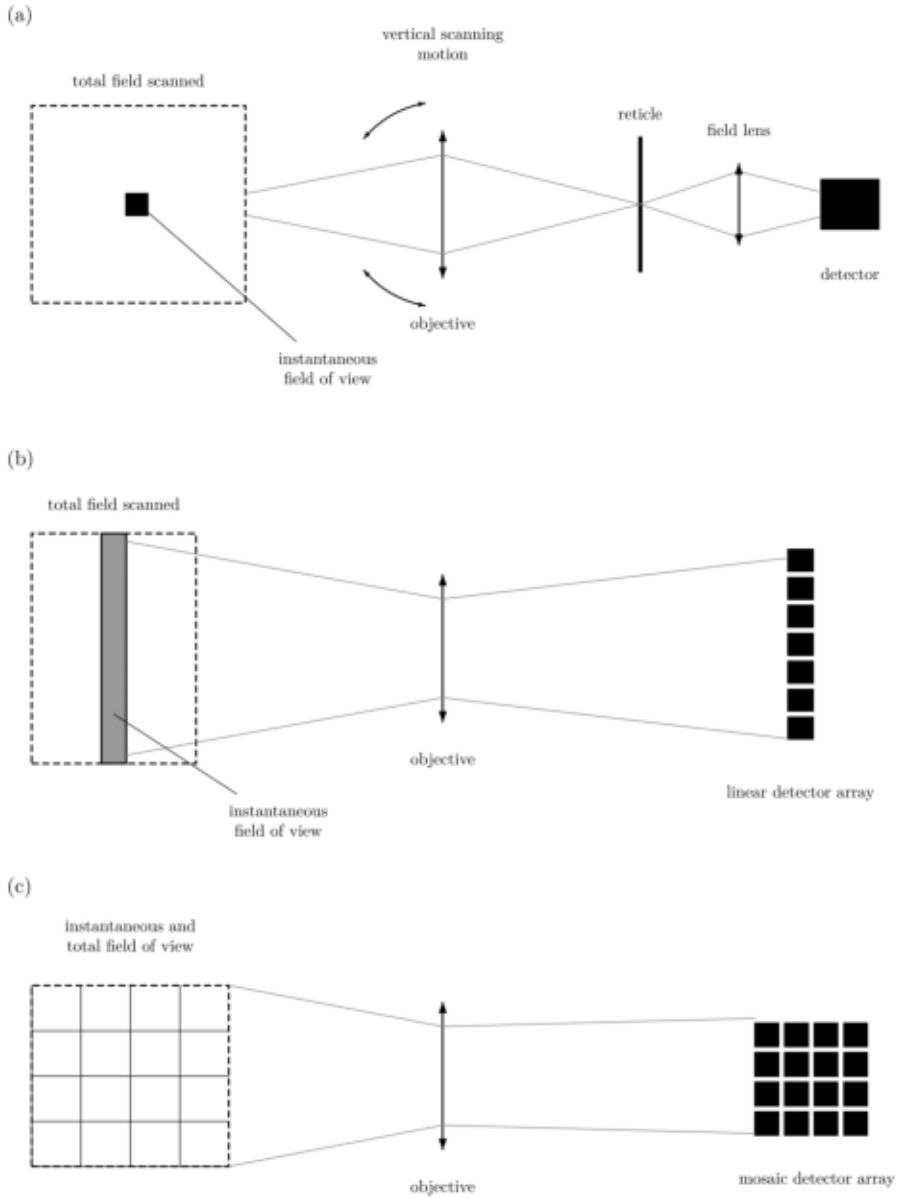


Figure 23.4 — DONE

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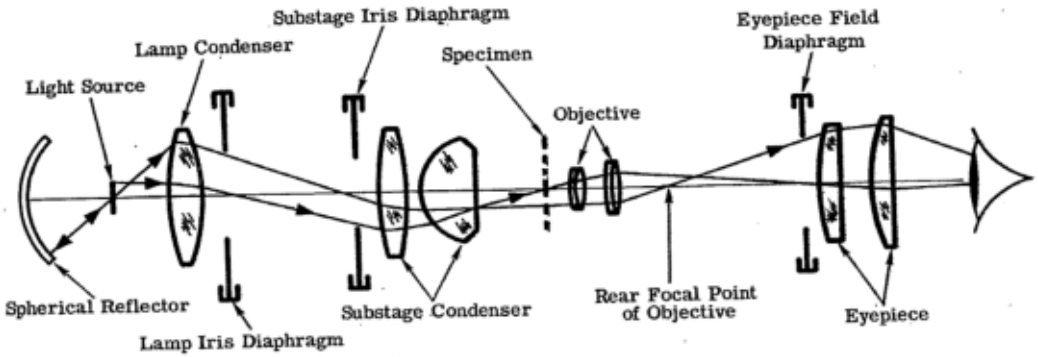


Figure 23.4- Kohler Illumination, schematic diagram.

23-5

Figure 23.4- Kohler Tumination, schematic diagram.

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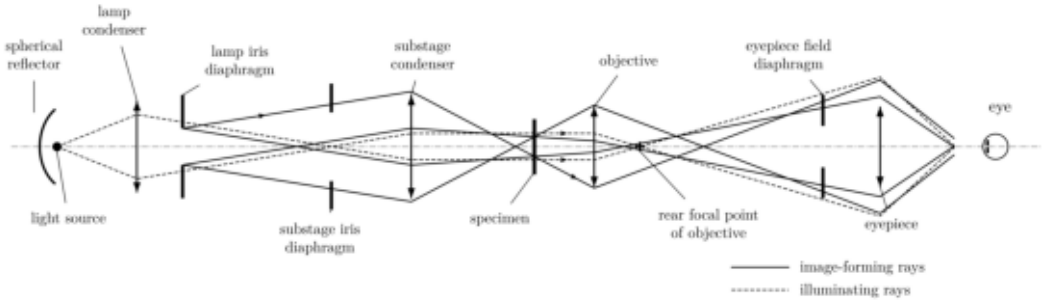


Figure 23.5 — DONE

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23.3.2.3.3 An iris diaphragm is often located very close to the lamp condenser. With this design, the condenser (in Kohler illumination) and the iris diaphragm are imaged on the specimen, and if the iris diaphragm is adjustable in diameter, its image can be made to precisely fill the field. An adjustable iris diaphragm will prevent illumination of a greater area of the specimen than is necessary, and will also prevent scattered light, with its resultant loss of contrast, from entering the microscope.

23.3.2.3.4 The lamp condenser is usually a two lens, air-spaced system. It should be as well corrected for spherical aberration as is possible. The focal length must be correctly determined in order to image the filament of the lamp large enough to fill the iris diaphragm of the substage condenser (Kohler illumination) at a convenient distance (approximately 15 inches). The diameter of the condenser must be large enough so that its image, as formed by the substage condenser, covers the specimen field, when viewed with a 16mm focal length,(10x) microscope objective. It is readily apparent then, that the smaller the light source, the greater must be the speed of the condenser.

23.3.2.4 Vertical illuminators. For opaque specimens, vertical illumination is required for seeing surface details. Vertical illumination requires that the specimen field be uniformly and intensely illuminated, and that the illuminated field be limited to that portion of the specimen which is in the field of view. If the illuminated field is not limited as mentioned, an undesirable amount of light is scattered by the unviewed portion of the specimen, by the edges of the objective lenses, or by the walls of the objective. This scattering will reduce the contrast in the visual field.

23.3.2.4.1 Vertical illuminator, type A. In this type of vertical illuminator, as shown in Figure 23.5, the incident light is focussed on the specimen by being passed through the objective, in a reverse direction, and onto the specimen. The light source in this type is usually a low voltage, concentrated filament bulb, located in a housing extending laterally to the axis of the microscope. The system consists of the light source, a condenser with an iris diaphragm mounted near the light source, a second condenser, and a semi-reflector at 45° to the microscope's axis for throwing light into the rear of the microscope's objective. The two condensers image the lamp filament at the exit pupil of the objective. The second condenser images the iris diaphragm (and the first condenser) at a virtual distance of about 160mm from the microscope's objective. Therefore, an image of the iris is formed by the microscope's objective on the specimen and it is uniformly illuminated. The field covered by the illuminated spot on the specimen can be regulated in size by adjusting the diameter of the field iris diaphragm, thereby preventing the scattering of light.

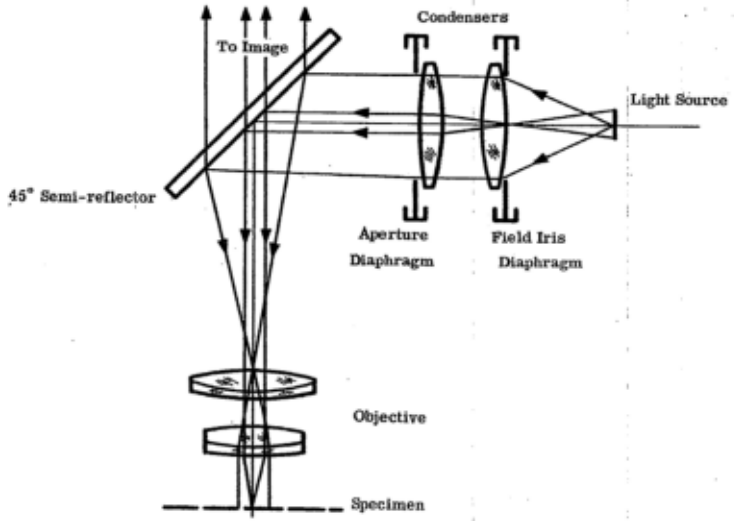


Figure 23.5- Vertical illumination.

Figure 23.5- Vertical illumination.

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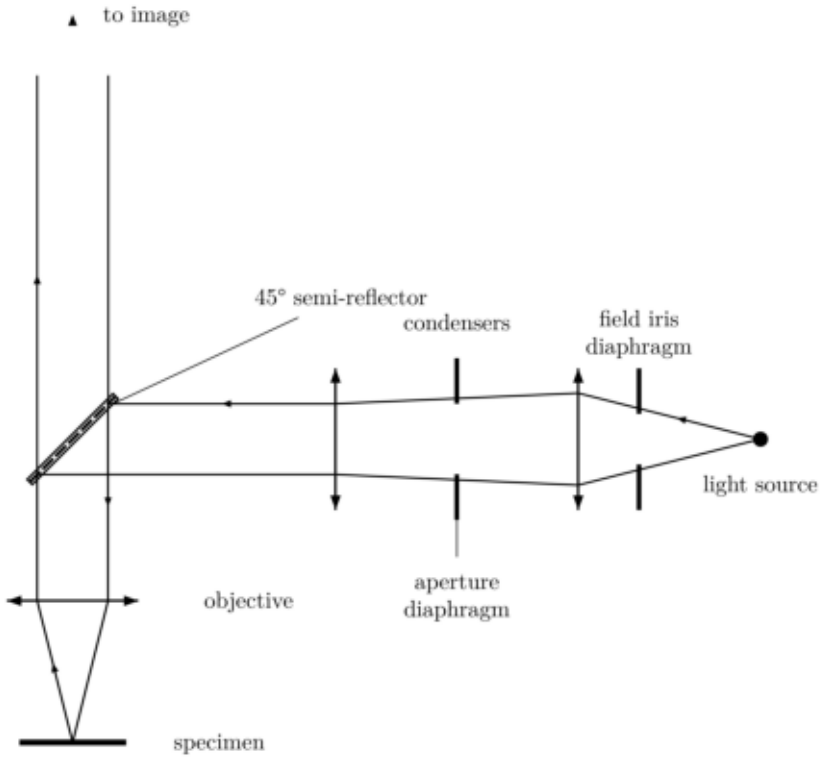


Figure 23.12 — DONE

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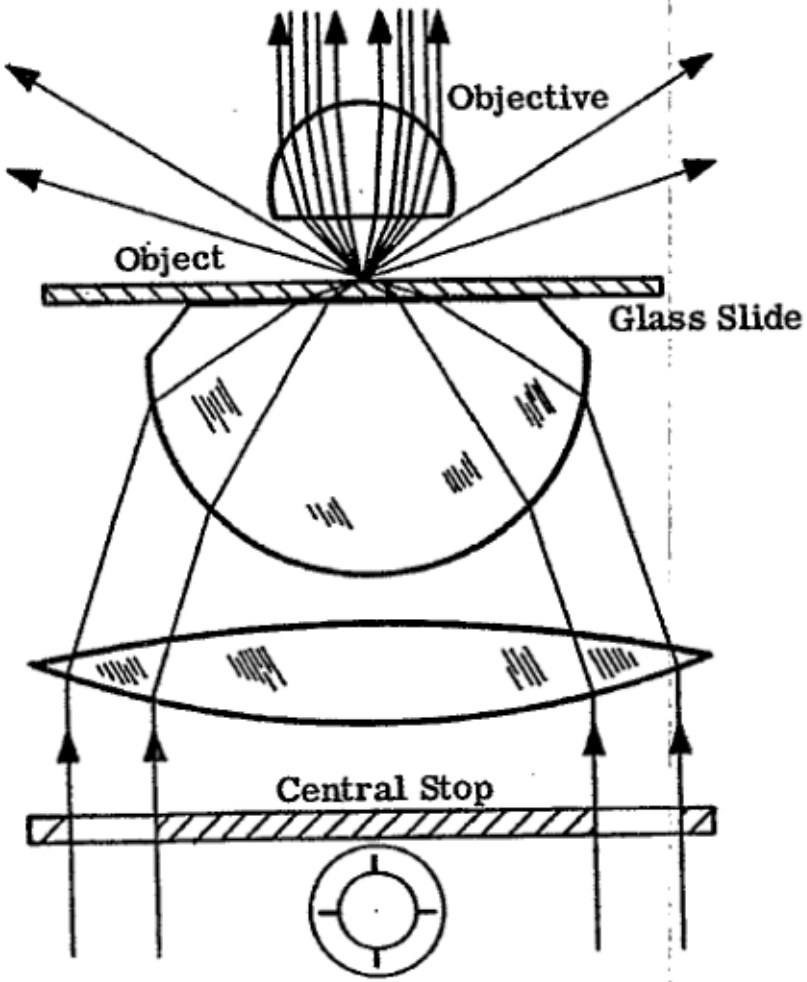


Figure 23. 12- Refracting condenser with a central stop for dark-field illumination.

Figure 23, 12-Refracting condenser with a cen-

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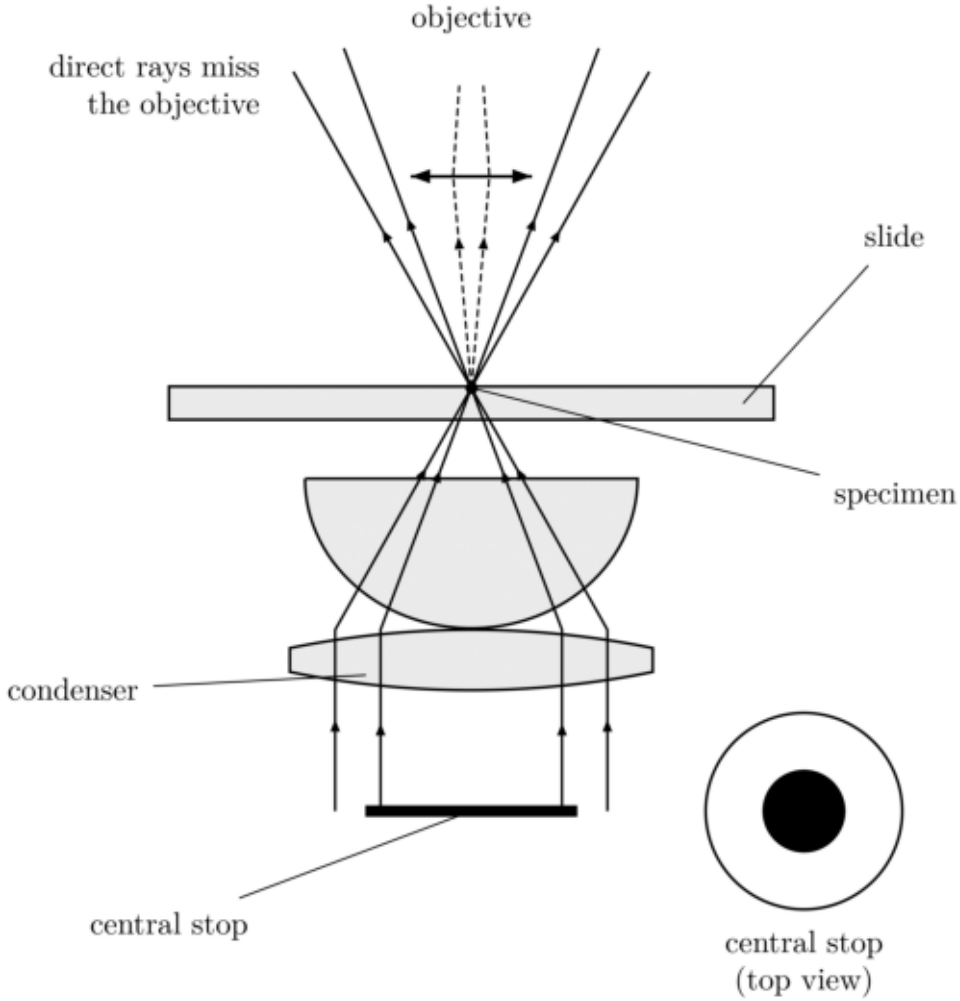


Figure 23.15 — DONE

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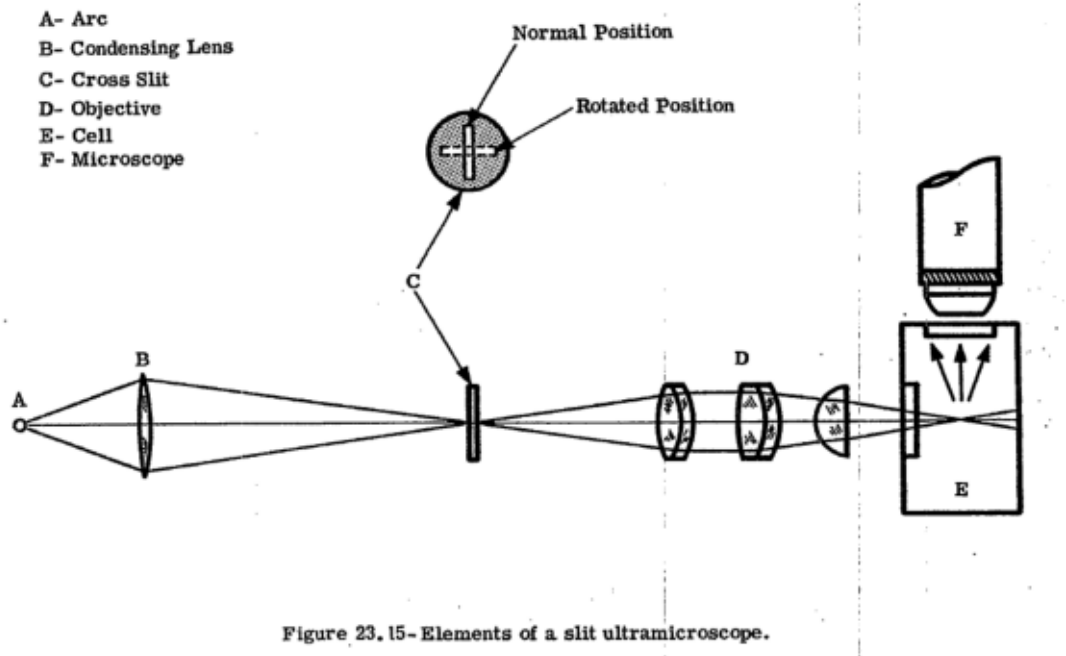


Figure 23.15- Elements of a slit ultramicroscope.

piece scale, the width and length of the beam can be measured. Therefore, if the cross slits (c) are rotated

Figure 23, 15- Elements of a slit ultramicroscope.

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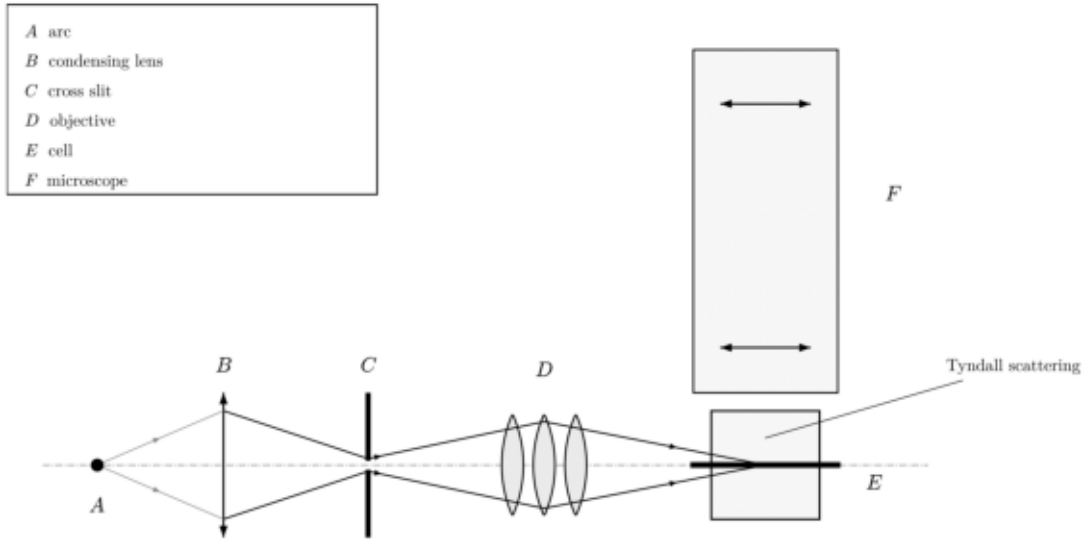
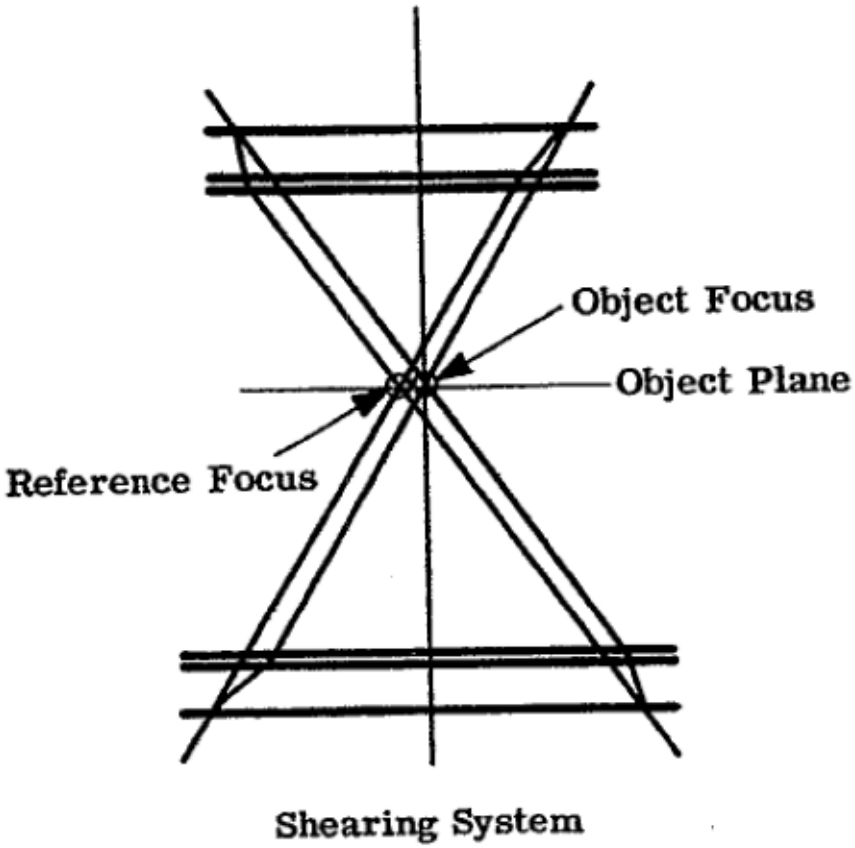


Figure 23.20 — DONE

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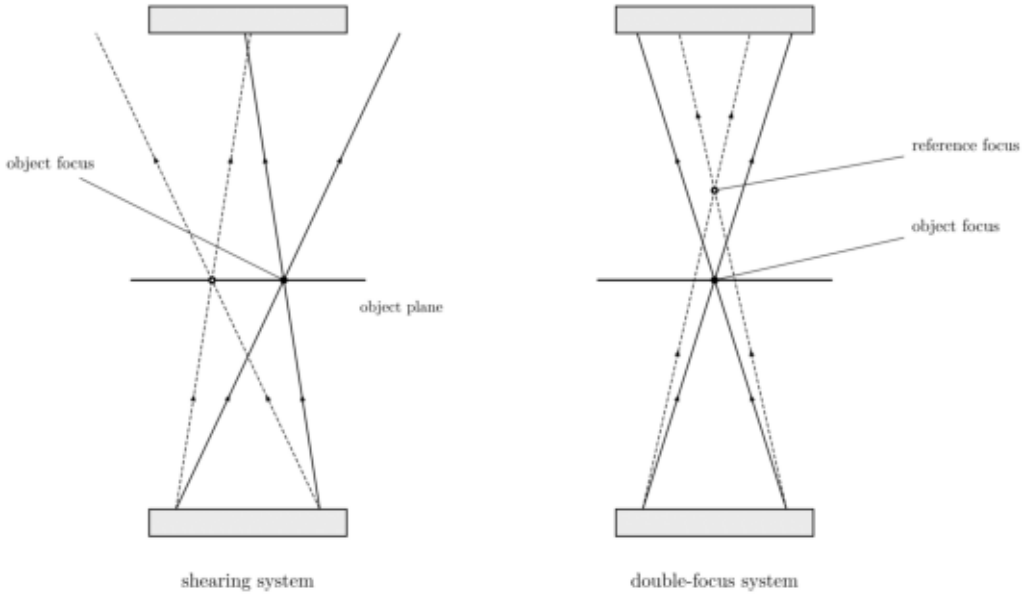


Figure 23. 20- Path of light rays thr

23. 7. 3. 1. 2 The birefringent plate on the front of t

Figure 23. 20- Path of light rays through a Shearing and Double-Focus system.

ch23_shearing_doublefocus | scan ch23 p21

Figure 24.2 — DONE

1962 scan

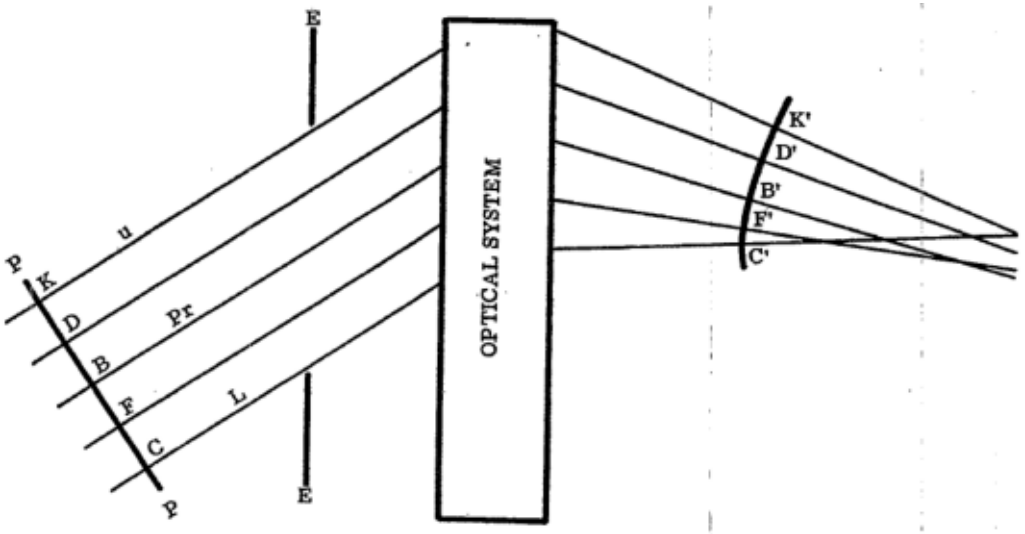


Figure 24. 2- Determination of constant phase front from ray tracing data.

Conder and Jacquinot, "Méthode pour l'observation des radiations de faible intensite au voisinage d'une raie brillante" Compte Rendus de

Figure 24, 2- Determination of constant phase front from ray tracing data.

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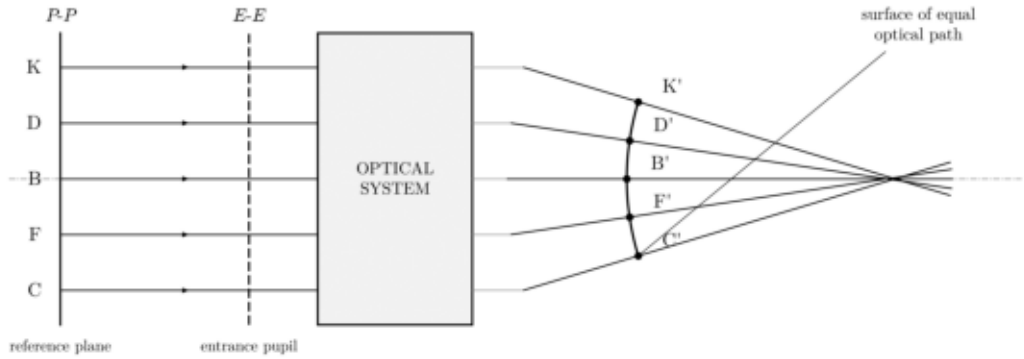


Figure 25.2 — DONE

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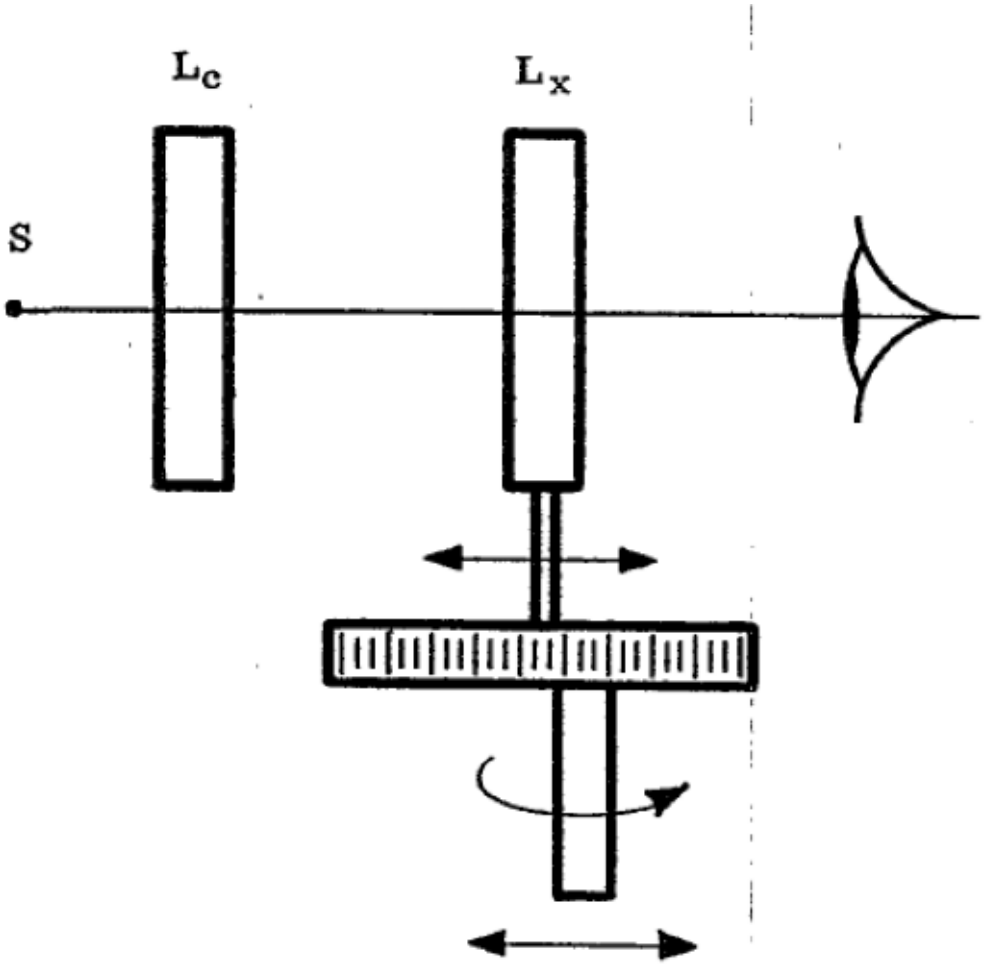


Figure 25. 2- Measurement of focal length by use of visual nodal slide.

Figure 25. 2- Measurement of focal length by use

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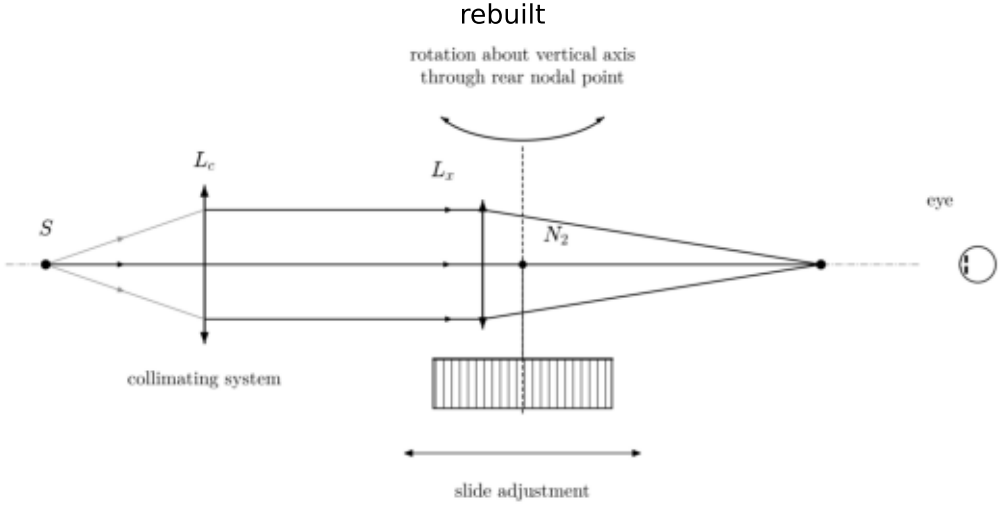


Figure 25.4 — DONE

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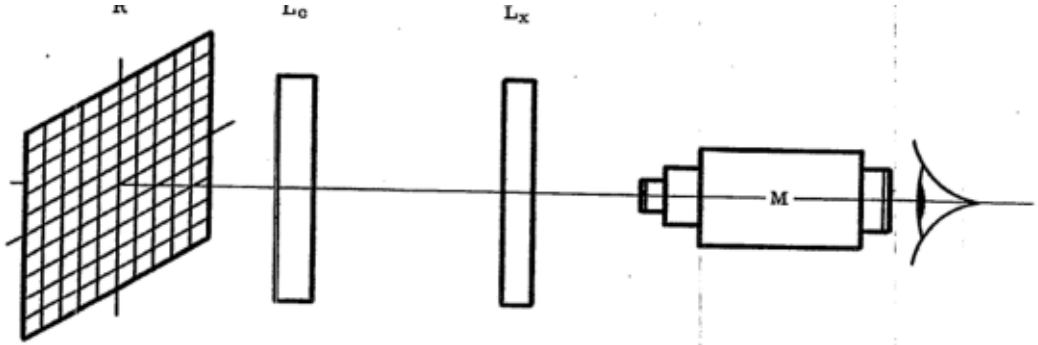


Figure 25. 4- Measurement of astigmatism with a grid reticle.

(10) Hardy and Perrin, The Principles of Optics. 382. McGraw-Hill (1932)

Figure 25, 4- Measurement of astigmatism with a grid reticle.

rebuilt

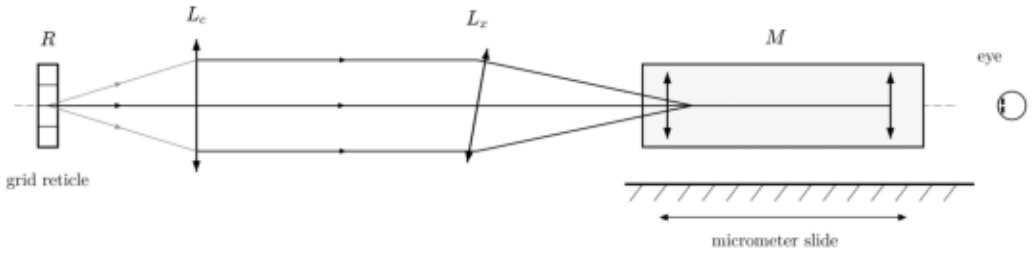


Figure 25.5 — DONE

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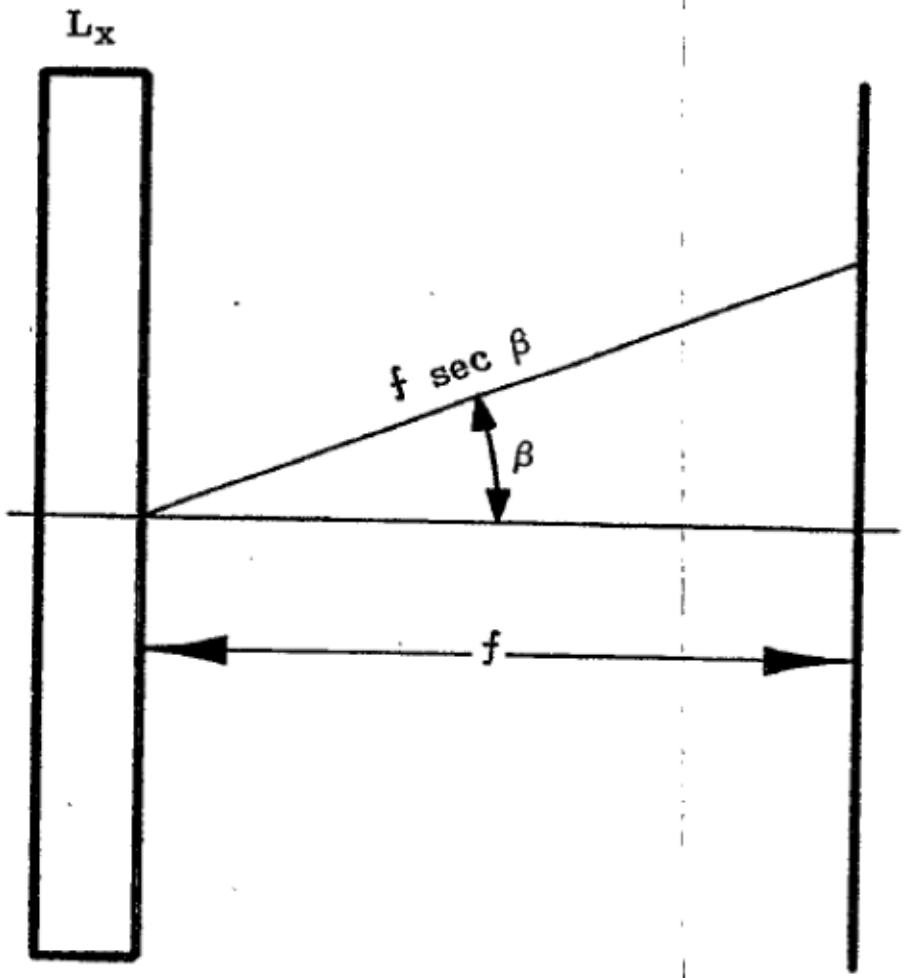


Figure 25. 5– Basic diagram for measurement of distortion by nodal slide.

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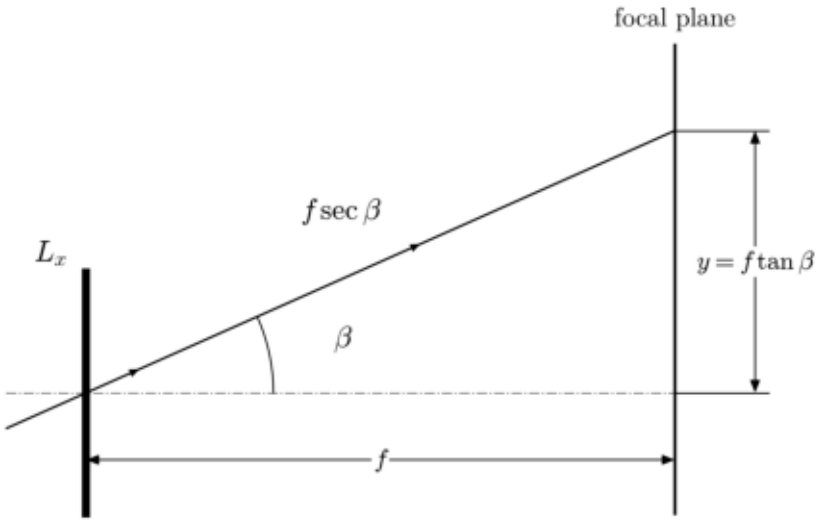


Figure 25. 5- Basic diagram for measurement of

Figure 25.9 — DONE

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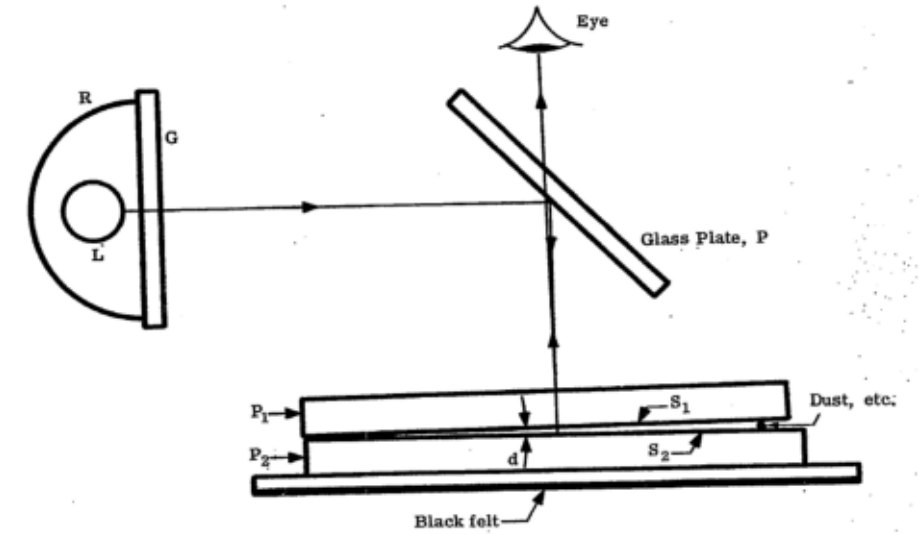


Figure 25.9 - Simple interferometer for viewing Fizeau fringes or Newton's fringes at normal incidence.

Figure 25.9 - Simple interferometer for viewing Fizeau fringes or Newton's fringes at normal incidence.

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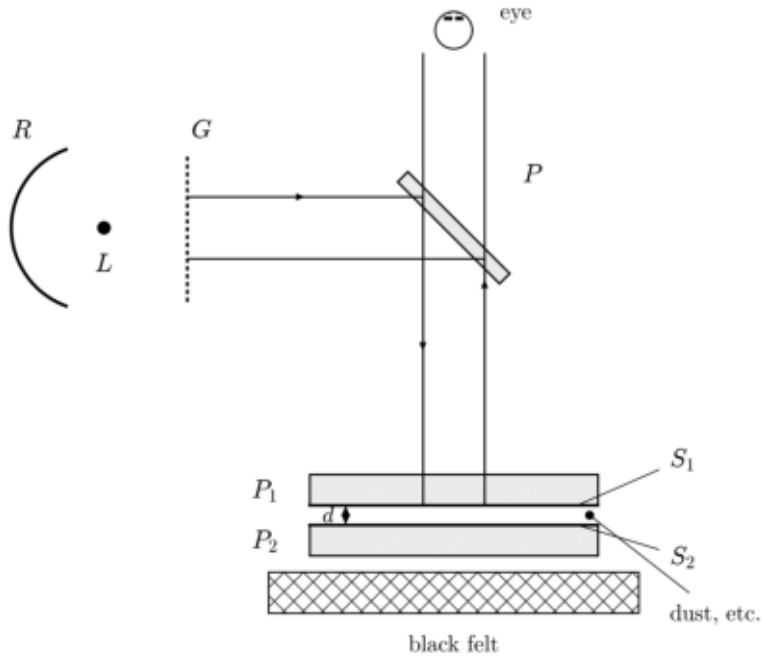
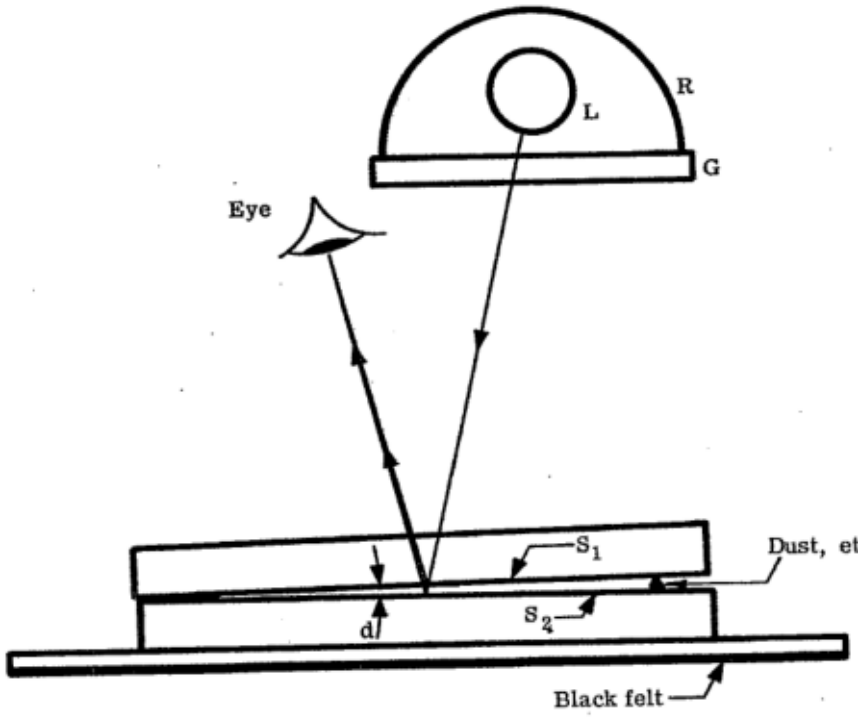


Figure 25.10 — DONE

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ıre 25. 10– Most commonly used method for illuminating the interferometer. The between surfaces S_1 and S_2 are observed at near – normal incidenc

25–9

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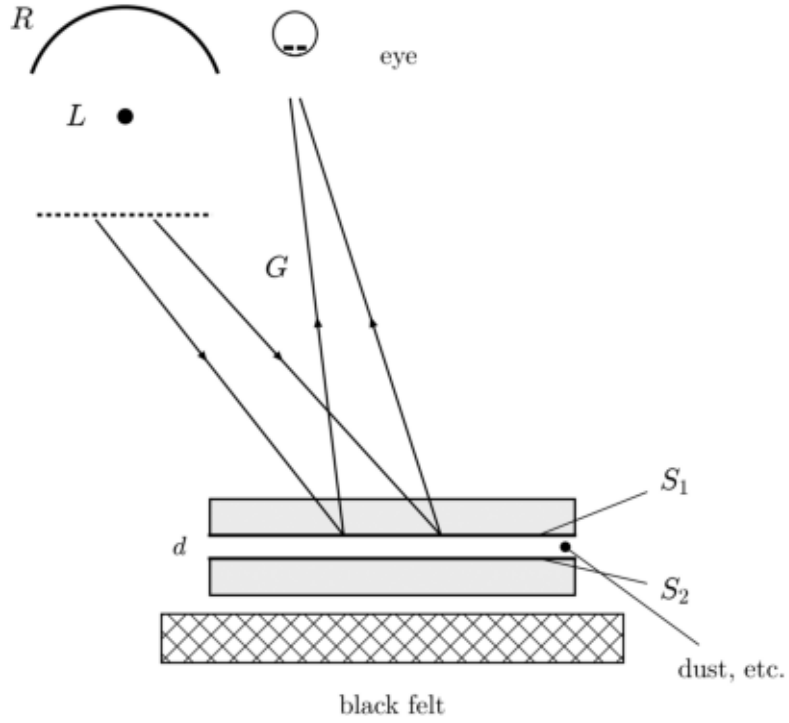


Figure 25. 10- Most commonly used method for illuminating the interferometer. The fringes formed

Figure 25.11 — DONE

1962 scan

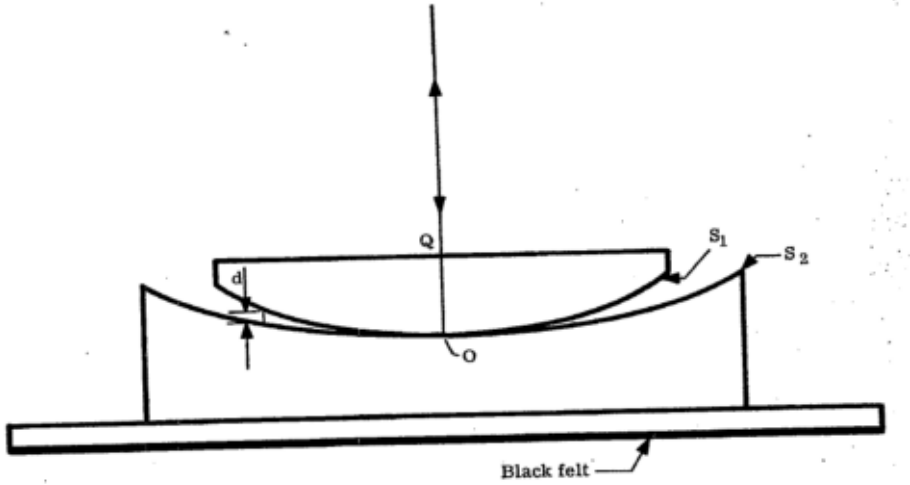


Figure 25. 11– Arrangement of the interferometer surfaces S_1 and S_2 for obtaining Newton's fringe

possible, as in paragraph 25.8.1.4 , to place surfaces S_1 and S_2 into sufficiently close contact so t
gle fringe spreads over surface S_1. For work of highest quality, it is customary to specify that this

Figure 25. 11- Arrangement of the interferometer surfaces S, and Sp for obtaining Newton's fringes.

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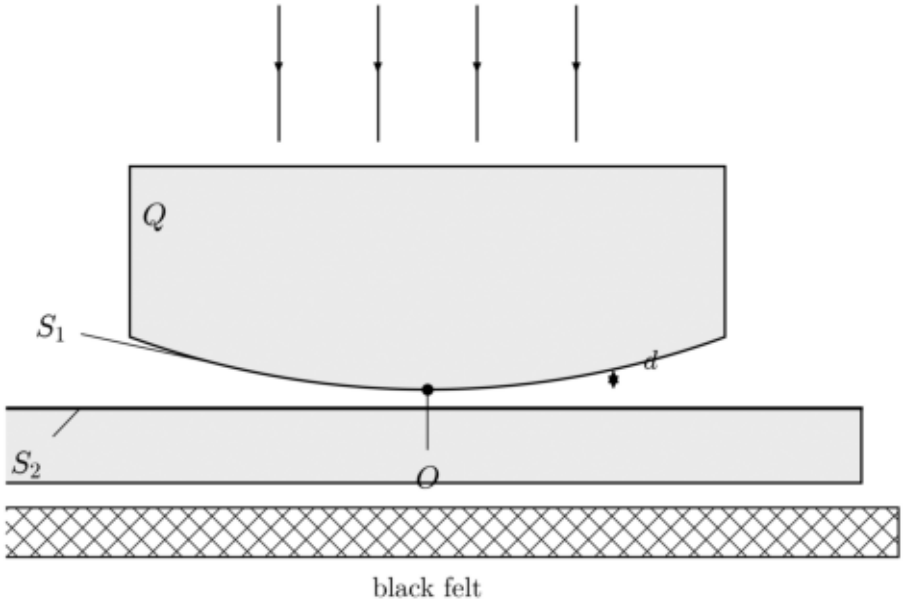


Figure 25.13 — DONE

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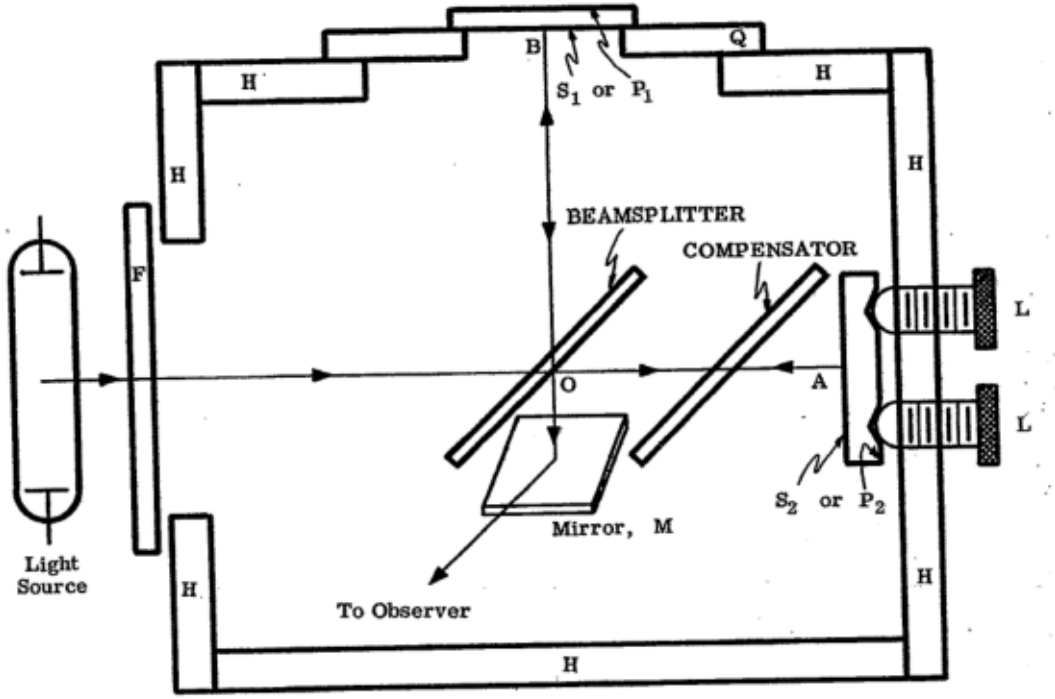
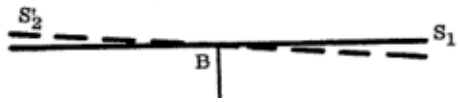


Figure 25.13- A modified Michelson interferometer.



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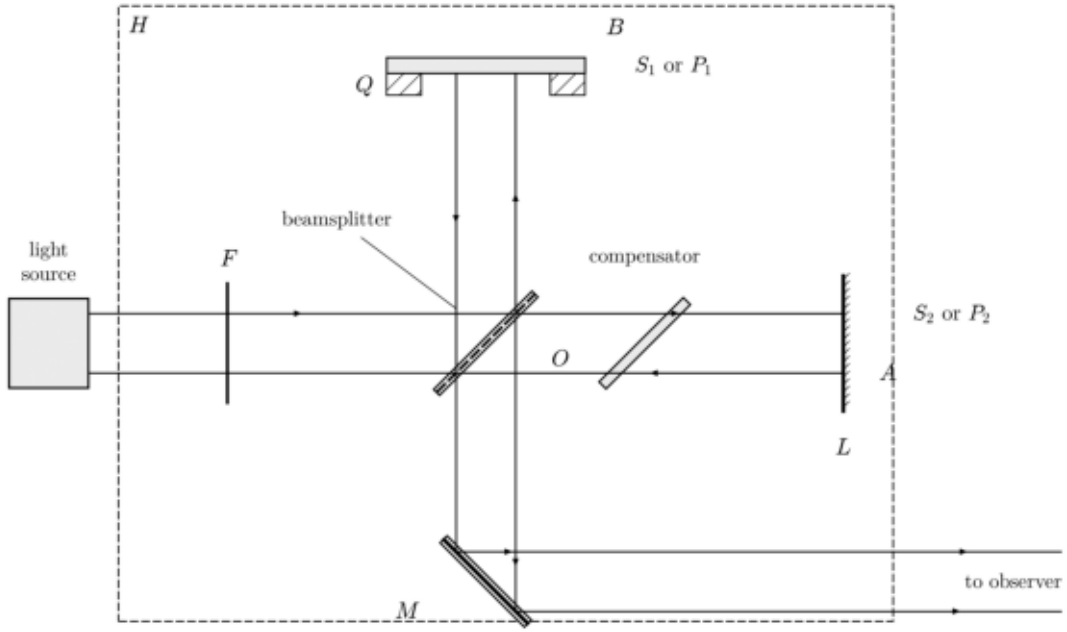


Figure 25.14 — DONE

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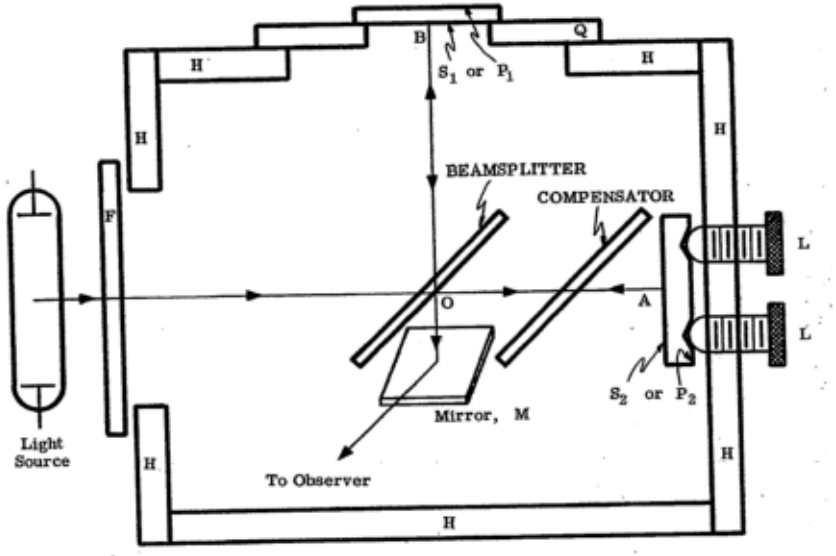


Figure 25.13- A modified Michelson interferometer.

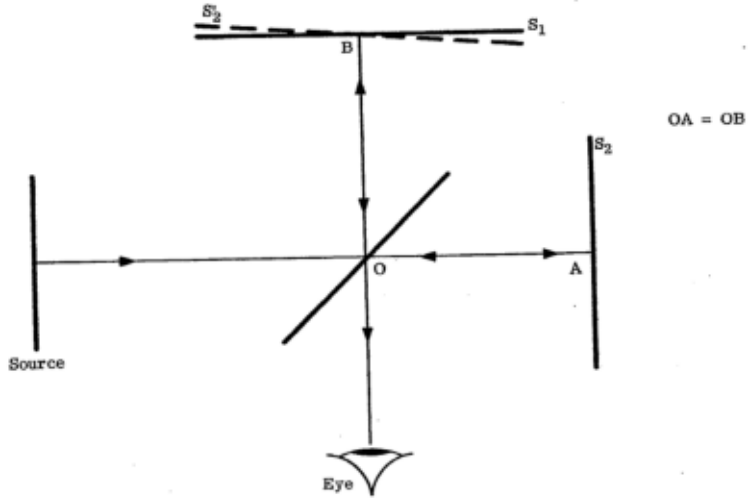


Figure 25.14- Michelson's interferometer as a method of Fizeau fringes. The observer sees sur as though it were located at S_2', consequently, the fringes are formed as by refl from two surfaces, S_1 and S_2', that are in close contact.

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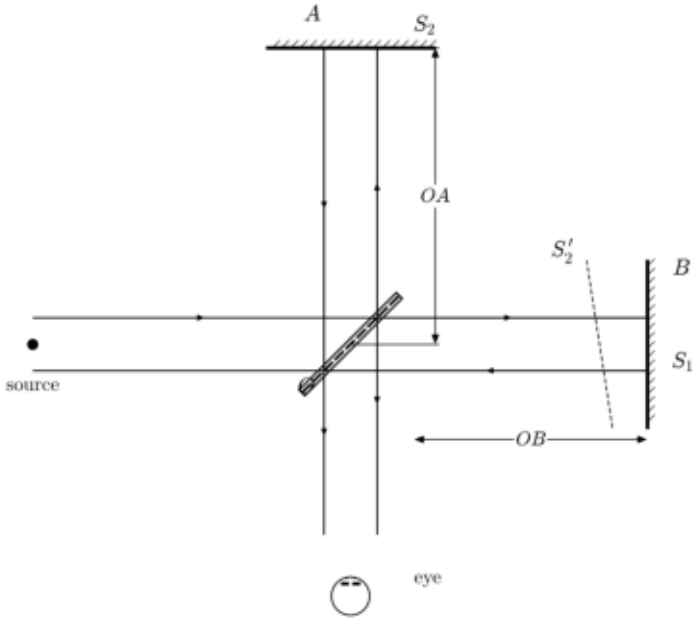


Figure 25.14- Michelson's interferometer as a method of Fizeau fringes. The observer sees surface Sz

Figure 25.15 — DONE

1962 scan

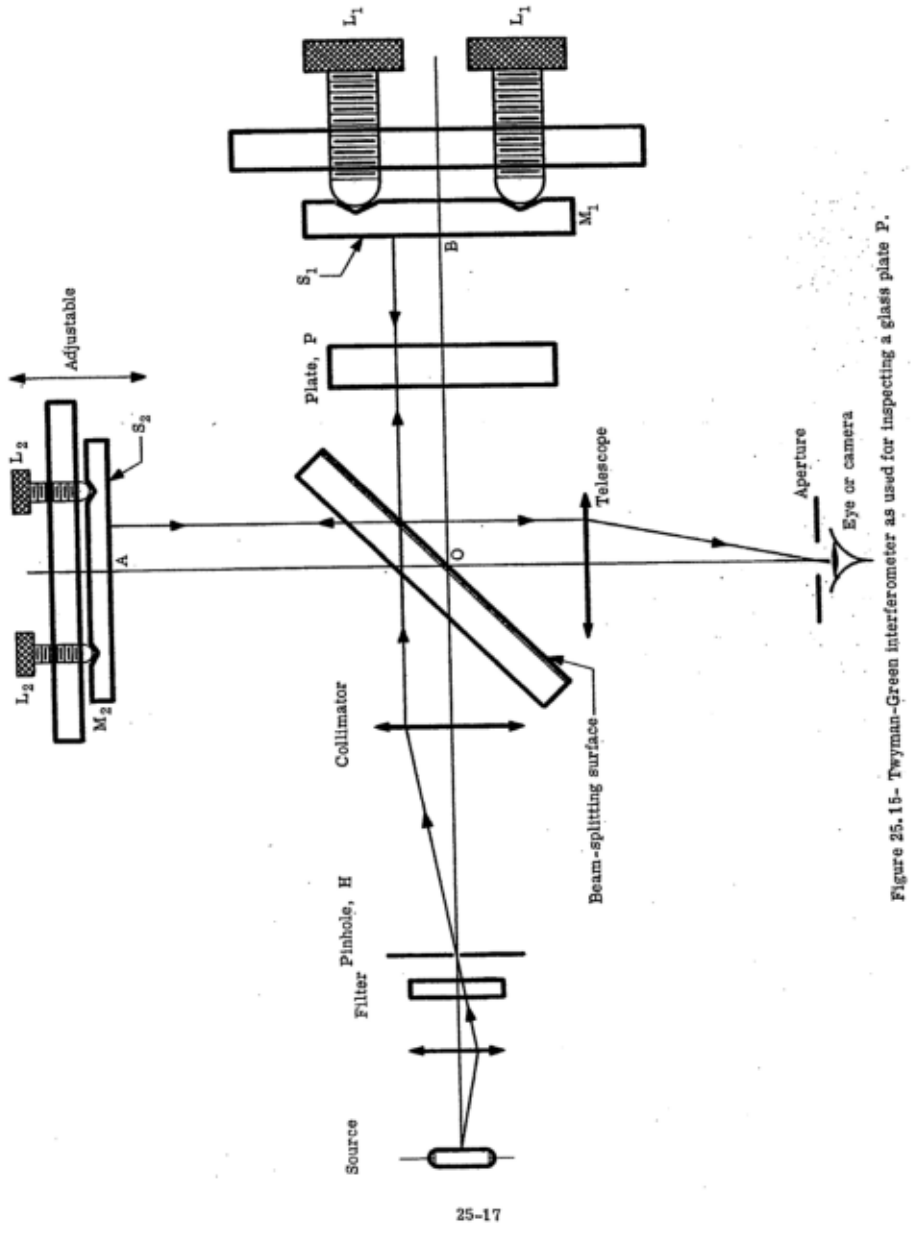
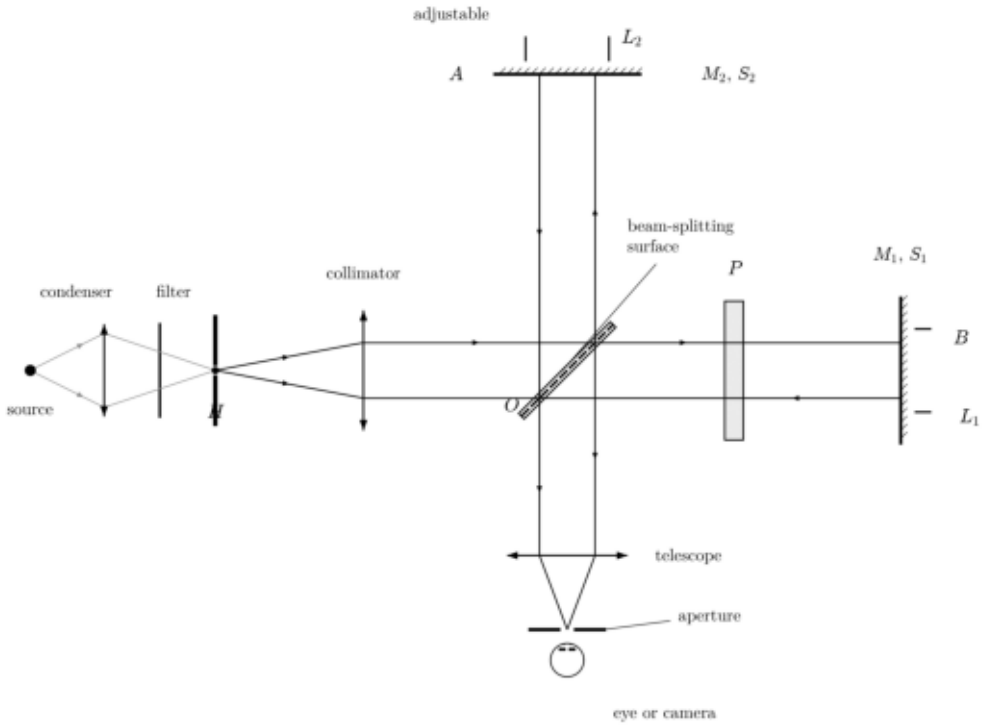


Figure 25.15- Twyman-Green interferometer as used for inspecting a glass plate P.

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Figure 25.16 — DONE

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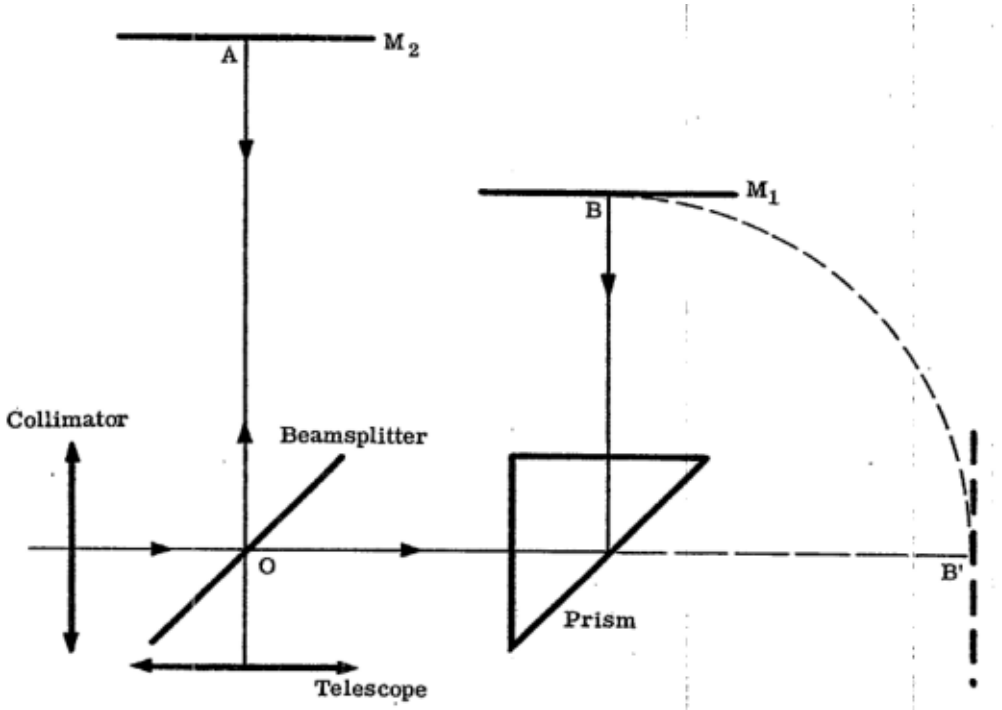


Figure 25.16- Modification of the location of end mirror M_1 for testing prisms in the Twyman-Green interferometer.

Figure 25.16- Modification of the location of end mirror M, for testing prisms in

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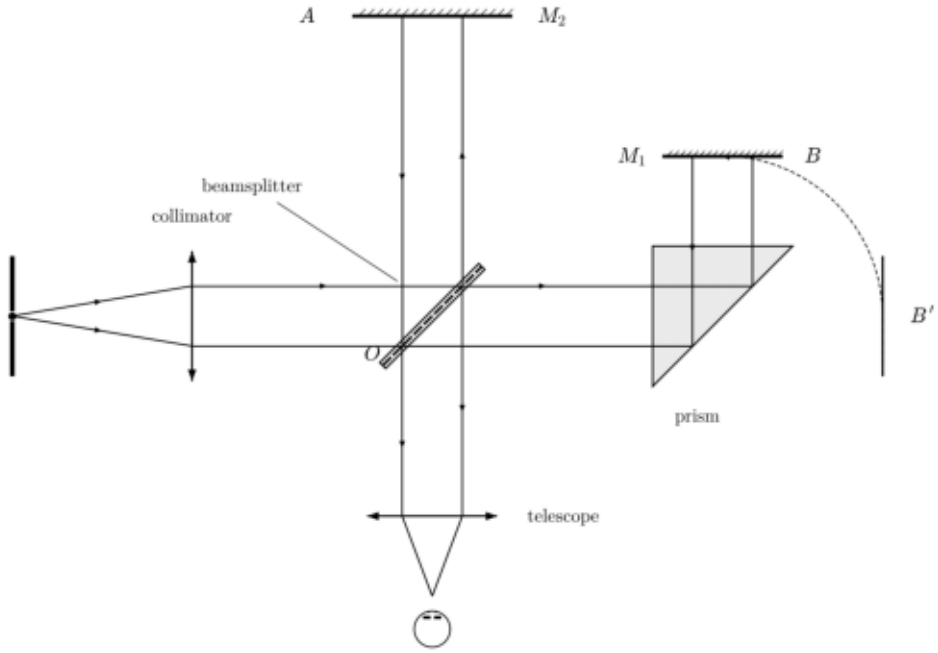


Figure 25.17 — DONE

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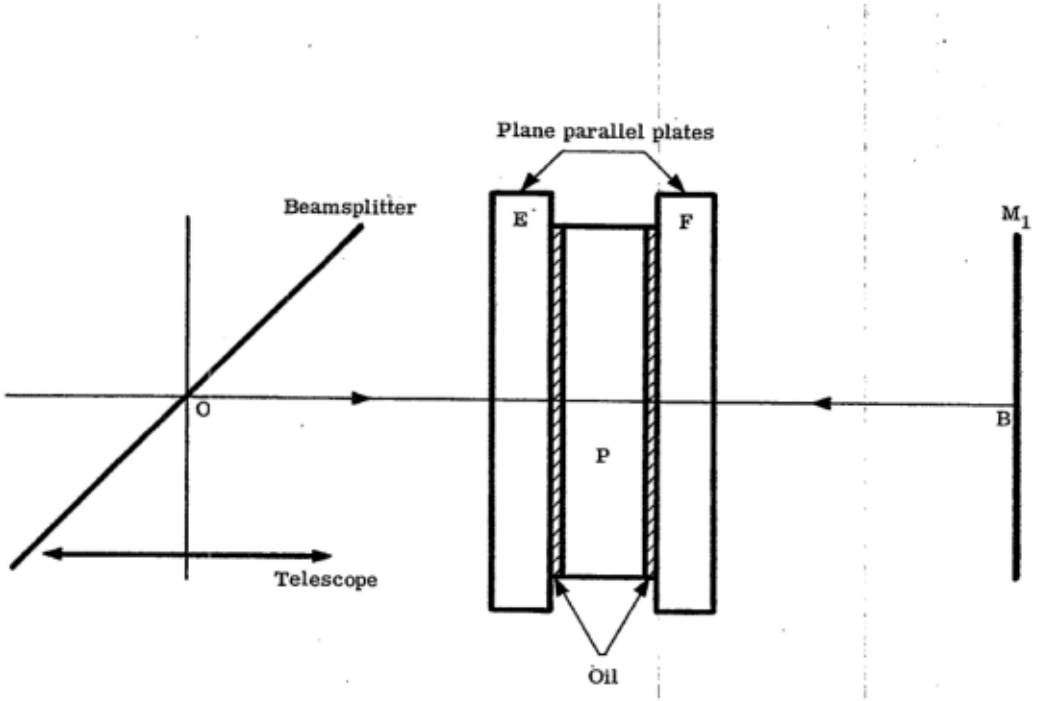


Figure 25.17- Methoi for obviating surface flatness of the test plate: P.

25-18

Figure 25.17- Methoi for obviating surface flatness of the test plate: P.

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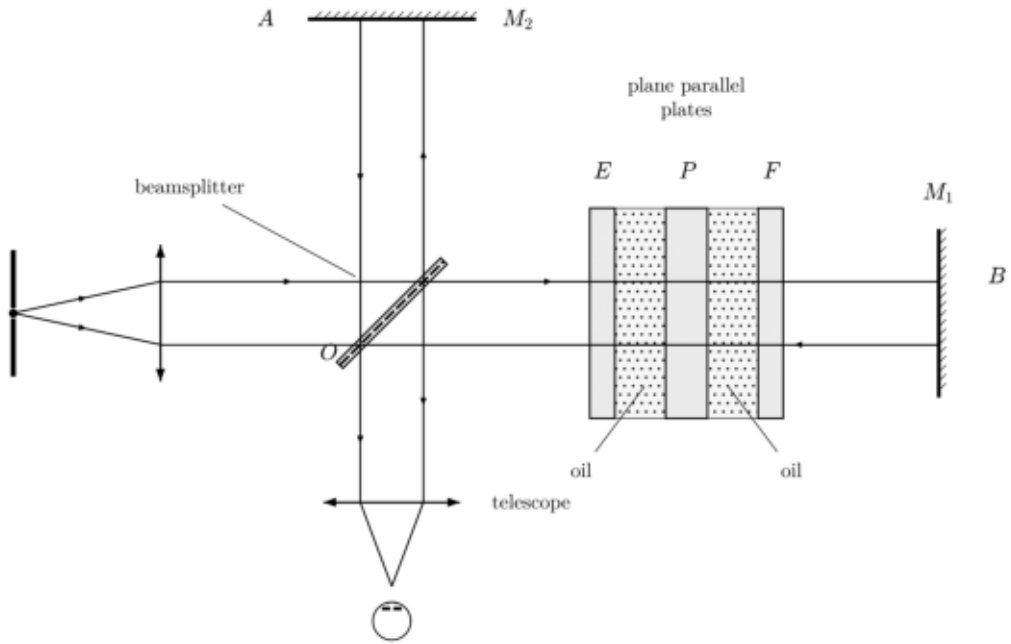


Figure 25.18 — DONE

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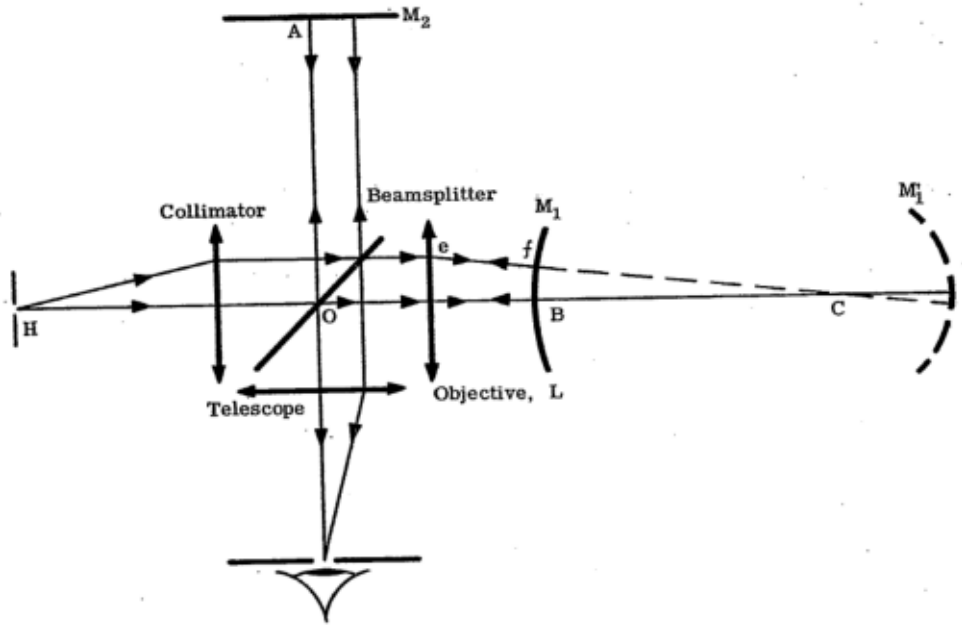


Figure 25.18- Adaptation of the Twyman-Green interferometer to the observation of spherical aberration of lenses, L.

25-19

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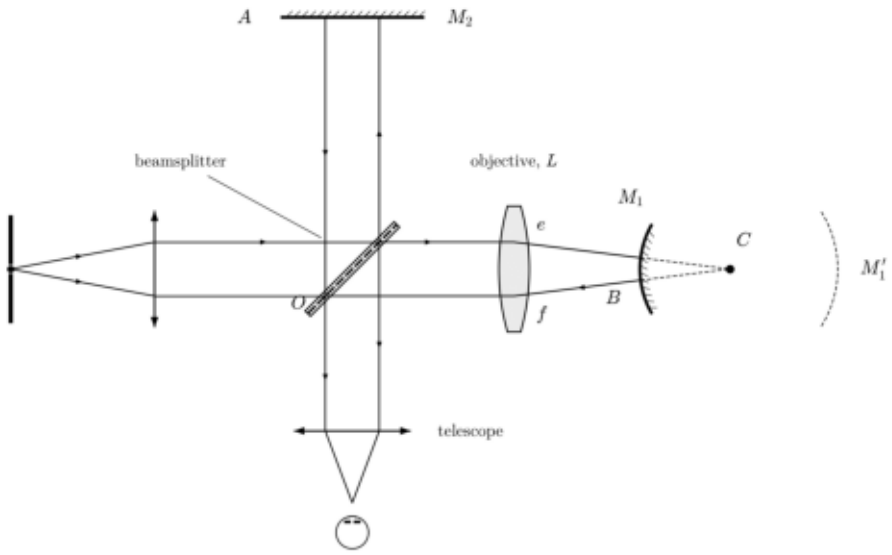


Figure 25.18- Adaptation of the 'Twyman-Green interferometer to the observation of

Figure 25.20 — DONE

1962 scan

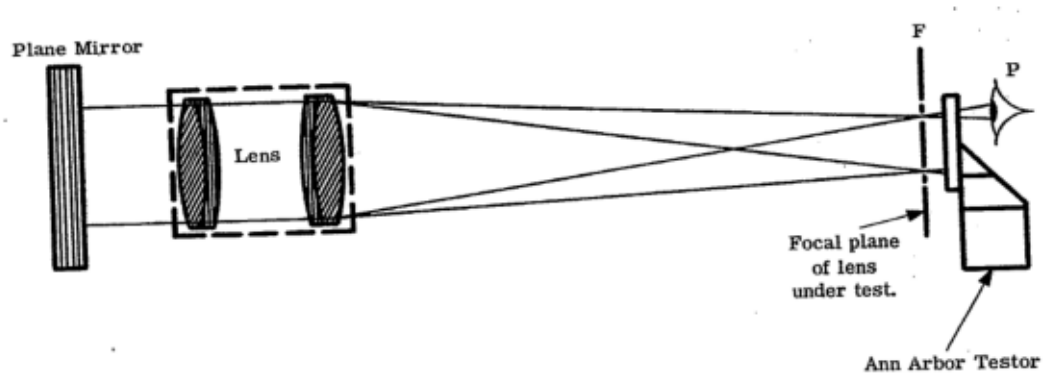


Figure 25.20- The Ann Arbor Tester.

(20) di Francia, Optical Image Evaluation, (NBS Circular 526), 161, U. S. Gov't. Printing Office, (1954).
(21) Martin, Technical Optics, vol. 1, p289, Pitman, (1948). also Jacobs, Fundamentals of Optical Engineering, McGraw

Figure 25.20- The Ann Arbor Tester.

rebuilt

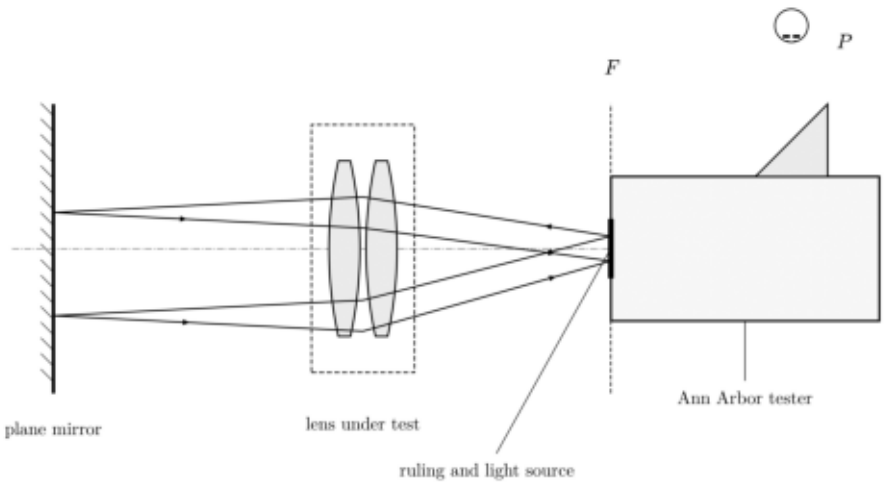
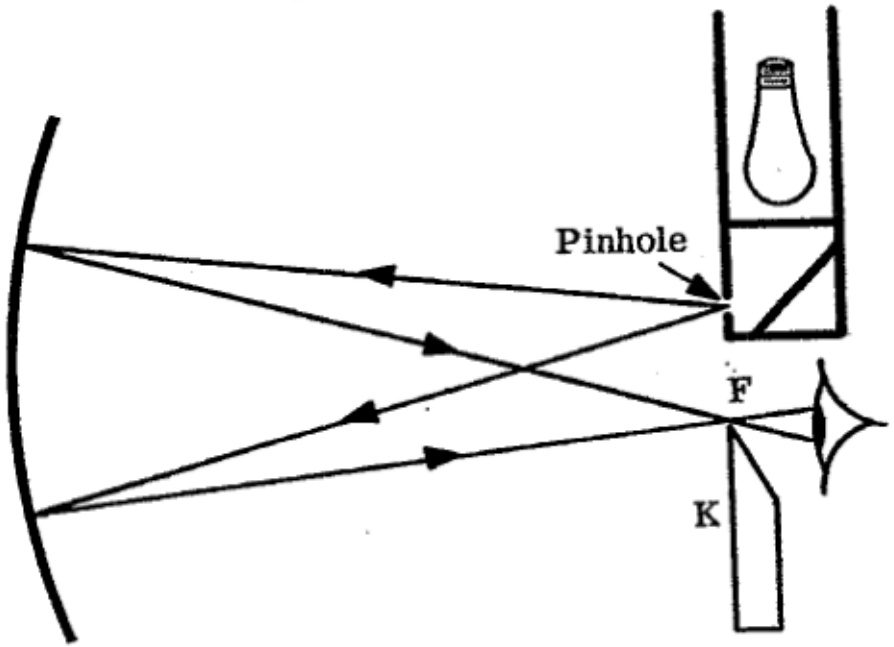


Figure 25.24 — DONE

1962 scan



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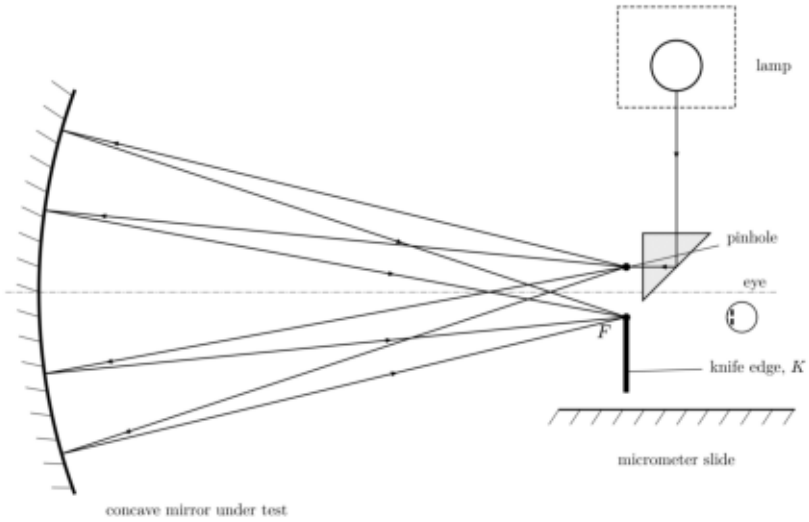


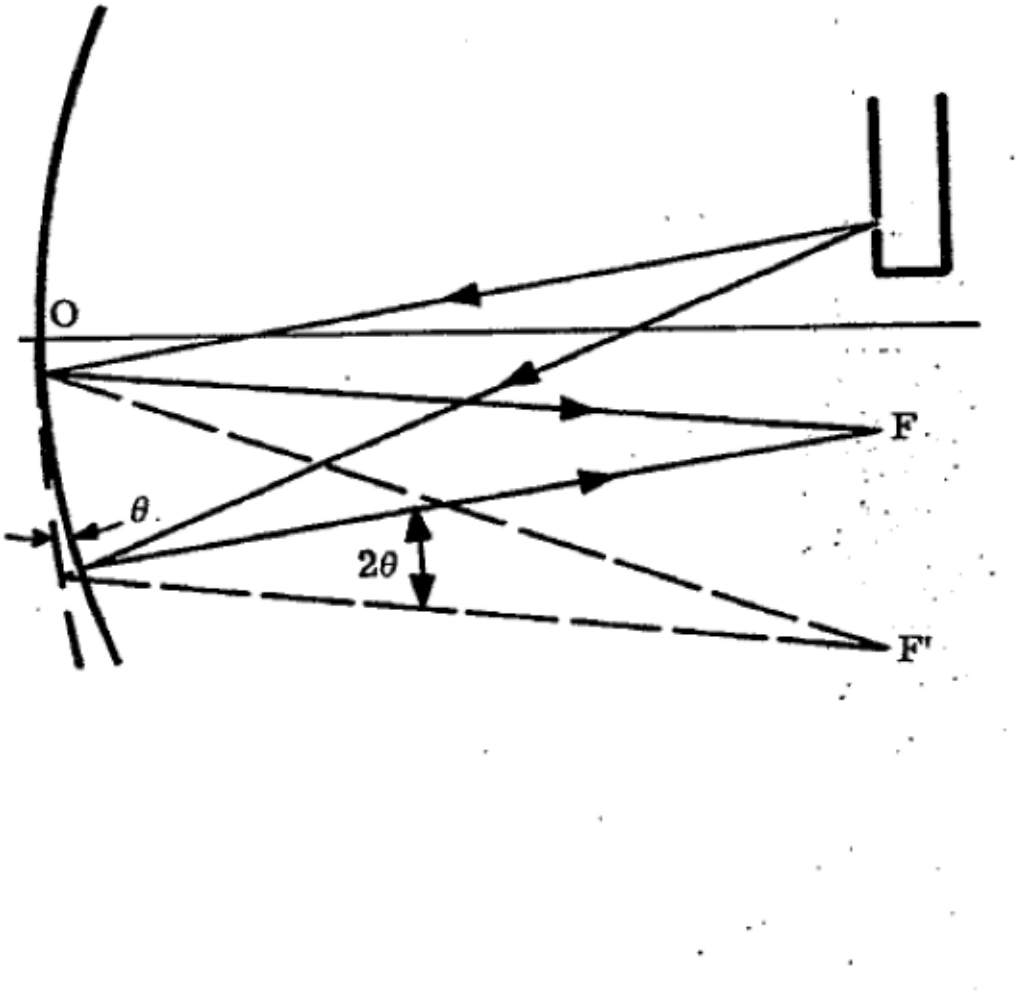
Figure 25. 24 – Experimental arrangement for a Foucault Test of a mirror.

this area will then be deflected from the correct focu

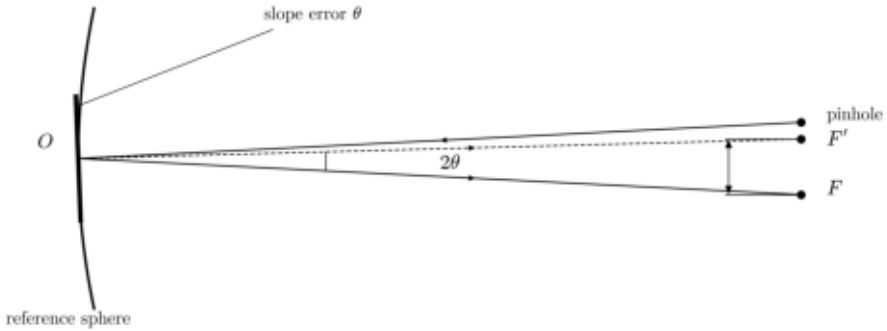
Figure 25. 24- 'Experimental arrangement for a Figure 25, 25- Schematic demonstration of

Figure 25.25 — DONE

1962 scan



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Figure 16.2 — DONE

1962 scan

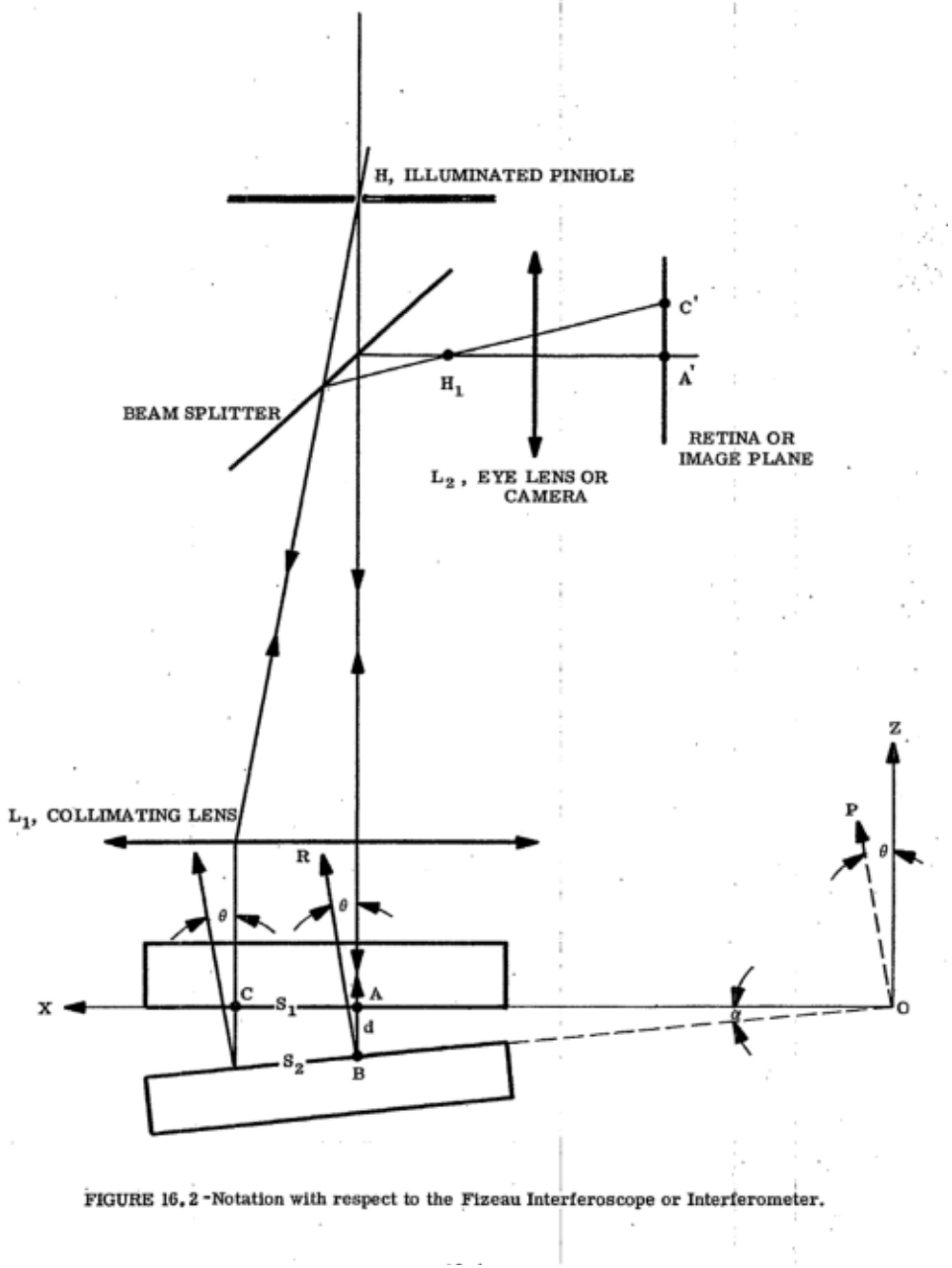


FIGURE 16, 2 -Notation with respect to the Fizeau Interferoscope or Interferometer.

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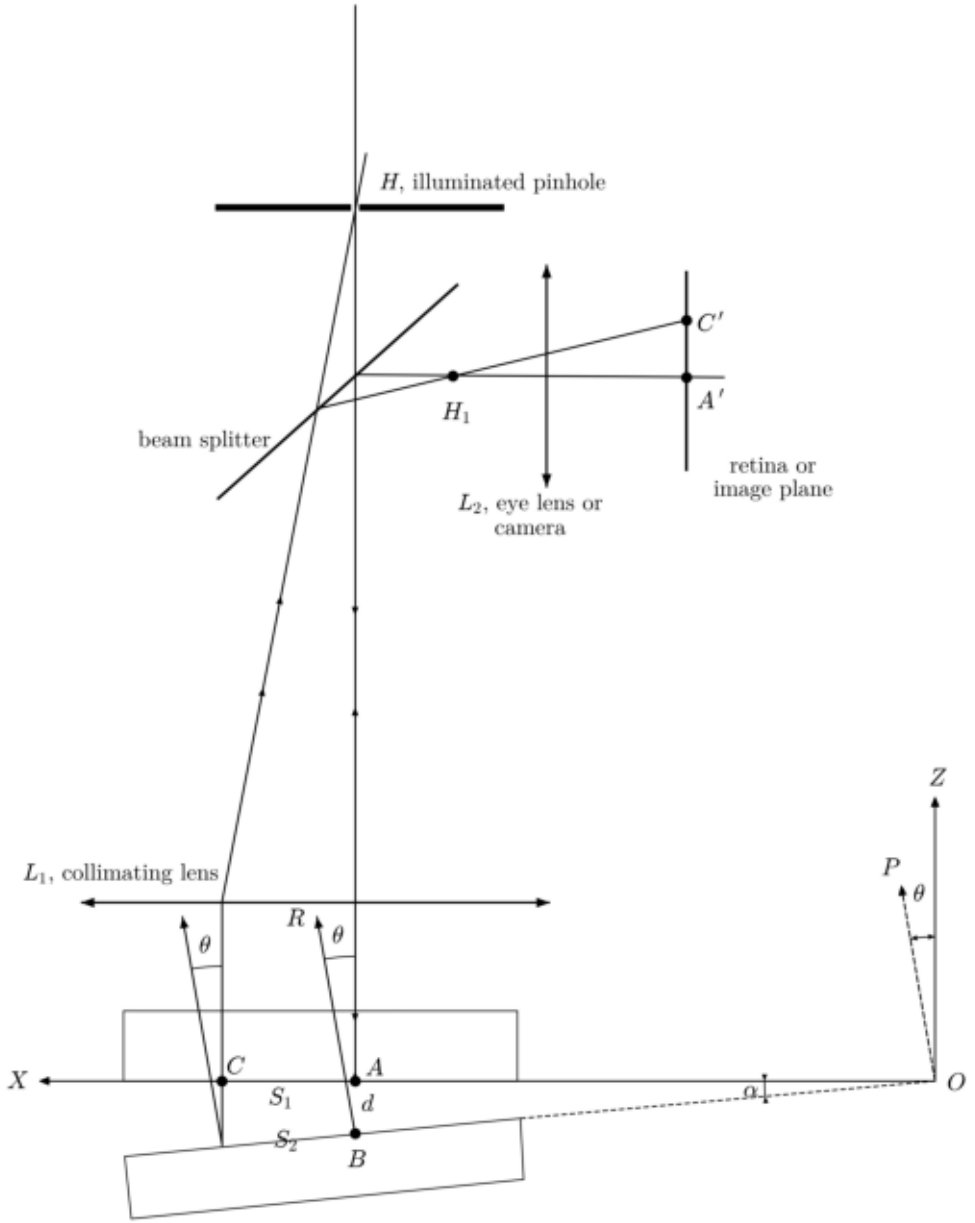


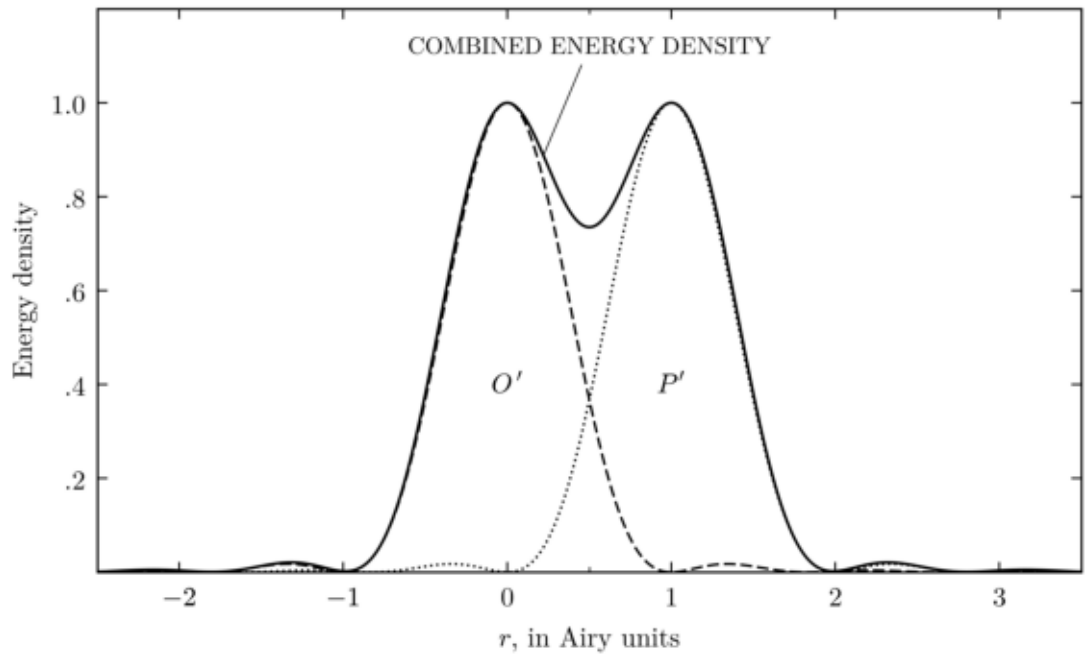
FIGURE 16, 2 ~Notation with respect to the Fizeau Interferoscope or Interferometer.

Figure Table 16.1 — DONE

1962 scan

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Figure 16.19 — DONE

1962 scan

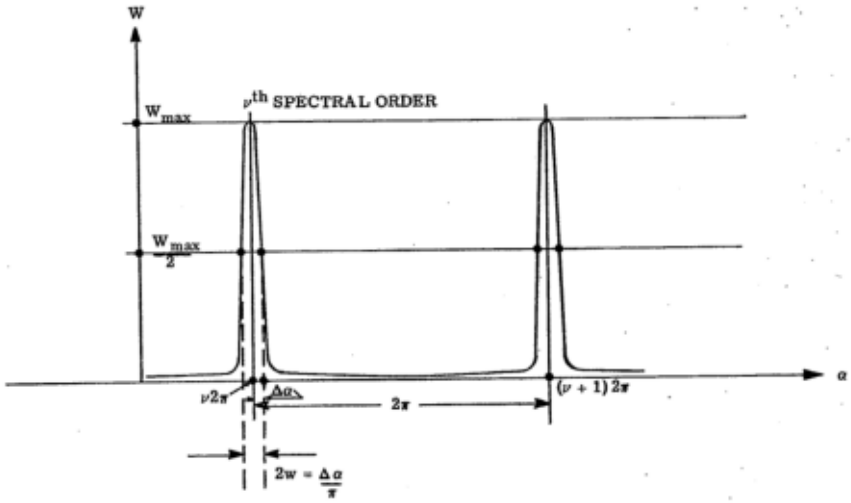


FIGURE 16. 19– The Sharpness Quality of Multiple Beam Interference Fringes.

rebuilt

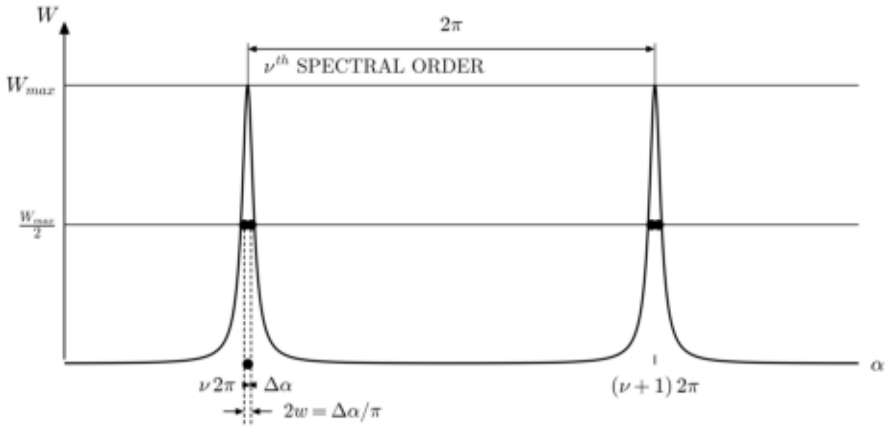


Figure 16. 19, is approximately 0.03 times the width from one bright fringe to the next. The fringes become

Figure 16.26 — DONE

1962 scan

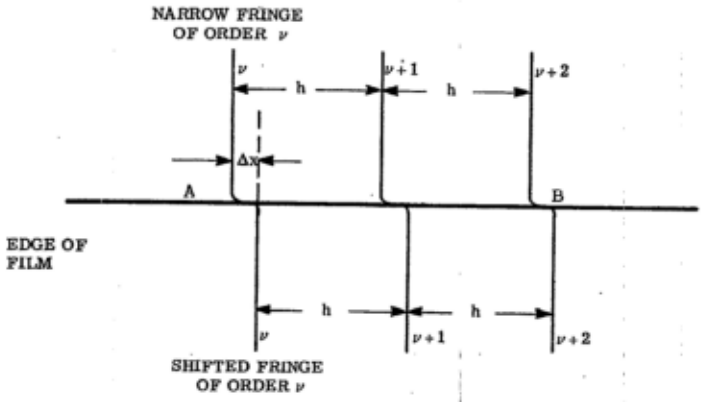


FIGURE 16. 26– Appearance of the narrow fringes when the thickness of the film is a small fraction of a wavelength and the film occupies the portion below the edge AB. If the surfaces S_1 and S_2 are optically flat, the indicated fringe widths, h, will be alike within the expiramental error of measuring h.

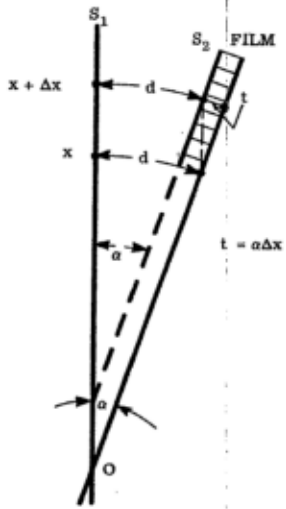
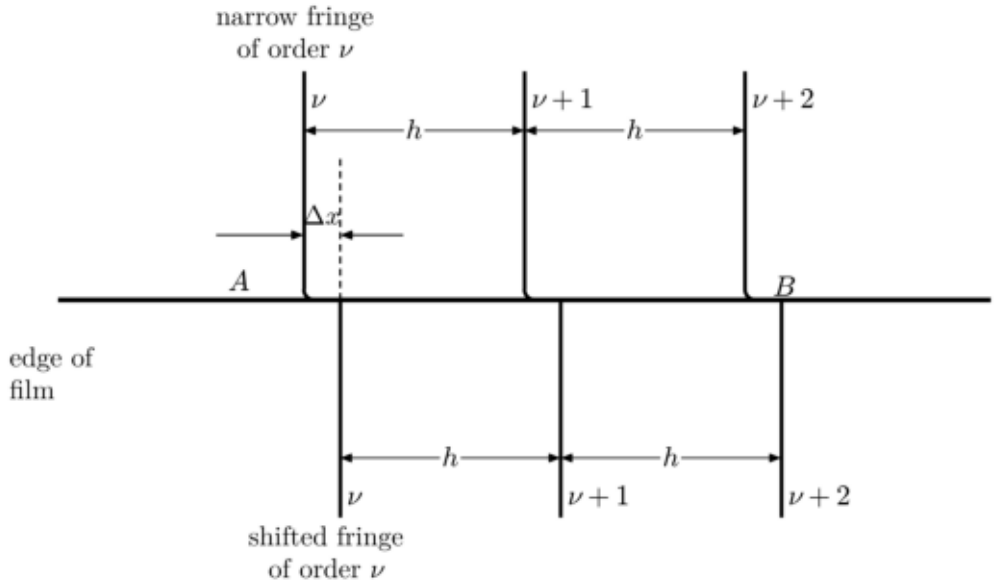


FIGURE 16. 27– Movement of an interference frings, x, to the position, x + Δ x, by the introduction of a film at thickness, t.

16–40

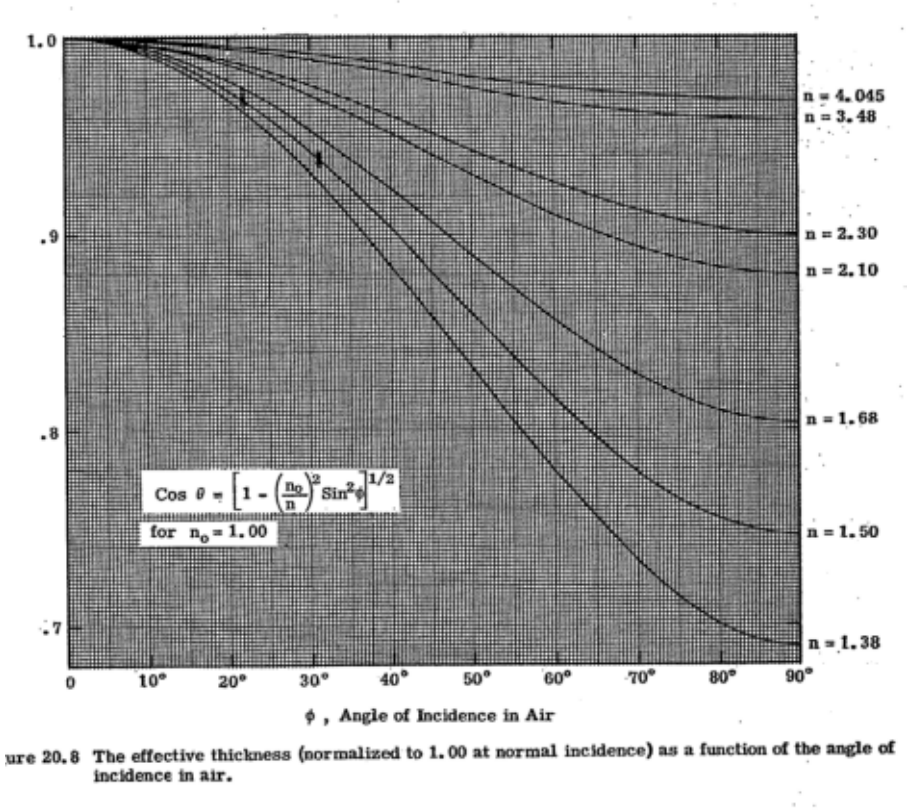
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Figure 20.8 — DONE

1962 scan



ure 20.8 The effective thickness (normalized to 1.00 at normal incidence) as a function of the angle of incidence in air.

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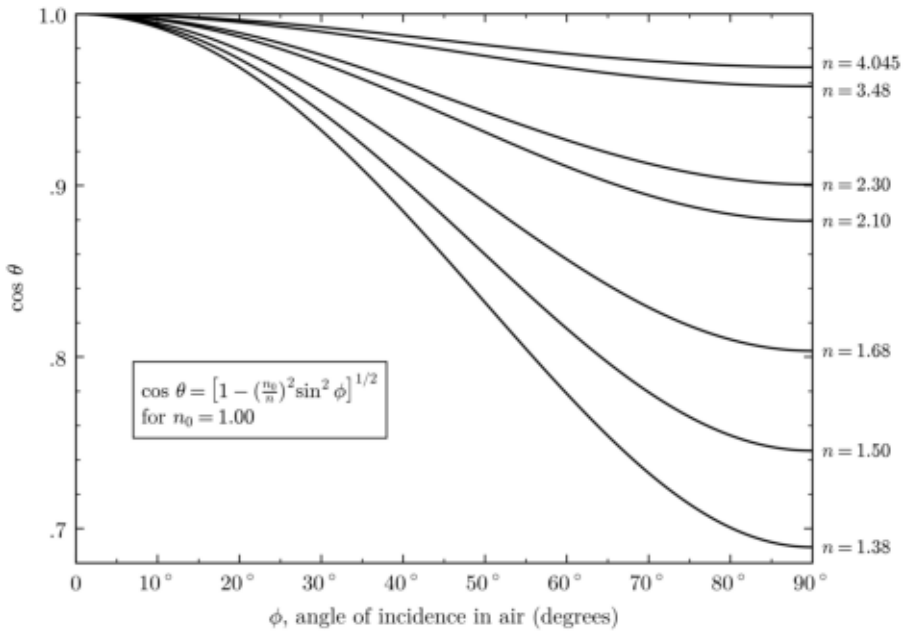
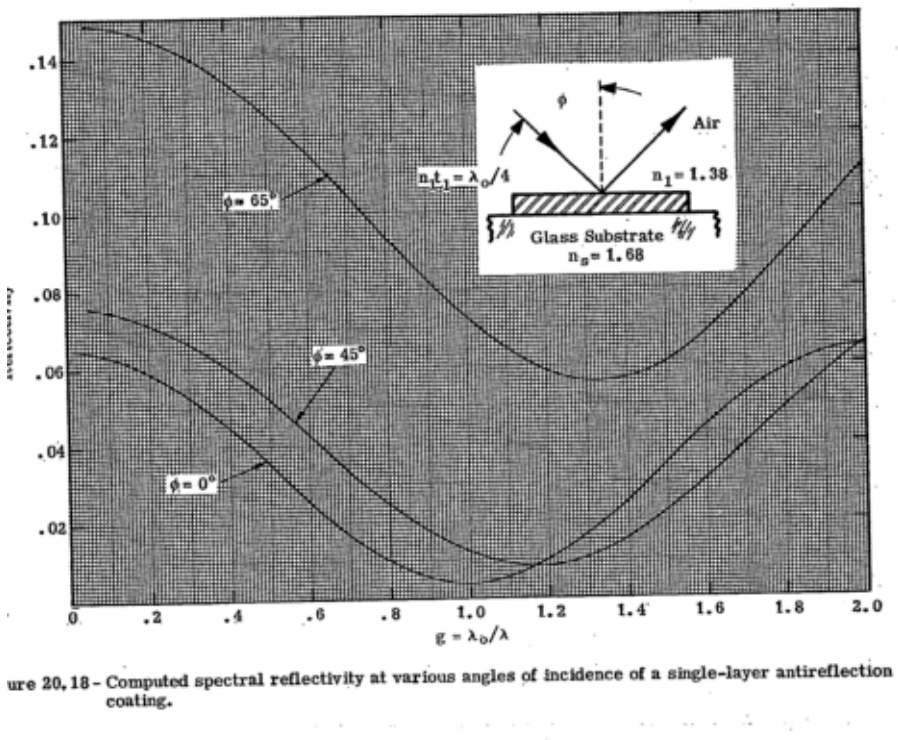


Figure 20.8 The effective thickness (normalized to 1.00 at normal incidence) as a function of the angle of

Figure 20.18 — DONE

1962 scan



ure 20, 18 – Computed spectral reflectivity at various angles of incidence of a single-layer antireflection coating.

rebuilt

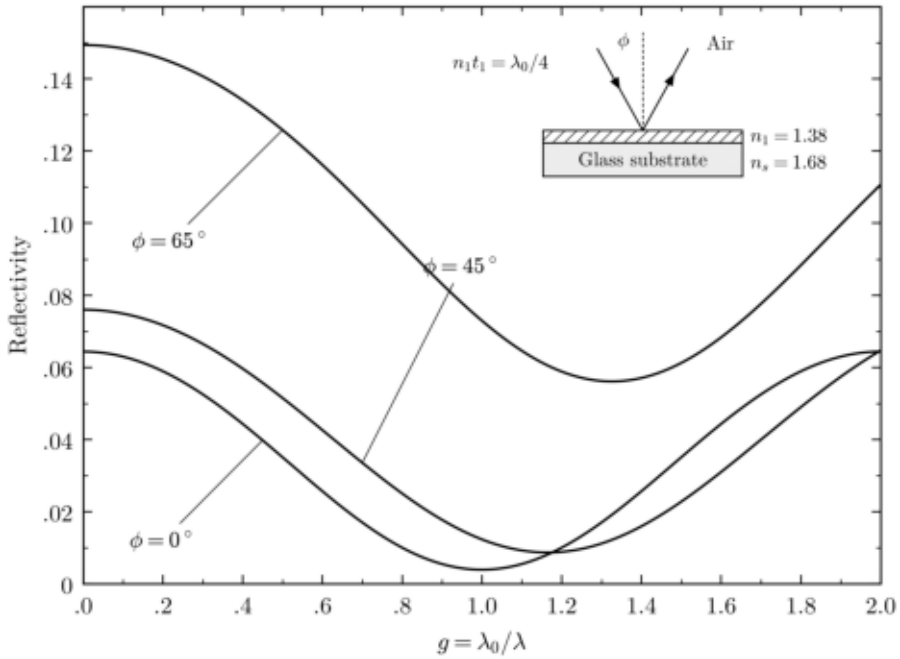


Figure 20, 18- Computed spectral reflectivity at various angles of incidence of a single-layer antireflection

Figure 20.19 — DONE

1962 scan

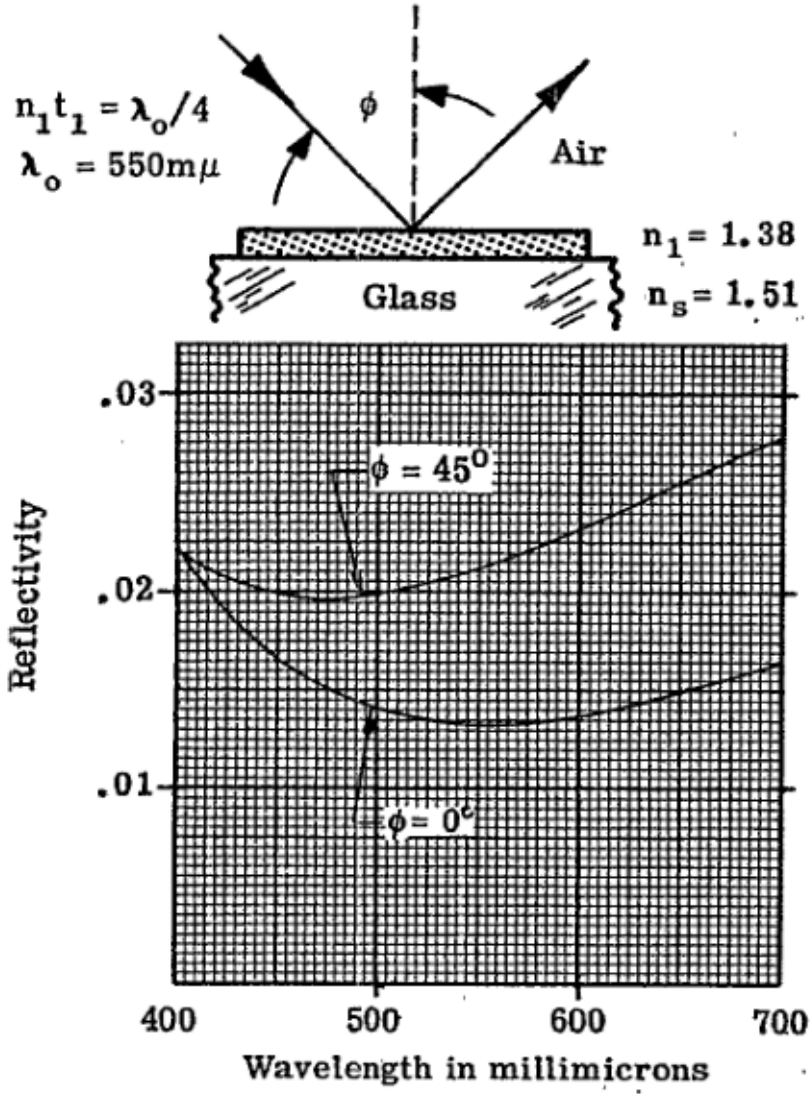


Figure 20.19 – Computed spectral reflectivity of a single layer at $\phi = 0$ and $\phi = 45°$.

rebuilt

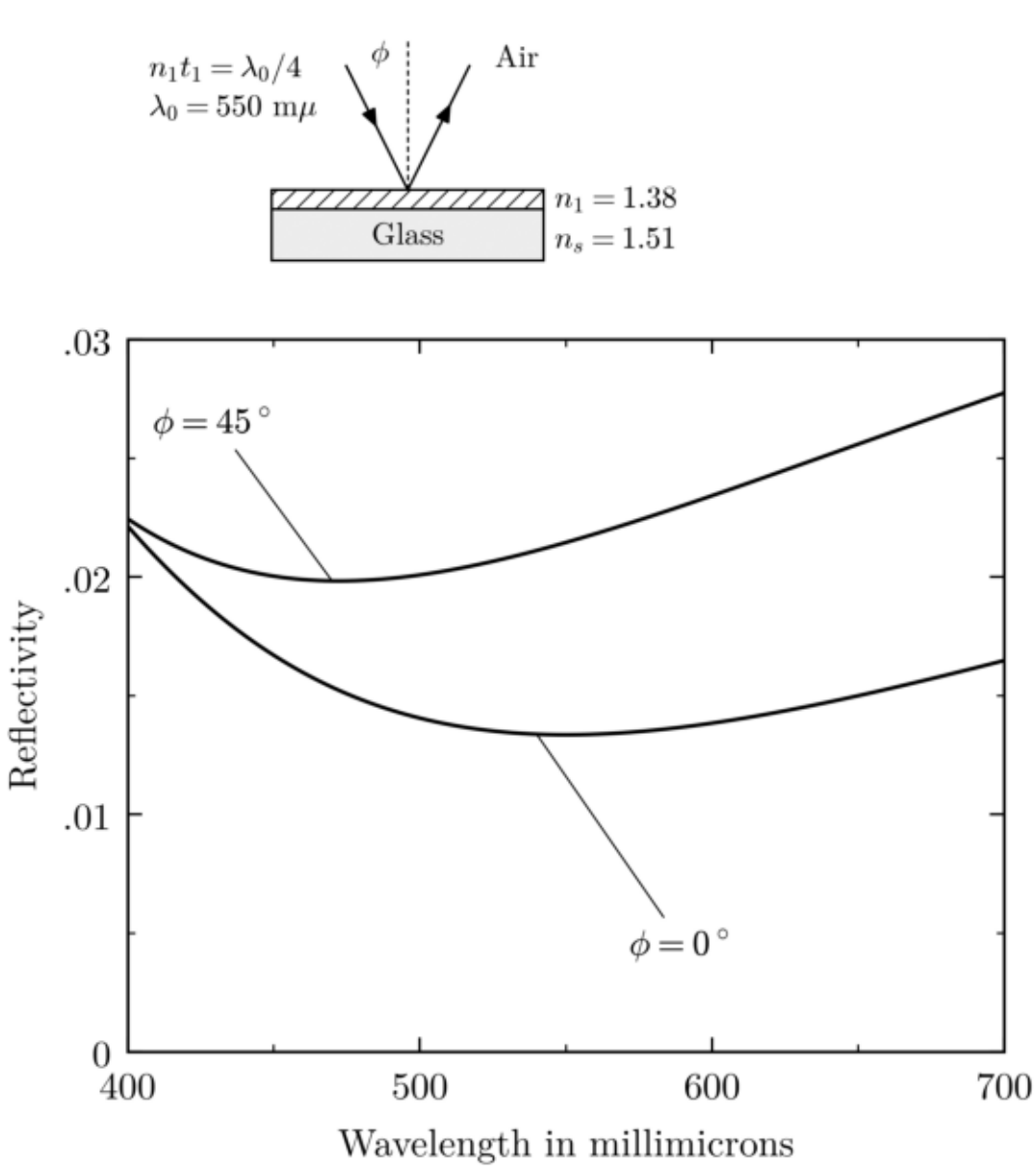
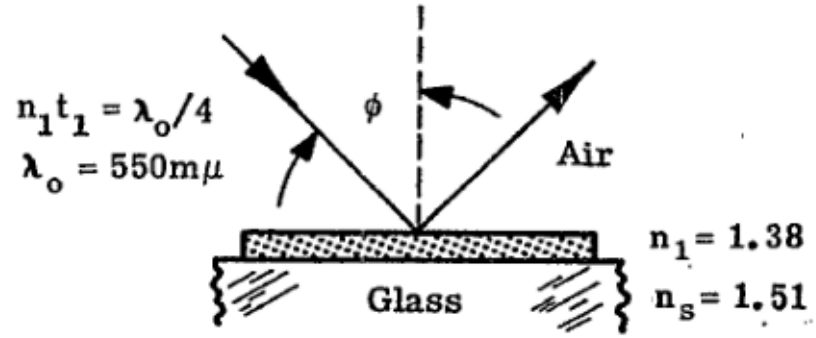


Figure 20. 19- Computed spectral reflectivity of a

Figure 20.20 — DONE

1962 scan



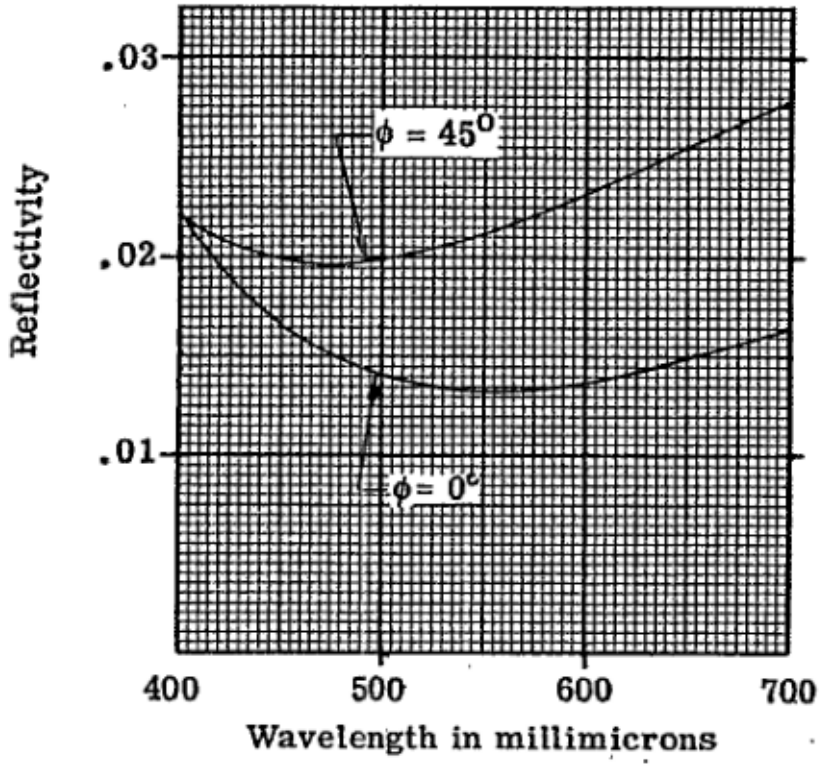
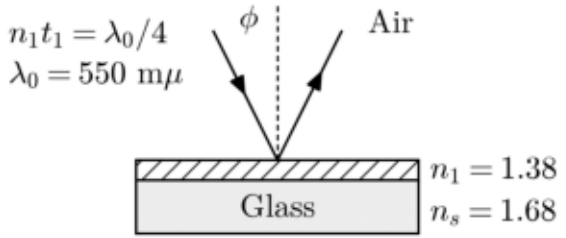


Figure 20.19 – Computed spectral reflectivity of a single layer at $\phi = 0$ and $\phi = 45°$.

rebuilt



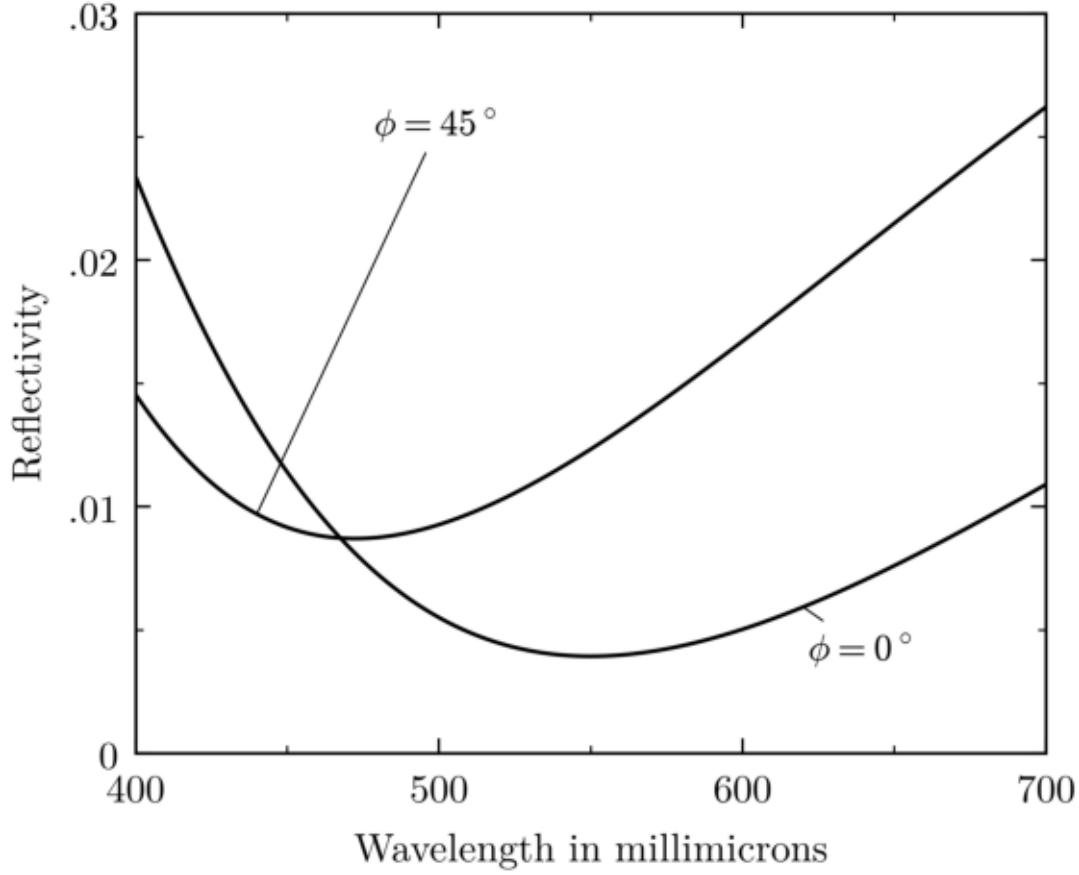


Figure 20.20 - Computed spectral reflectivity of

Figure 20.21 — DONE

1962 scan

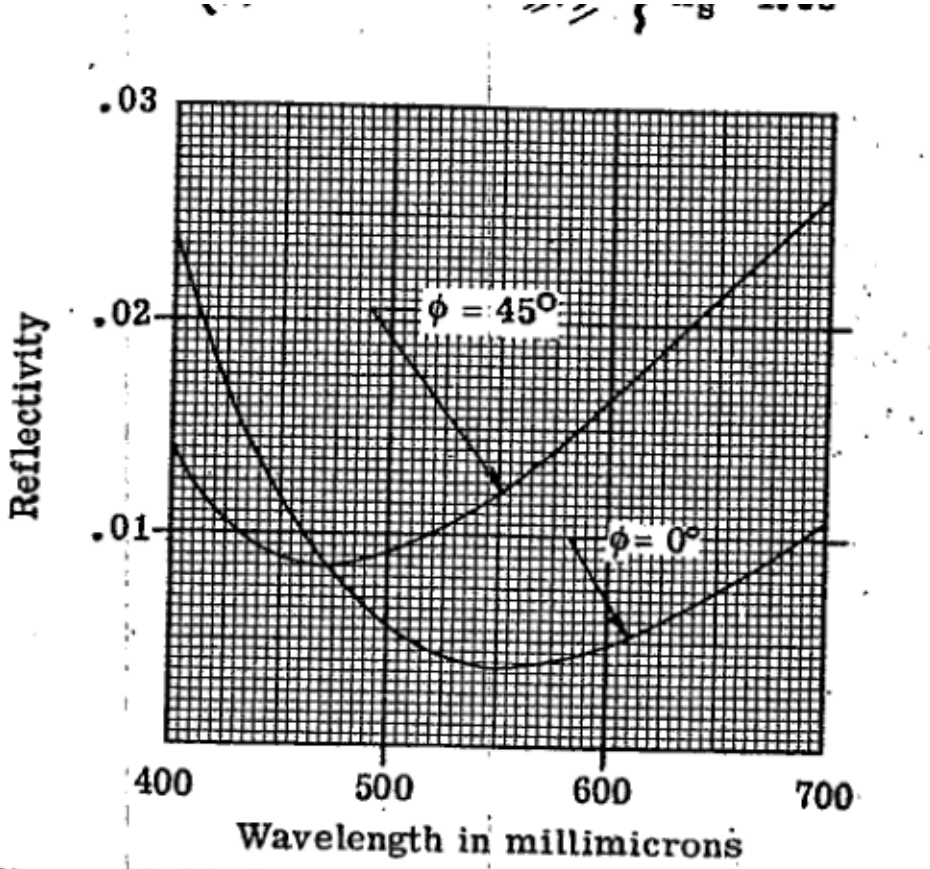
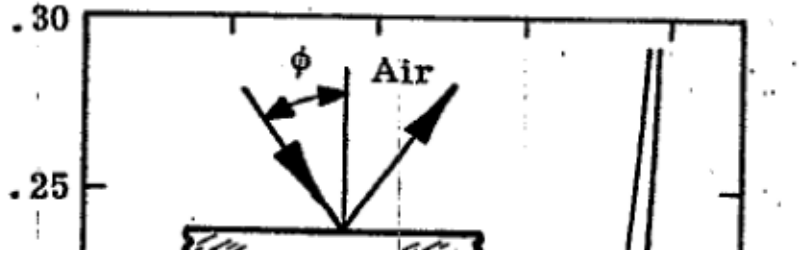


Figure 20.20 – Computed spectral reflectivity of a single layer at $\phi = 0$ and $\phi = 45°$.



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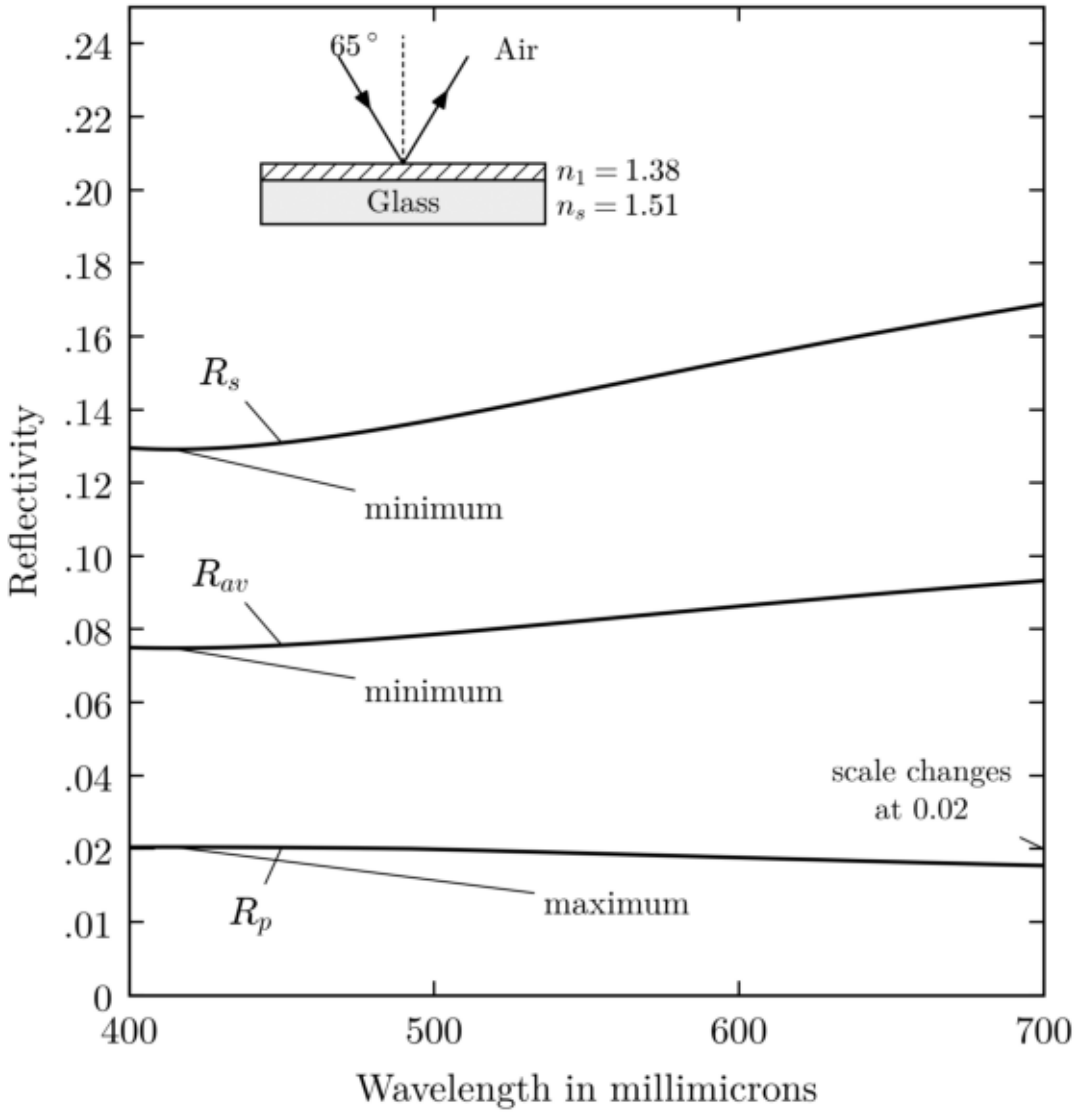


Figure 20.22 — DONE

1962 scan

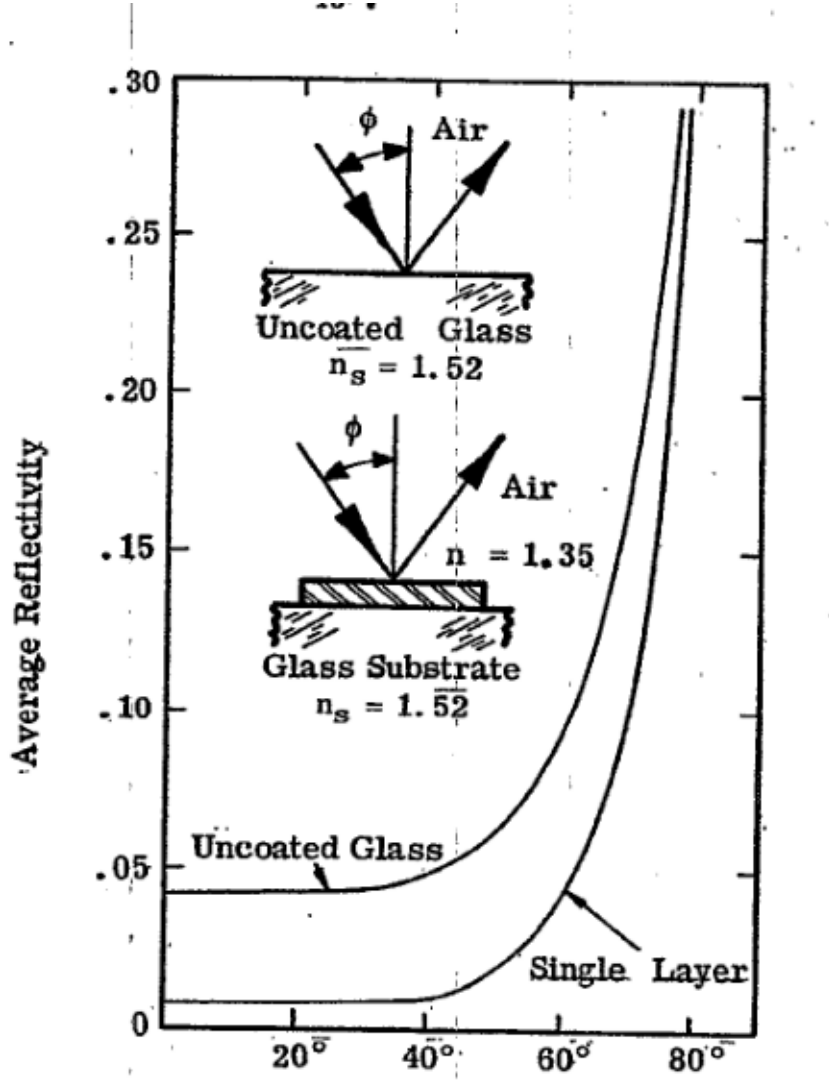


Figure 20.22 – Angle of incidence ϕ versus R_{av} of uncoated glass and the minimum R_{av} of the same glass coated with a single layer.

rebuilt

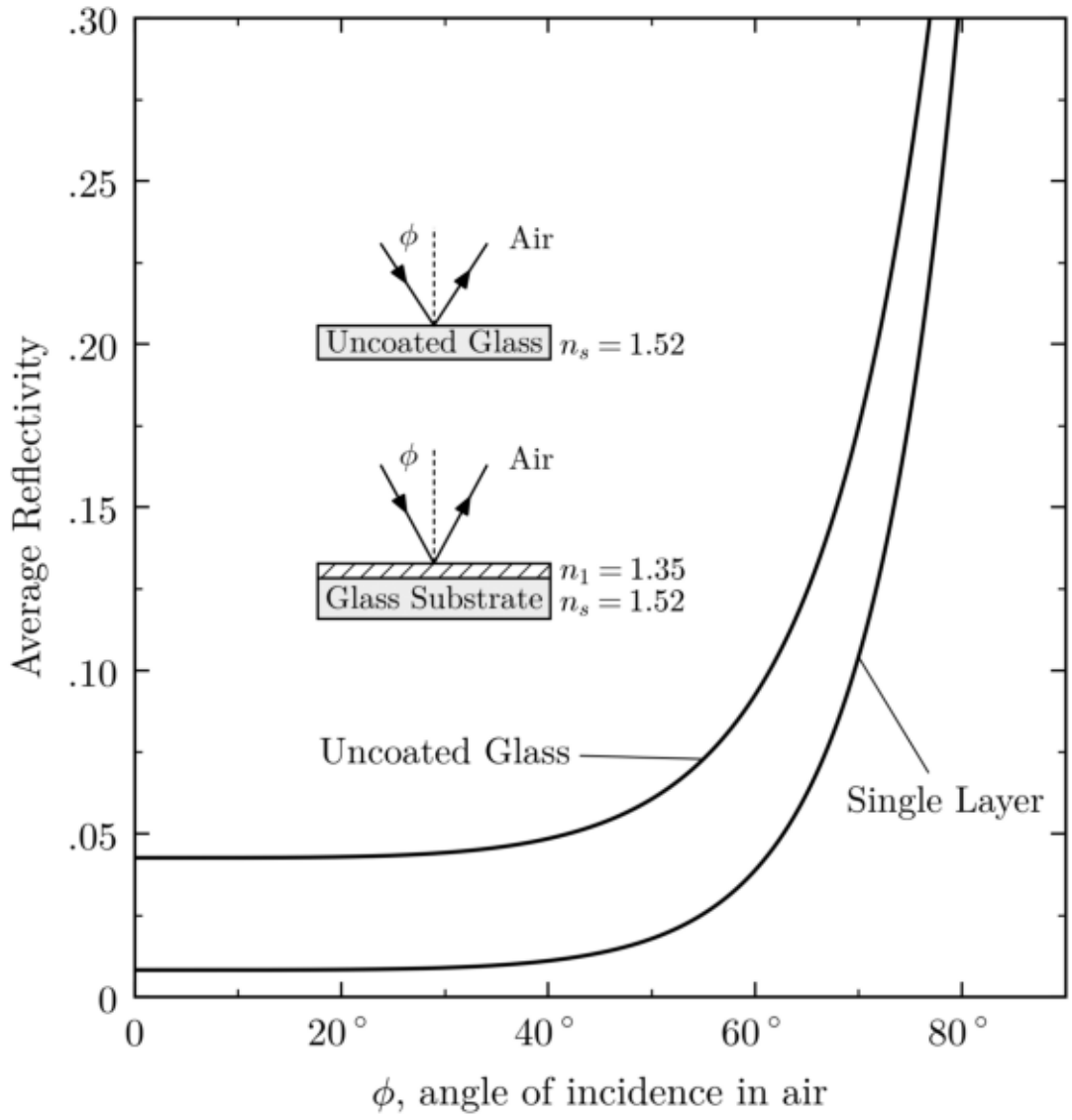


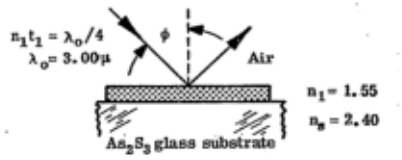
Figure 20,22 — Angle of incidence @ versus Ry

Figure 20.23 — DONE

1962 scan

LAYER	INDEX AT NORMAL INCIDENCE	EFFECTIVE INDEX AT $\phi = 65^\circ$	
		POLARIZATION P	POLARIZATION S
Incident medium (Air)	1.00	2.366	0.423
Magnesium fluoride	1.38	1.83	1.041
Glass substrate	1.51	1.89	1.208

Table 20.3 – The effective indices at $\phi = 65^\circ$ incidence of a single-layer antireflection coating.



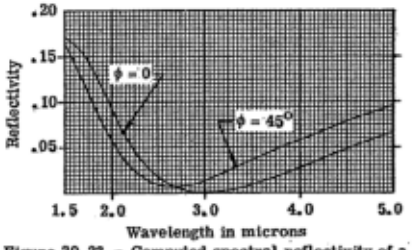
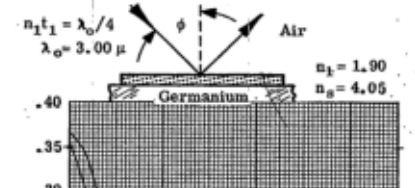


Figure 20.23 – Computed spectral reflectivity of a single-layer antireflection coating at $\phi = 0$ and $\phi = 45^\circ$.



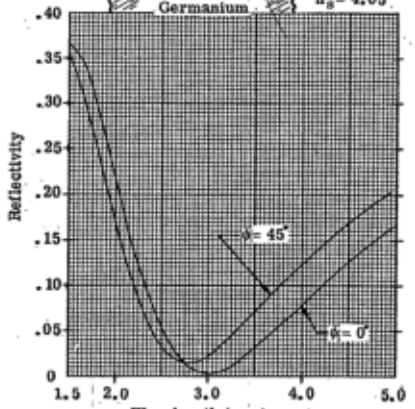
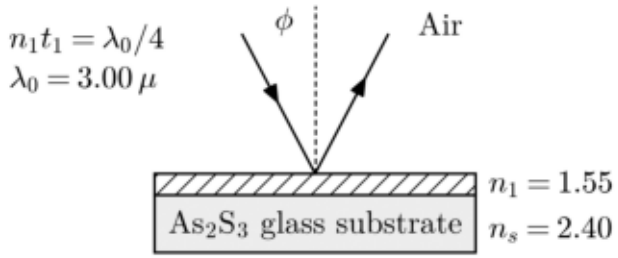


Figure 20.24 – Computed spectral reflectivity of a single-layer antireflection coating at $\phi = 0$ and $\phi = 45^\circ$.

rebuilt



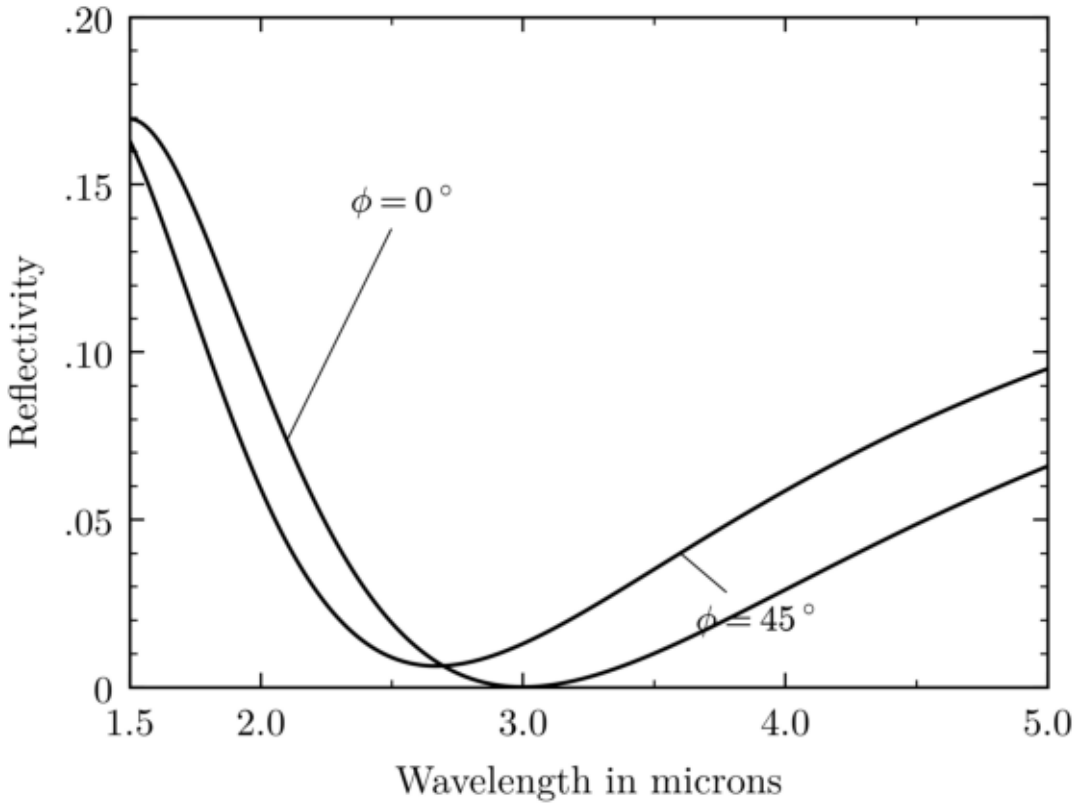


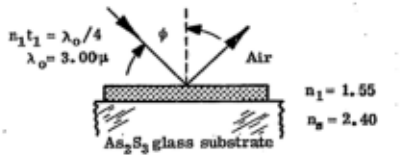
Figure 20.23 - Computed spectral reflectivity of a

Figure 20.24 — DONE

1962 scan

LAYER	INDEX AT NORMAL INCIDENCE	EFFECTIVE INDEX AT $\phi = 65°$	
		POLARIZATION P	POLARIZATION S
Incident medium (Air)	1.00	2.366	0.423
Magnesium fluoride	1.38	1.83	1.041
Glass substrate	1.51	1.89	1.208

Table 20.3 – The effective indices at $\phi = 65°$ incidence of a single-layer antireflection coating.



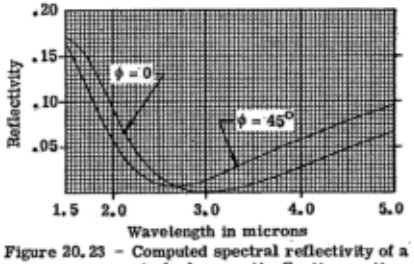


Figure 20.23 – Computed spectral reflectivity of a single-layer antireflection coating at $\phi = 0$ and $\phi = 45°$.

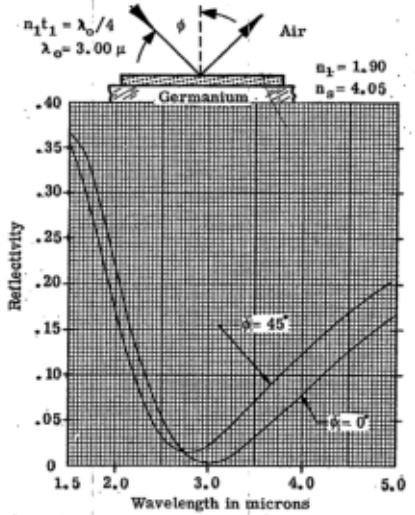
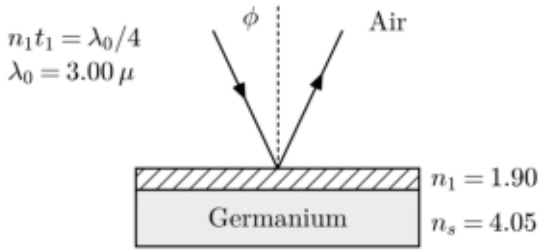


Figure 20.24 – Computed spectral reflectivity of a single-layer antireflection coating at $\phi = 0$ and $\phi = 45°$.

Figure 20,24 - Computed spectral reflectivity of a

rebuilt



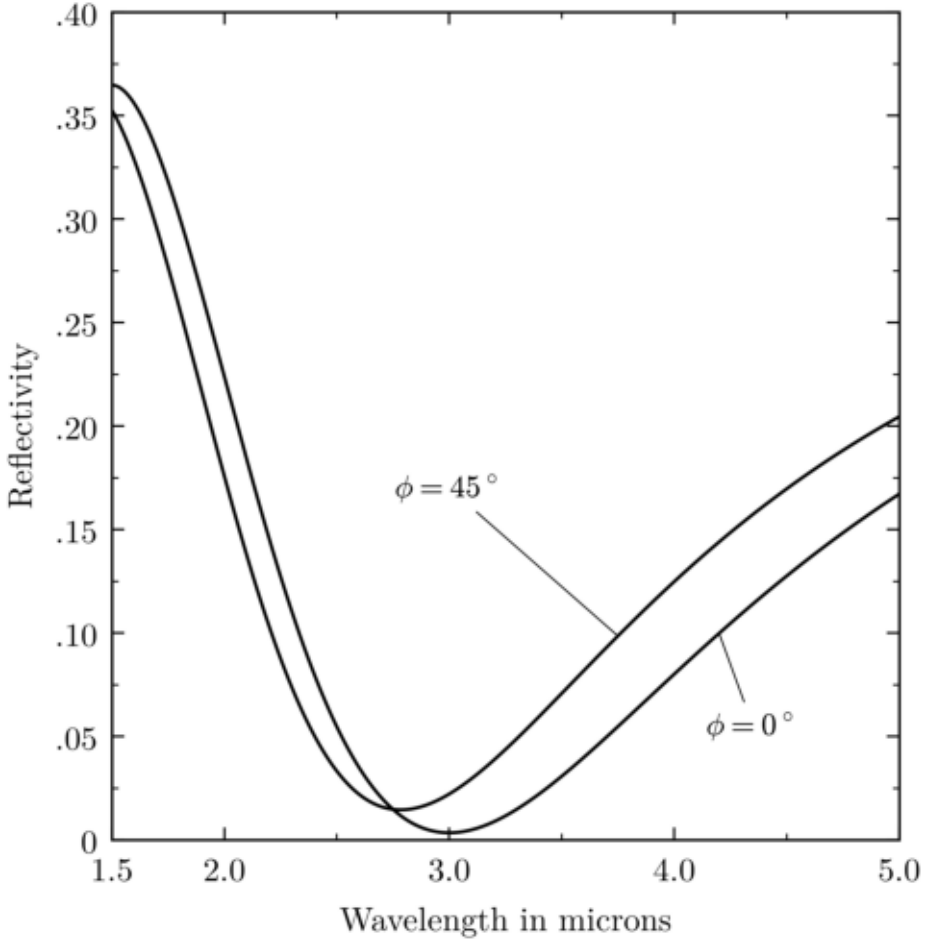


Figure 20.44 — DONE

1962 scan

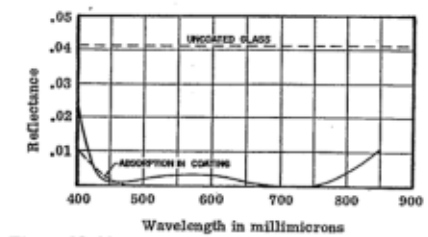


Figure 20.44 – Measured reflectance of a quarter-half-quarter antireflection coating consisting of MgF_2 + ZrO_2 + CeF_3 on glass. From ref. 31.

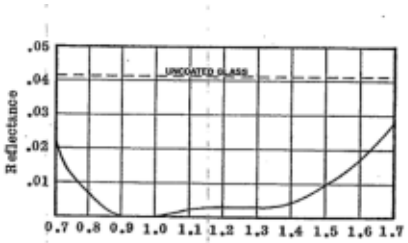


Figure 20.45 – Measured reflectance of a quarter-half-quarter antireflection coating consisting of MgF_2 + Nd_2O_3 + CeF_3 on glass. From ref. 31.

Figure 20.44 - Measured reflectance of a quarter- Figure '20. 45 ~Measured reflectance of a quarter-

rebuilt

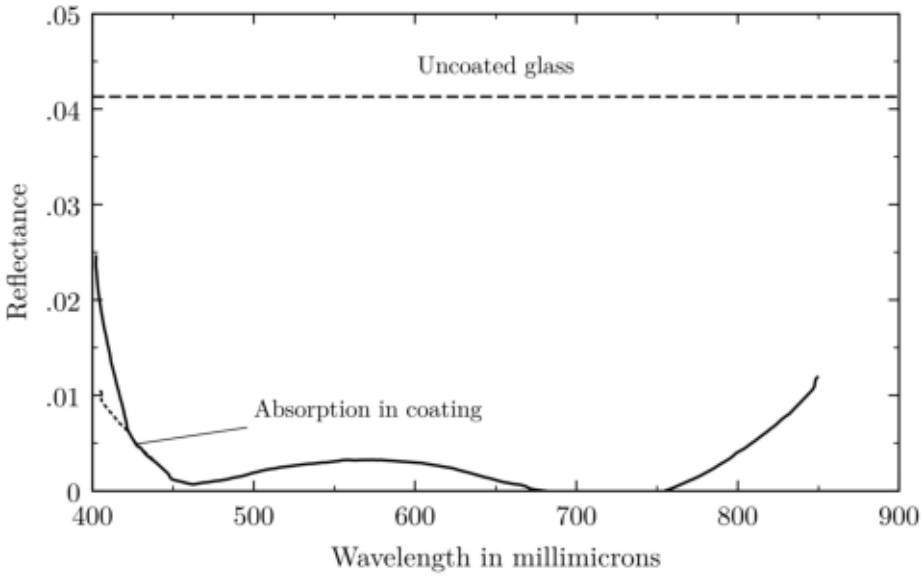


Figure 20.45 — DONE

1962 scan

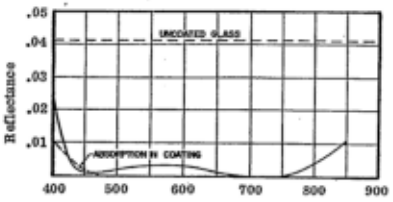


Figure 20.44 – Measured reflectance of a quarter-half-quarter antireflection coating consisting of MgF_2 + ZrO_2 + CeF_3 on glass. From ref. 31.

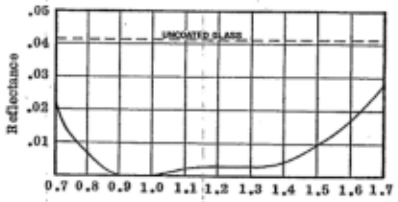


Figure 20.45 – Measured reflectance of a quarter-half-quarter antireflection coating consisting of MgF_2 + Nd_2O_3 + CeF_3 on glass. From ref. 31.

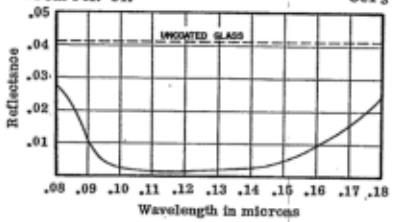


Figure 20.46 – Measured reflectance of a quarter-half-quarter antireflection coating consisting of MgF_2 + SiO + CeF_3 on glass. From ref. 31.

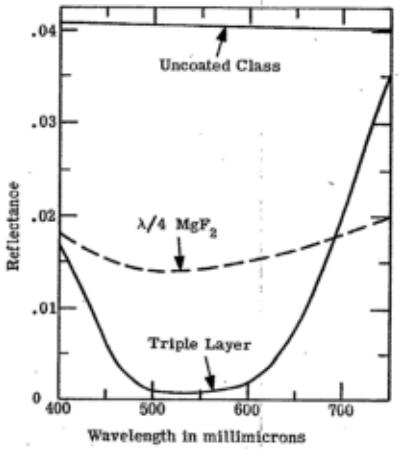


Figure 20.47 – Measured spectral reflectivity of single and triple-layer antireflection coatings. Courtesy of Fish-Schurman, Corporation.

20-36

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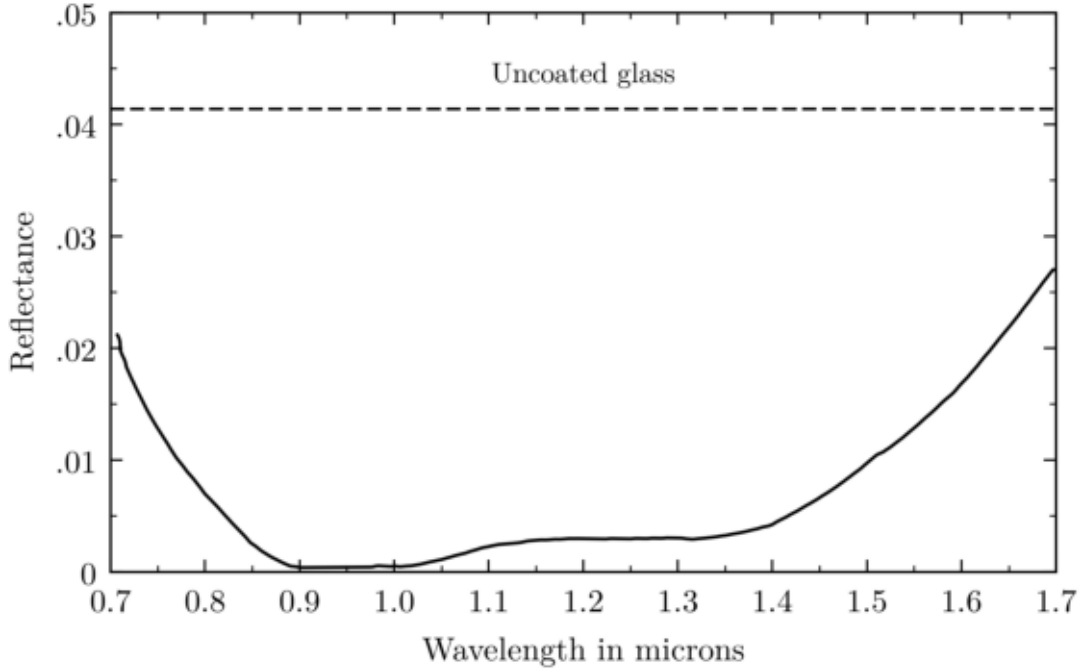


Figure 20.46 — DONE

1962 scan

consisting of MgF_2 + ZrO_2 + CeF_3 on glass. From ref. 31.

consisting of MgF_2 + Nd_2O_3 + CeF_3 on glass. From ref. 31.

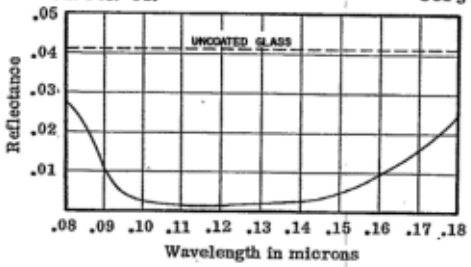


Figure 20.46 - Measured reflectance of a quarter-half-quarter antireflection coating consisting of MgF_2 + SiO + CeF_3 on glass. From ref. 31.

Figure 20.46 - Measured reflectance of a quarter-

rebuilt

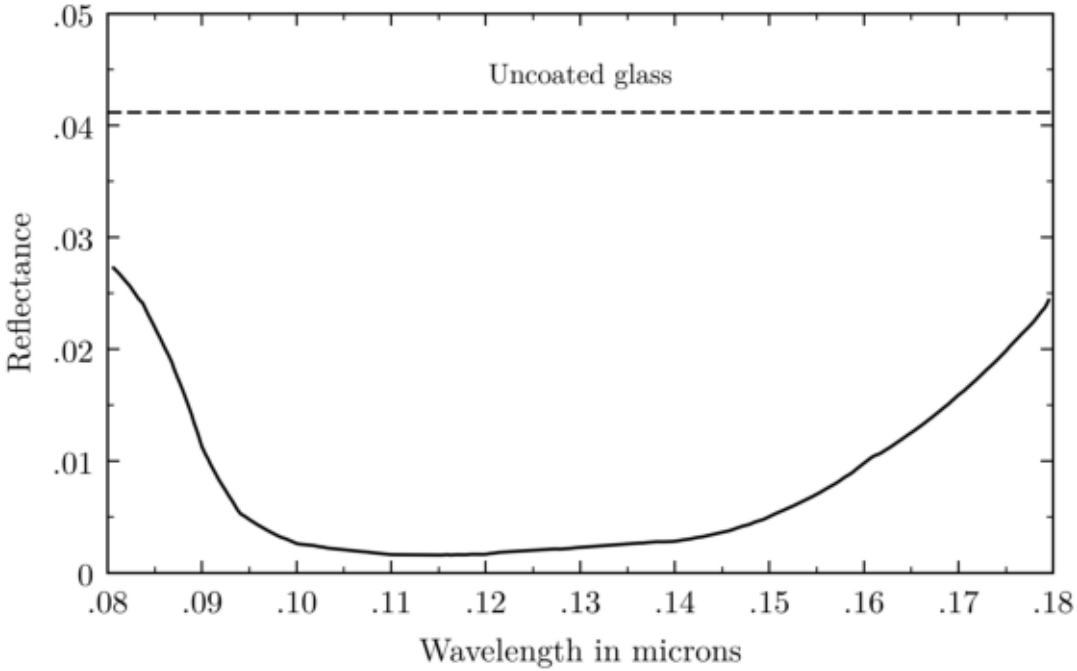


Figure 20.47 — DONE

1962 scan

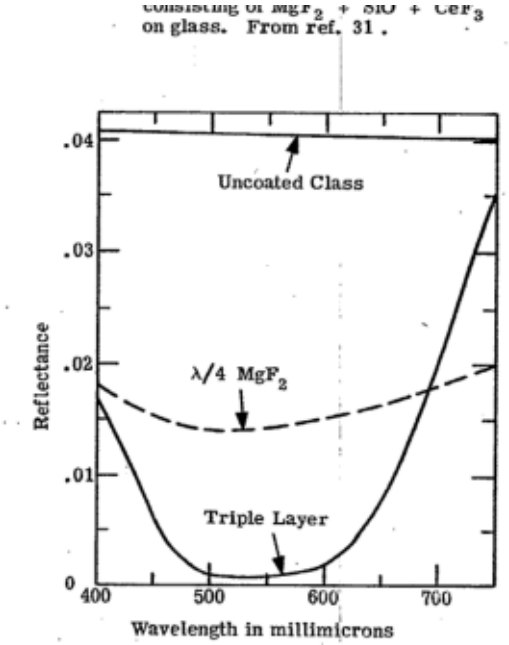


Figure 20. 47 – Measured spectral reflectivity of single and triple-layer antireflection coatings. Courtesy of Fish-Schurman, Corporation.

Figure 20. 47— Measured spectral reflectivity of

rebuilt

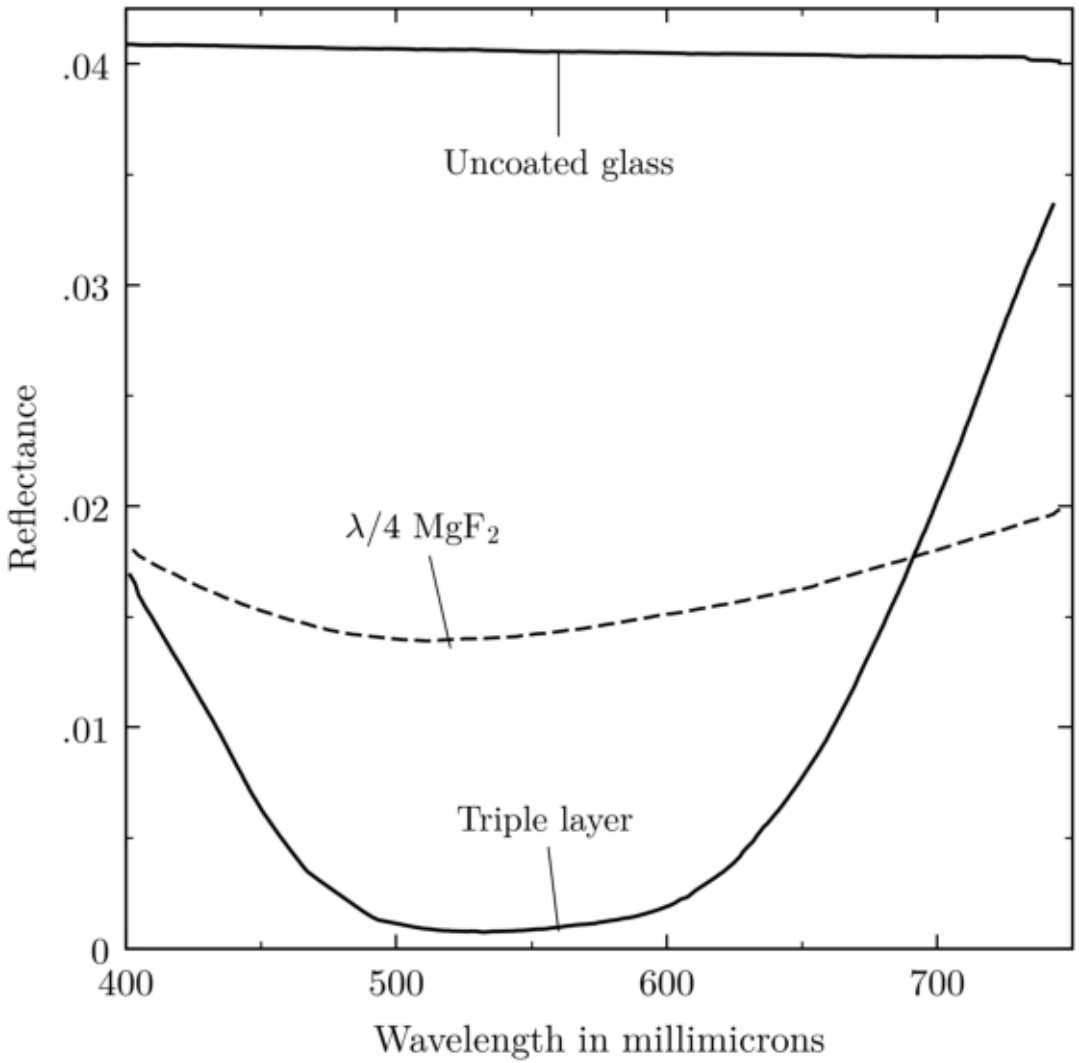


Figure 20.59 — DONE

1962 scan

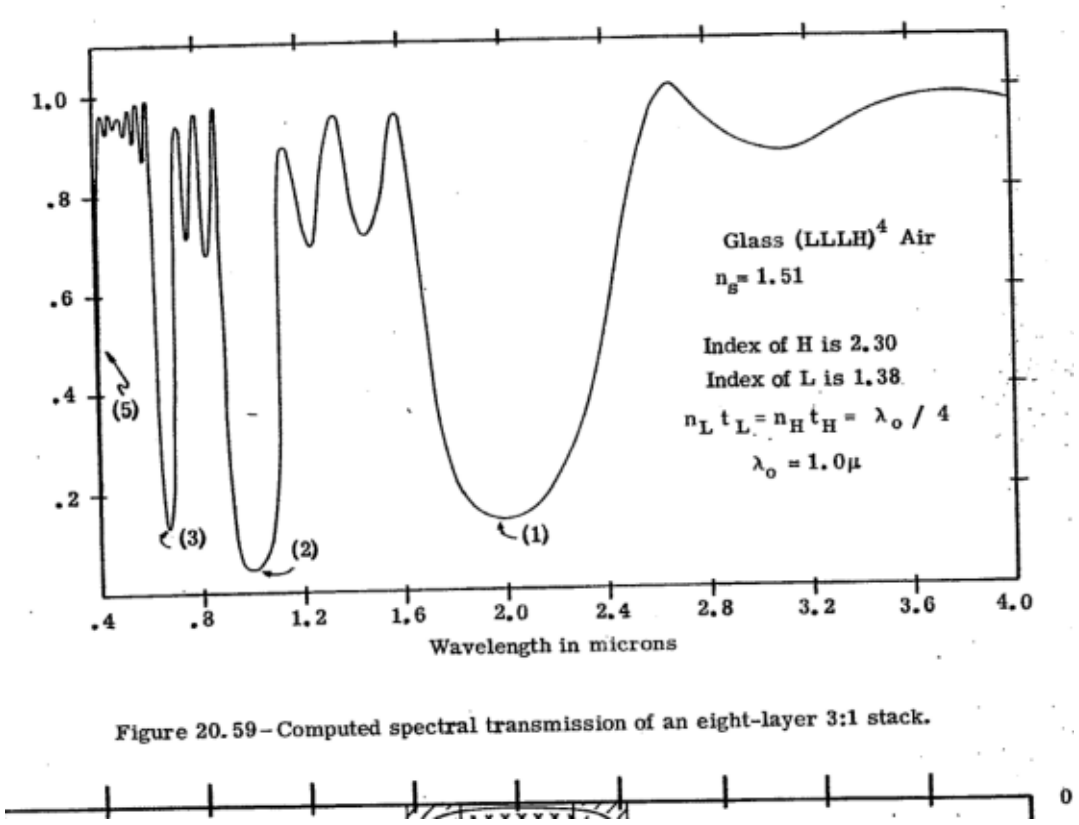


Figure 20.59 – Computed spectral transmission of an eight-layer 3:1 stack.

Figure 20. 59- Computed spectral transmission of an eight-layer 3:1 stack.

rebuilt

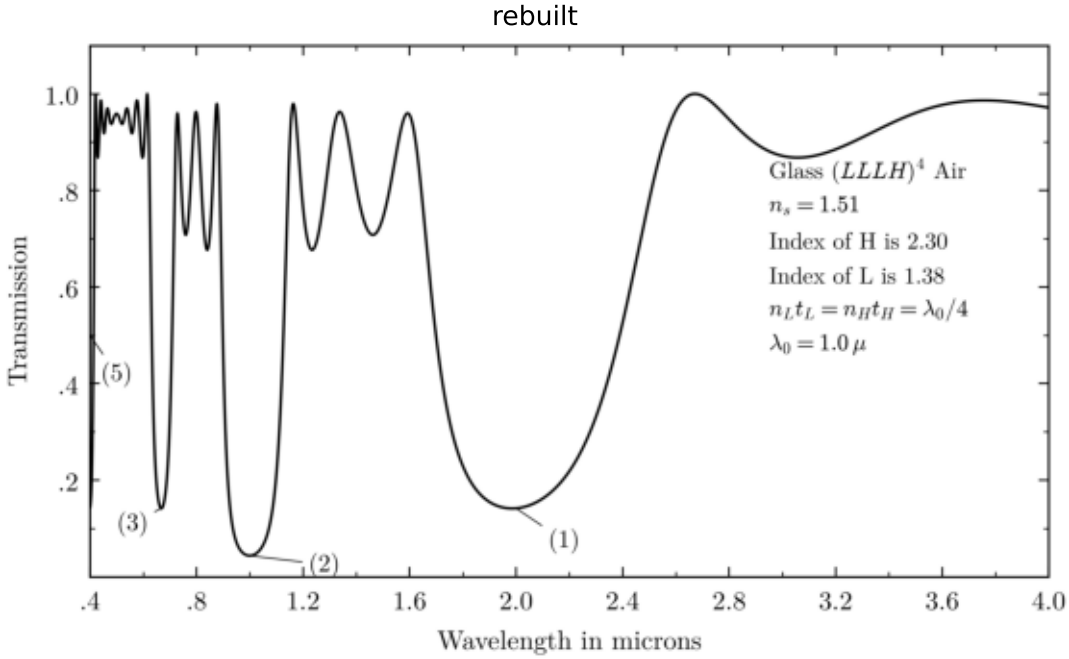


Figure 20.60 — DONE

1962 scan

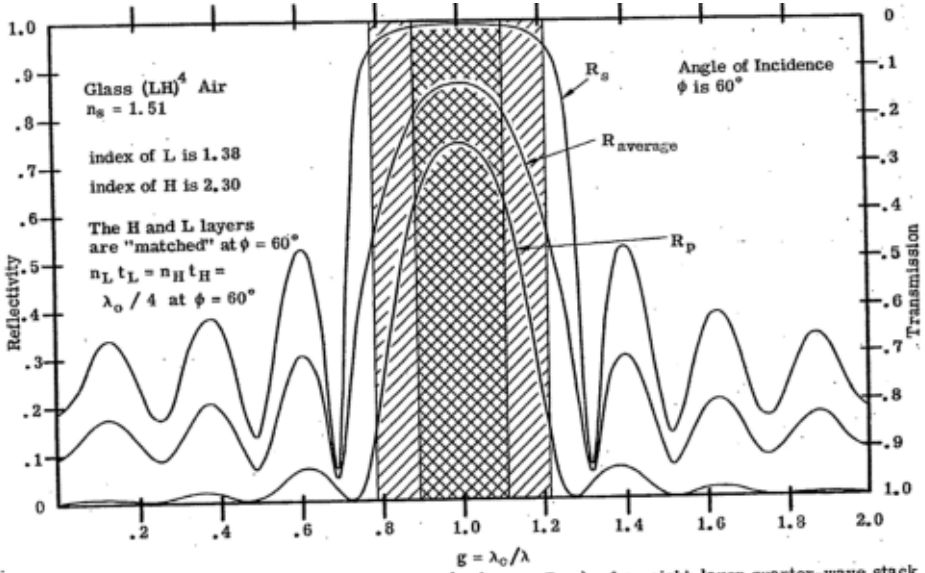


Figure 20.60 – Computed R_p, R_s, and R_{av} = 1/2 (R_p + R_s) of an eight-layer quarter-wave stack at ϕ = 60°. The optical thickness of the H and L layers is matched 1:1 at ϕ = 60°.

rebuilt

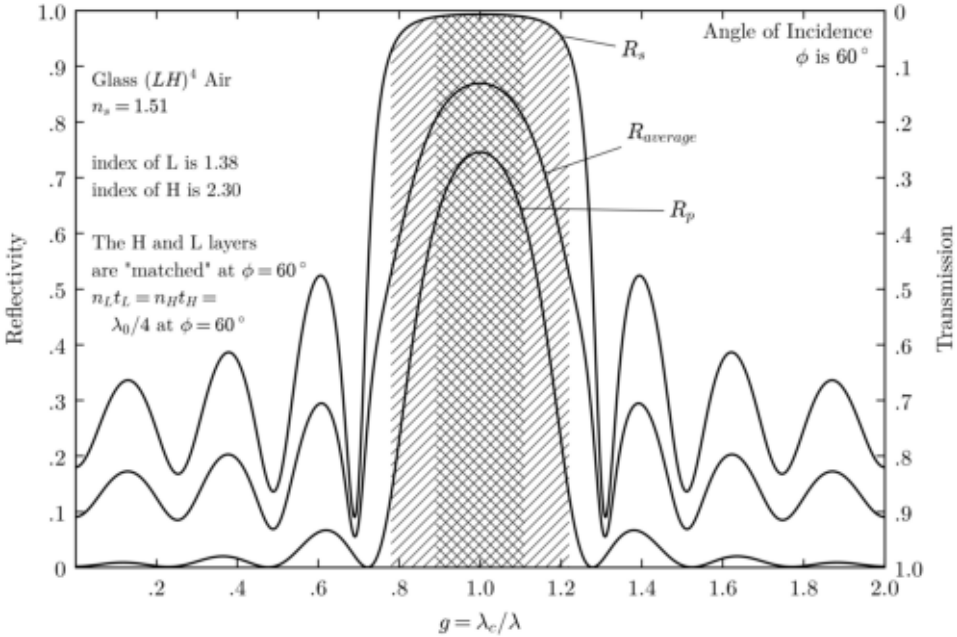


Figure 20,60-Computed Rp, Rs, and Ray = 1/2 (Rp + Rg) of an eight-layer quarter-wave stack

Figure 20.85 — DONE

1962 scan

of a short-wave pass heat reflecting filter. Courtesy of Eastman Kodak Company.

Figure 20. 85-

blue and near infrared. Courtesy of Fish-Schurman, Corporation.

rebuilt

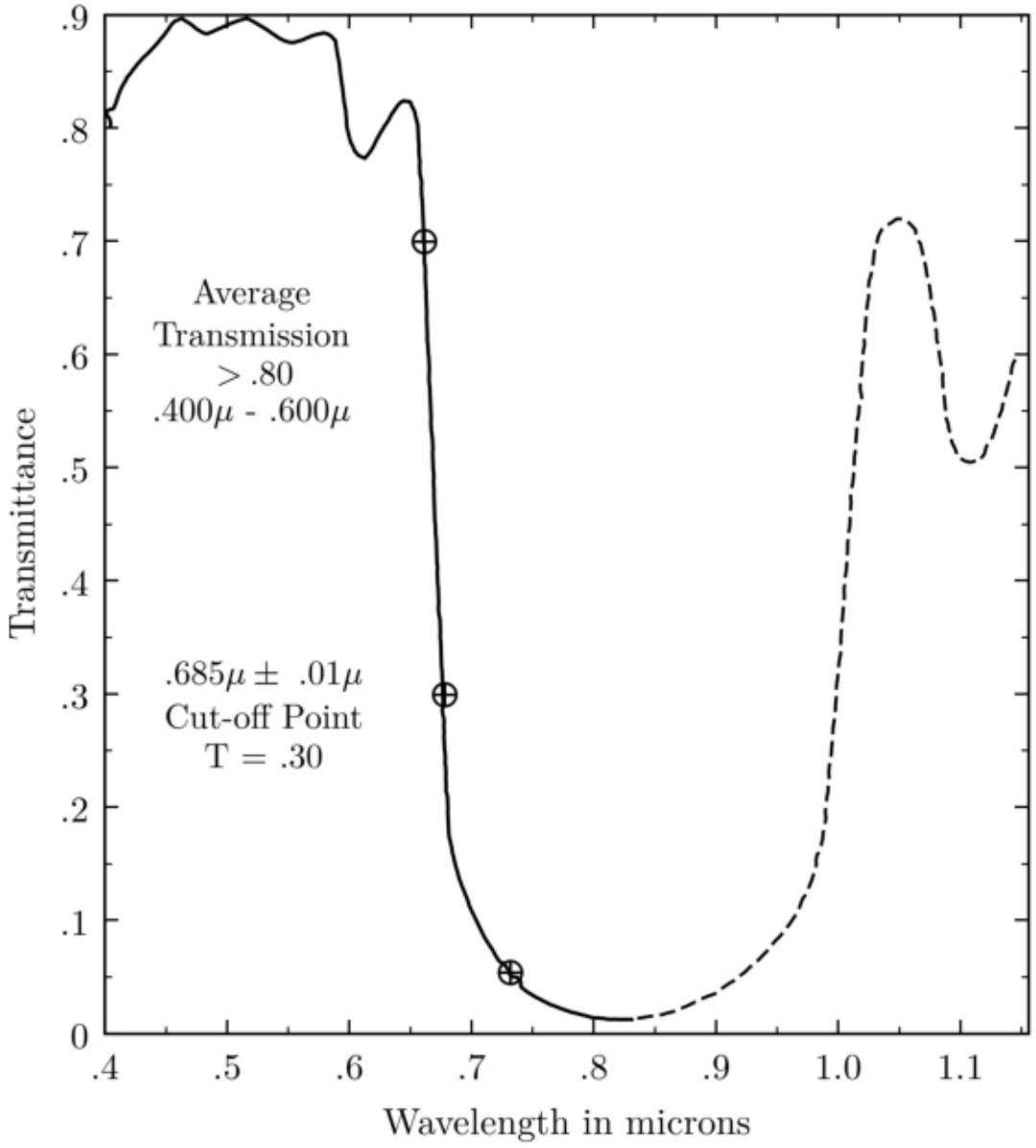


Figure 20.86 — DONE

1962 scan

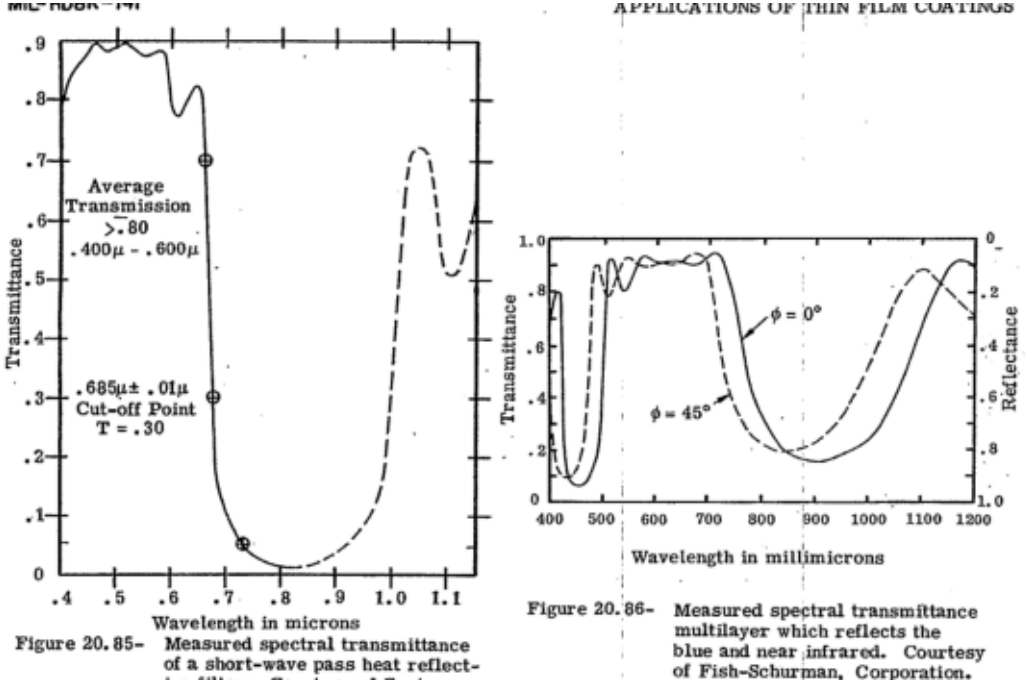


Figure 20.85- Measured spectral transmittance of a short-wave pass heat reflect-

Figure 20.86- Measured spectral transmittance multilayer which reflects the blue and near infrared. Courtesy of Fish-Schurman, Corporation.

Figure 20.'86- Measured spectral transmittance

rebuilt

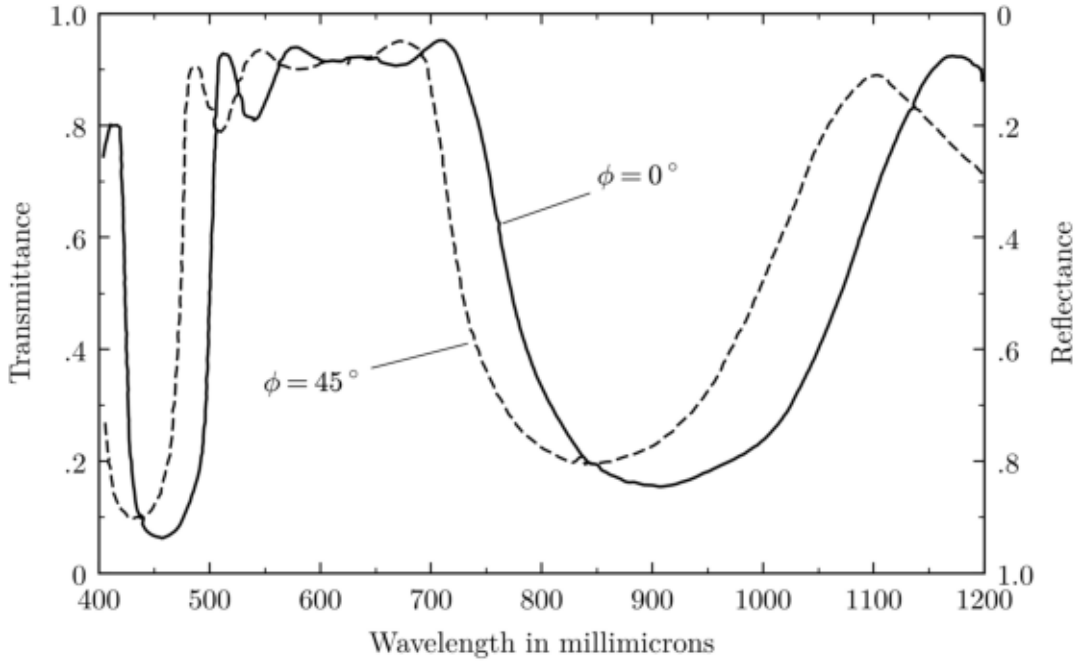


Figure 20.87 — DONE

1962 scan

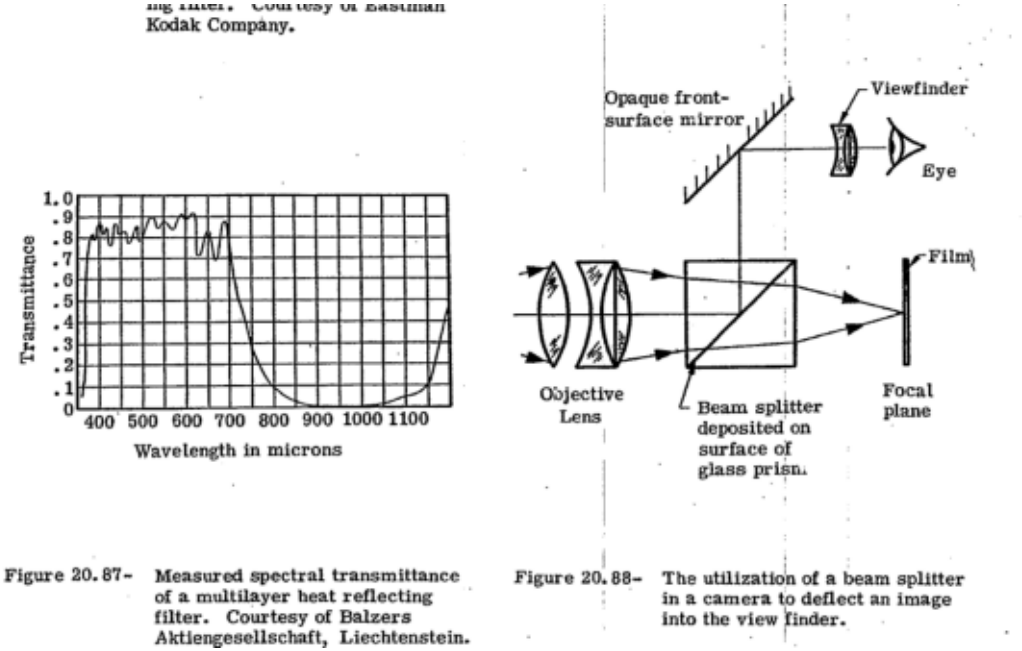


Figure 20.87– Measured spectral transmittance of a multilayer heat reflecting filter. Courtesy of Balzers Aktiengesellschaft, Liechtenstein.

Figure 20.88– The utilization of a beam splitter in a camera to deflect an image into the view finder.

Figure 20. 87~

rebuilt

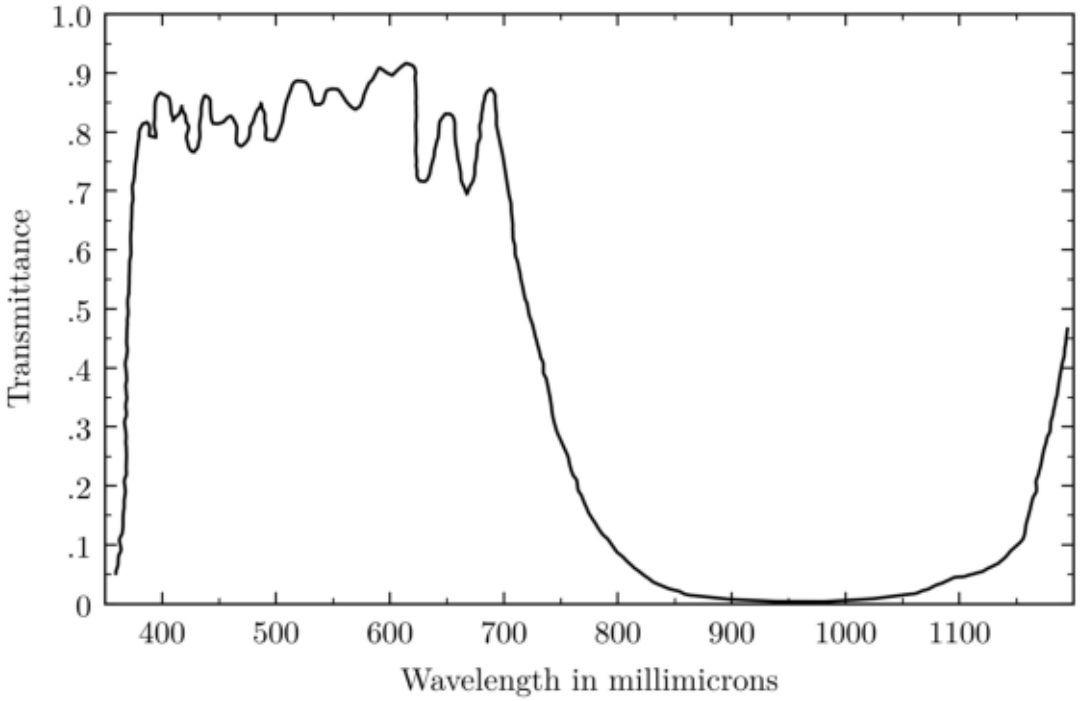


Figure 20.95 — DONE

1962 scan

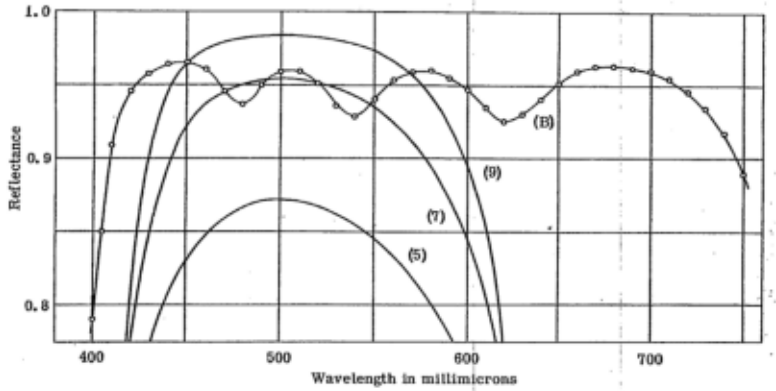


Figure 20.95- Measured spectral reflectivity of quarter-wave stacks with 5, 7, and 9 layers and also a broad-band reflector consisting of 15 layers of ZnS and cryolite. From ref. 49.

MEASURED TRANSMITTANCE OF A SERIES OF BAND PASS FILTERS COMPLETE WITH AUXILIARIES

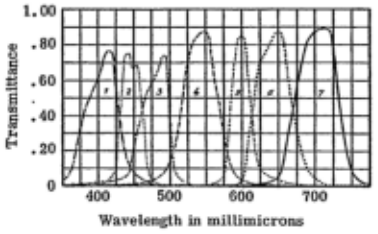


Figure 20.96- Measured transmittance of band pass filters which utilize both multilayer and glass absorption filters. Courtesy of Balzers Aktiengesellschaft, Liechtenstein.

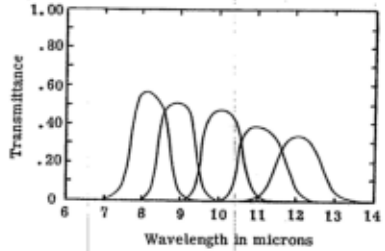
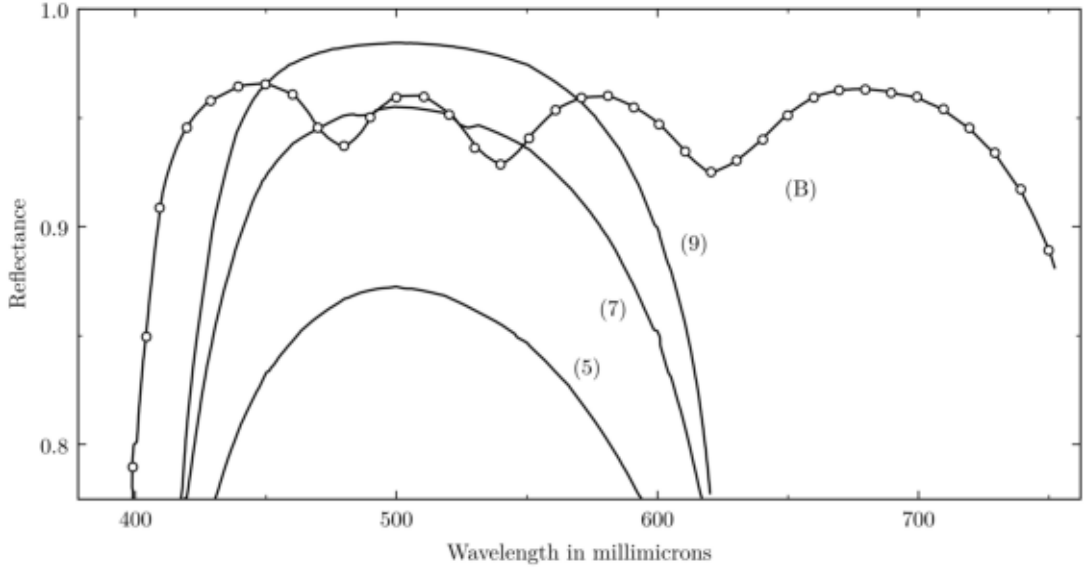


Figure 20.97- Measured spectral transmittance of some infrared band pass filters, complete with auxilliaries. Courtesy of Bausch and Lomb, Inc.

rebuilt



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Figure 20.96 — DONE

1962 scan

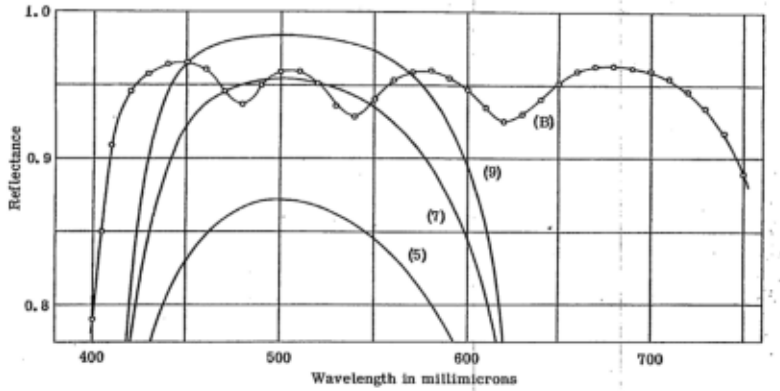


Figure 20.95- Measured spectral reflectivity of quarter-wave stacks with 5, 7, and 9 layers and also a broad-band reflector consisting of 15 layers of ZnS and cryolite. From ref. 49.

MEASURED TRANSMITTANCE OF A SERIES OF BAND PASS FILTERS COMPLETE WITH AUXILIARIES

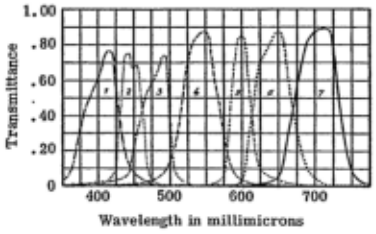


Figure 20.96- Measured transmittance of band pass filters which utilize both multilayer and glass absorption filters. Courtesy of Balzers Aktiengesellschaft, Liechtenstein.

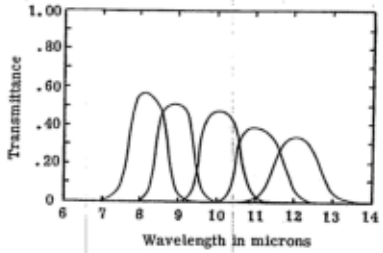
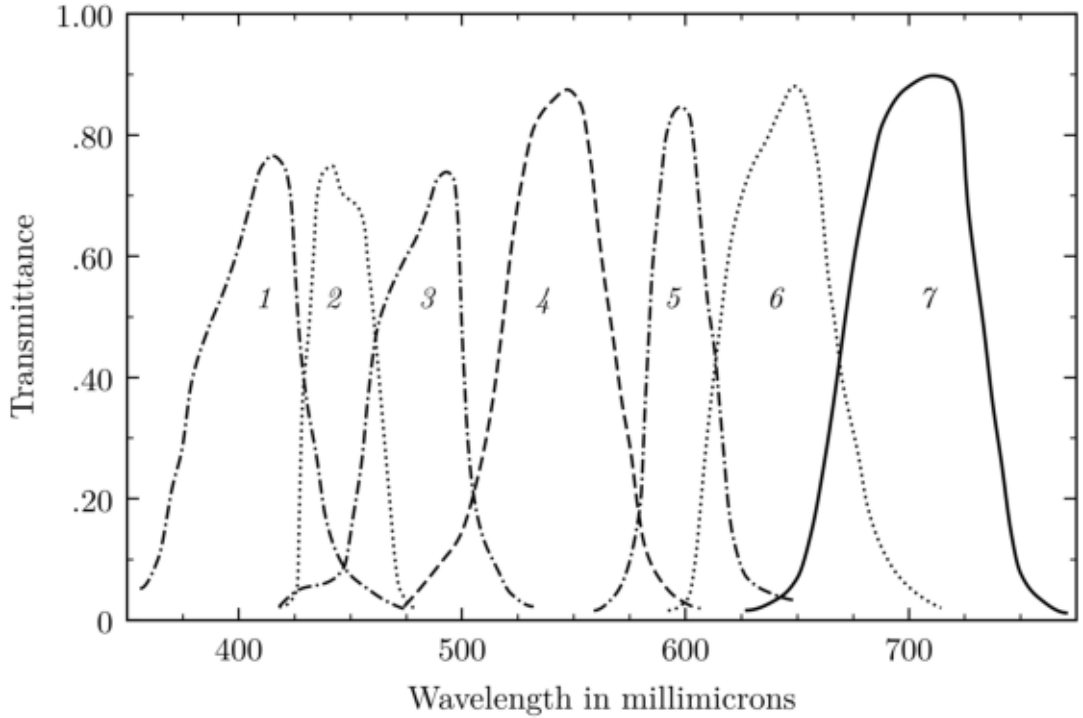


Figure 20.97- Measured spectral transmittance of some infrared band pass filters, complete with auxilliaries. Courtesy of Bausch and Lomb, Inc.

rebuilt



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Figure 20.97 — DONE

1962 scan

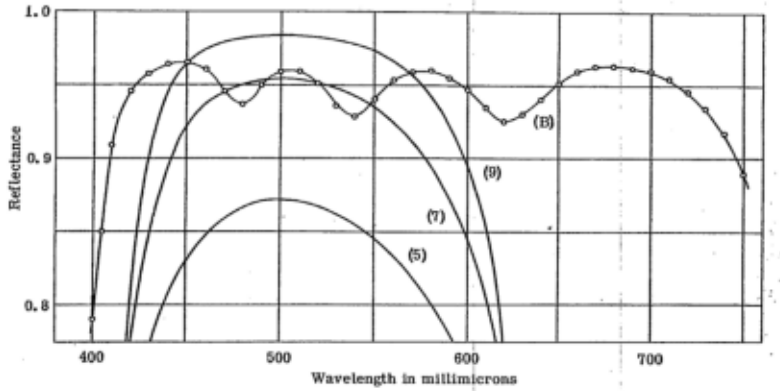


Figure 20.95- Measured spectral reflectivity of quarter-wave stacks with 5, 7, and 9 layers and also a broad-band reflector consisting of 15 layers of ZnS and cryolite. From ref. 49.

MEASURED TRANSMITTANCE OF A SERIES OF BAND PASS FILTERS COMPLETE WITH AUXILIARIES

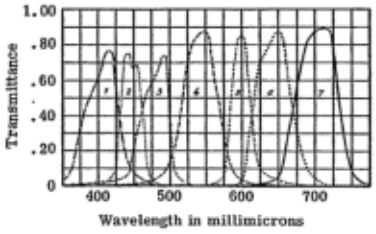


Figure 20.96- Measured transmittance of band pass filters which utilize both multilayer and glass absorption filters. Courtesy of Balzers Aktiengesellschaft, Liechtenstein.

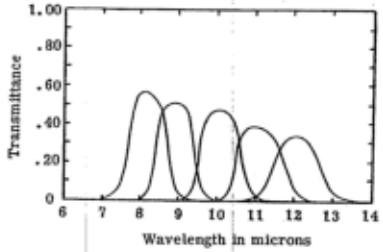
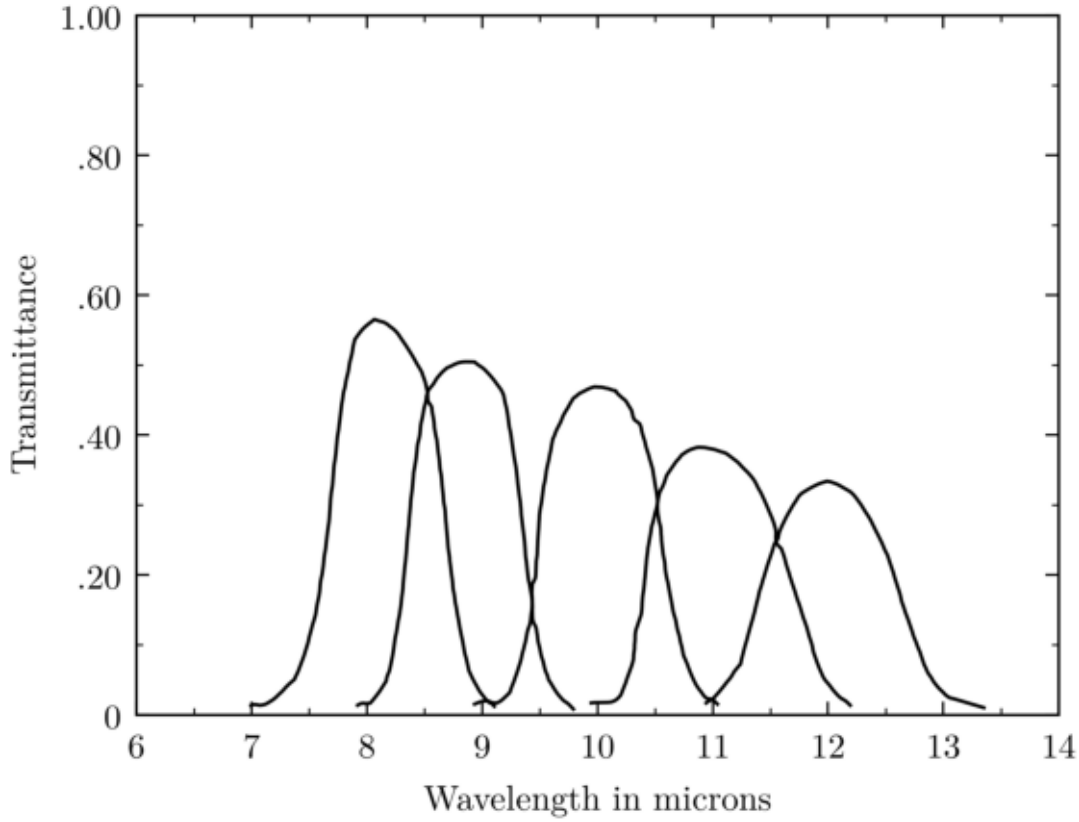


Figure 20.97- Measured spectral transmittance of some infrared band pass filters, complete with auxilliaries. Courtesy of Bausch and Lomb, Inc.

rebuilt



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Figure 20.100 — DONE

1962 scan

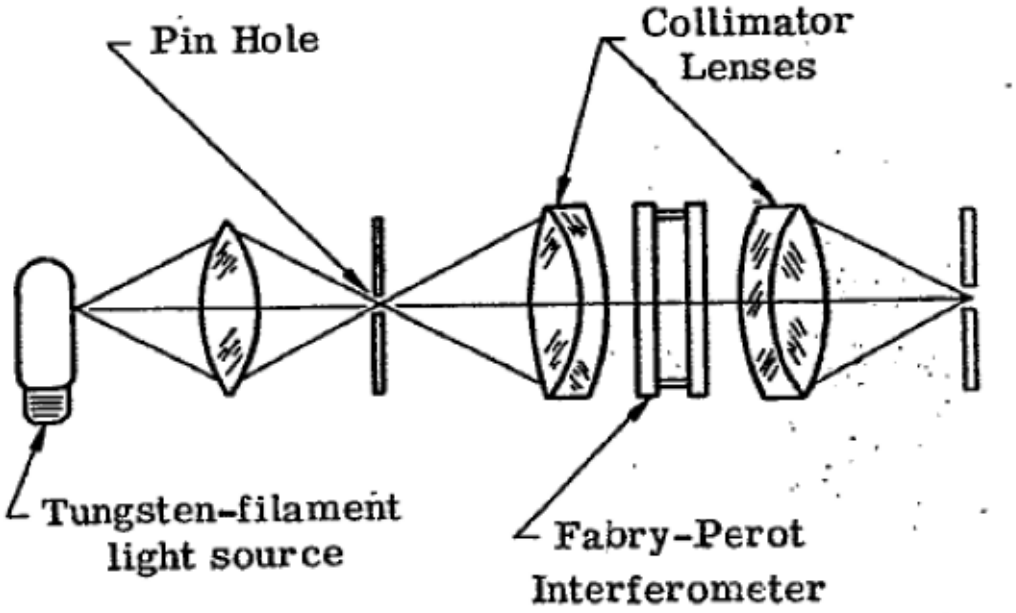


Figure 20.99- A method of illuminating the Fabry-

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rebuilt

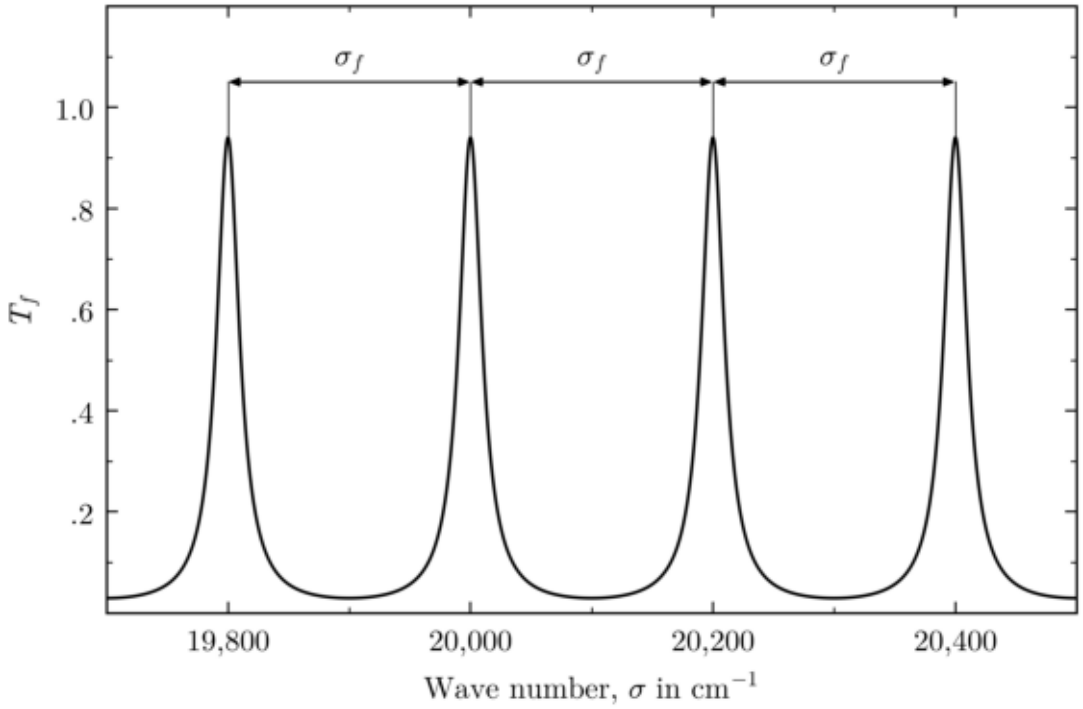


Figure 20.101 — DONE

1962 scan

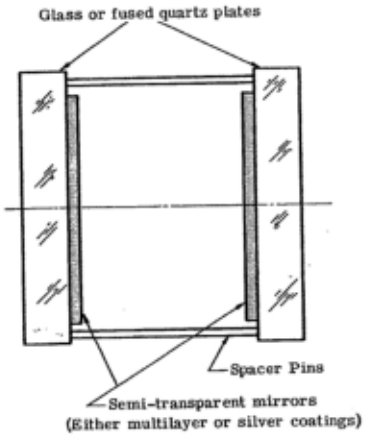


Figure 20.98- The essential parts of a Fabry-Perot interferometer.

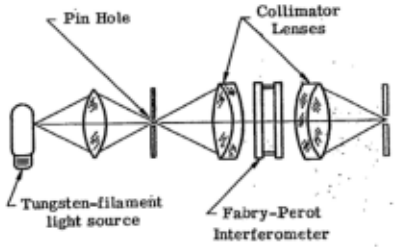


Figure 20.99- A method of illuminating the Fabry-Perot interferometer with collimated light so that it can be used as an optical filter.

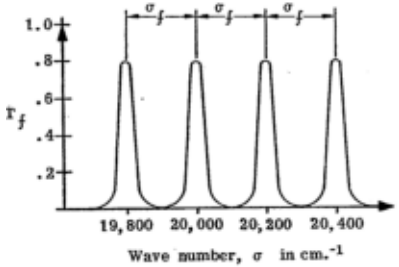
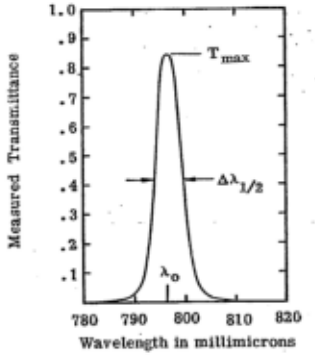


Figure 20.100 – The transmission as a function of wave number of a Fabry-Perot interferometer with t_g = 0.0025 cm .



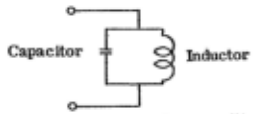
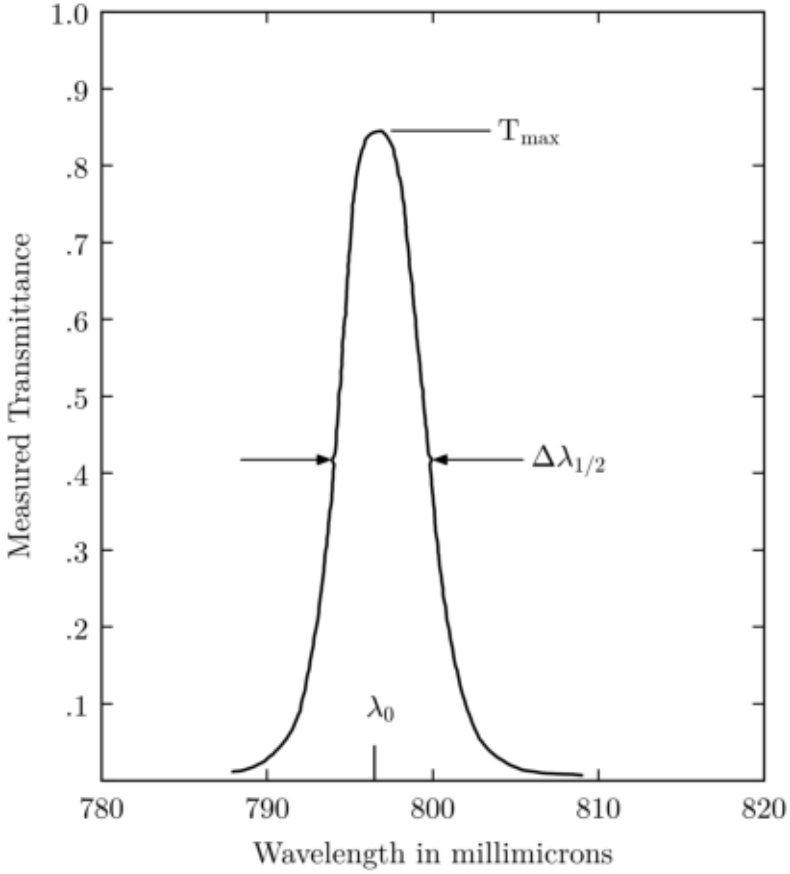
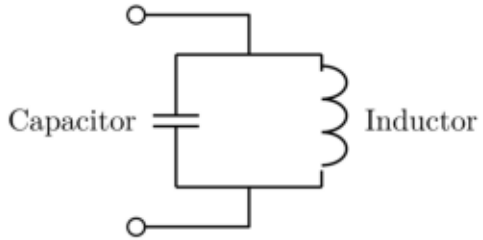


Figure 20.101- Measured spectral transmittance of Fabry-Perot type multilayer filter. Courtesy of Dr. C. Alley.
A parallel resonant circuit.

20-75

rebuilt





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Figure 20.120 — DONE

1962 scan

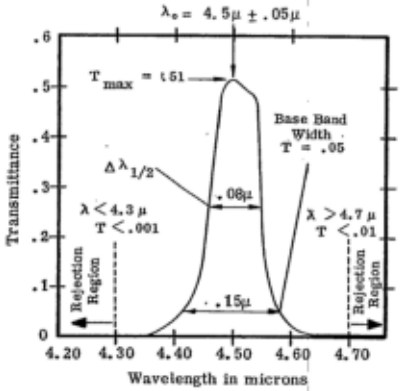


Figure 20.120- Measured spectral transmittance of a narrow band-pass filter which has a nearly rectangular shaped pass band. Courtesy of

Figure 20.120- Measured spectral transmittance

rebuilt

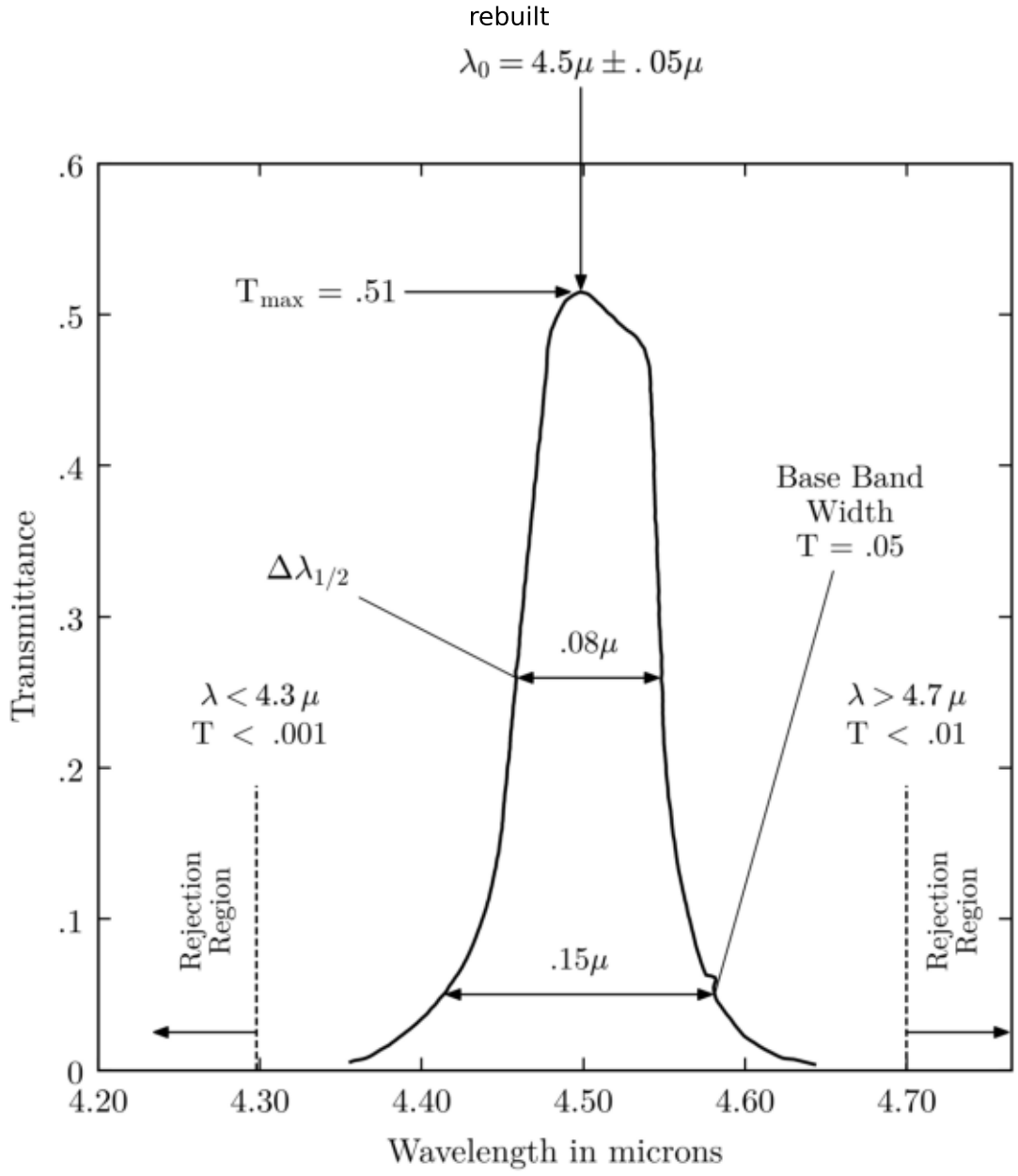


Figure 20.121 — DONE

1962 scan

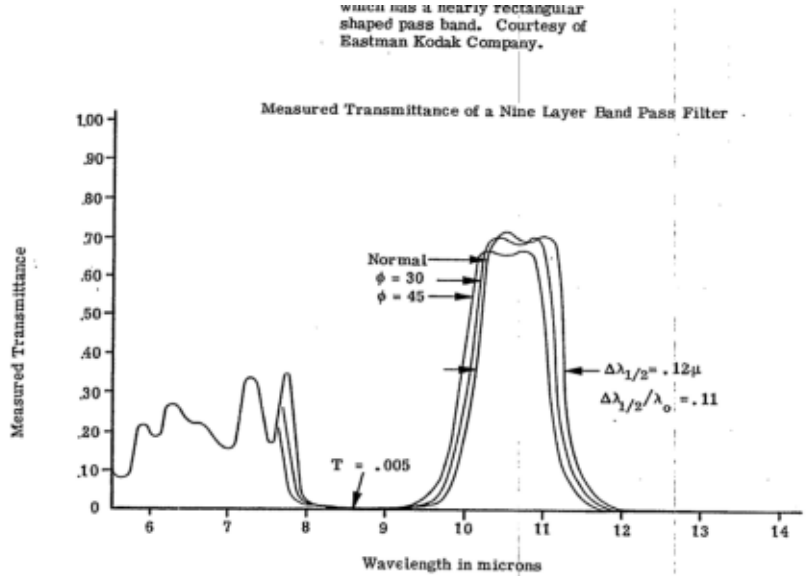


Figure 20.121- Measured spectral transmittance of a narrow band-pass filter which has a nearly rectangular shaped pass band. Courtesy of Bausch and Lomb, Inc.

Figure 20,121-

rebuilt

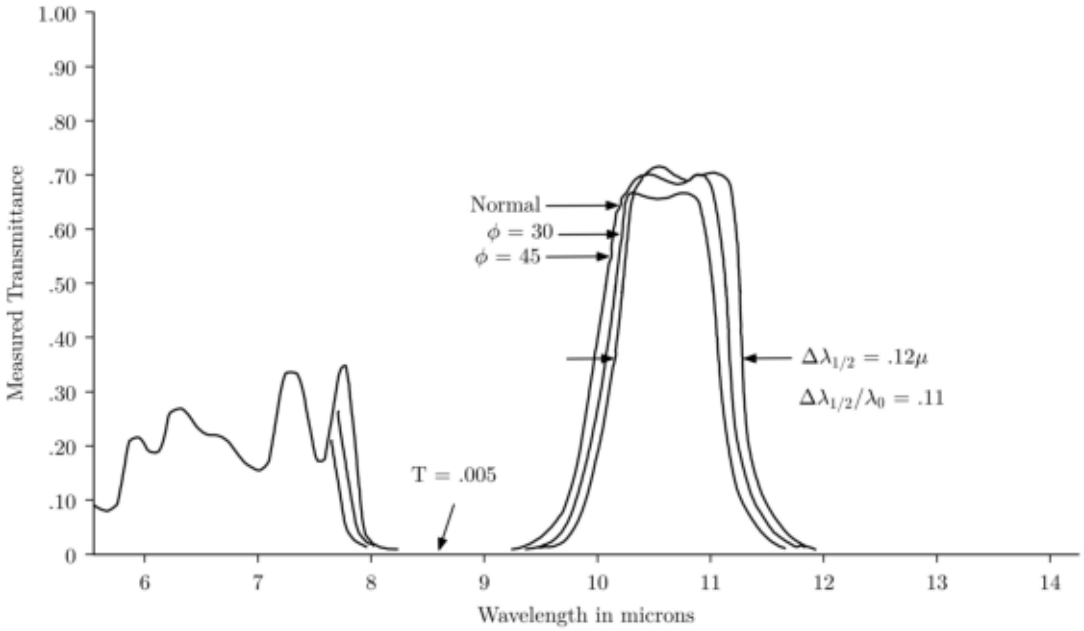
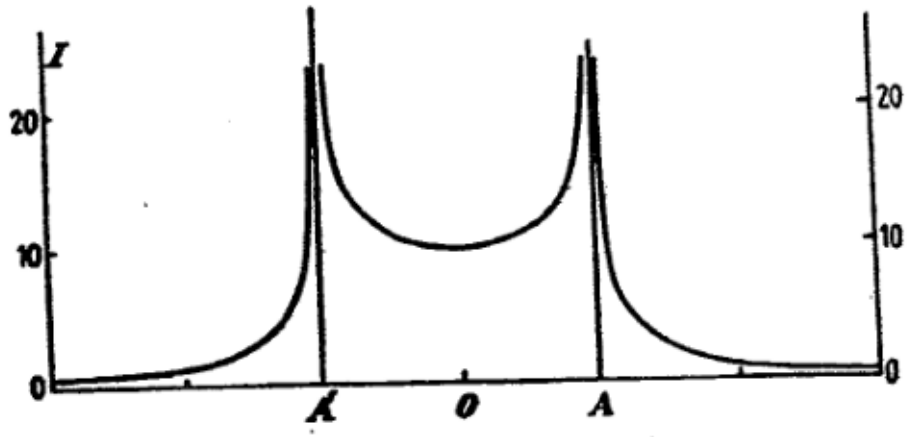


Figure 25.27 — DONE

1962 scan



ensity distribution along the horizontal diameter (after Lin

otal lines and intensity distribution for a mirror under the
From Linfoot's, Recent Advances in Optics, Oxford Univ. Press,

25. 27

Figure 25. 27- Isophotal lines and intensity distribution for a mirror under the Foucault test

rebuilt

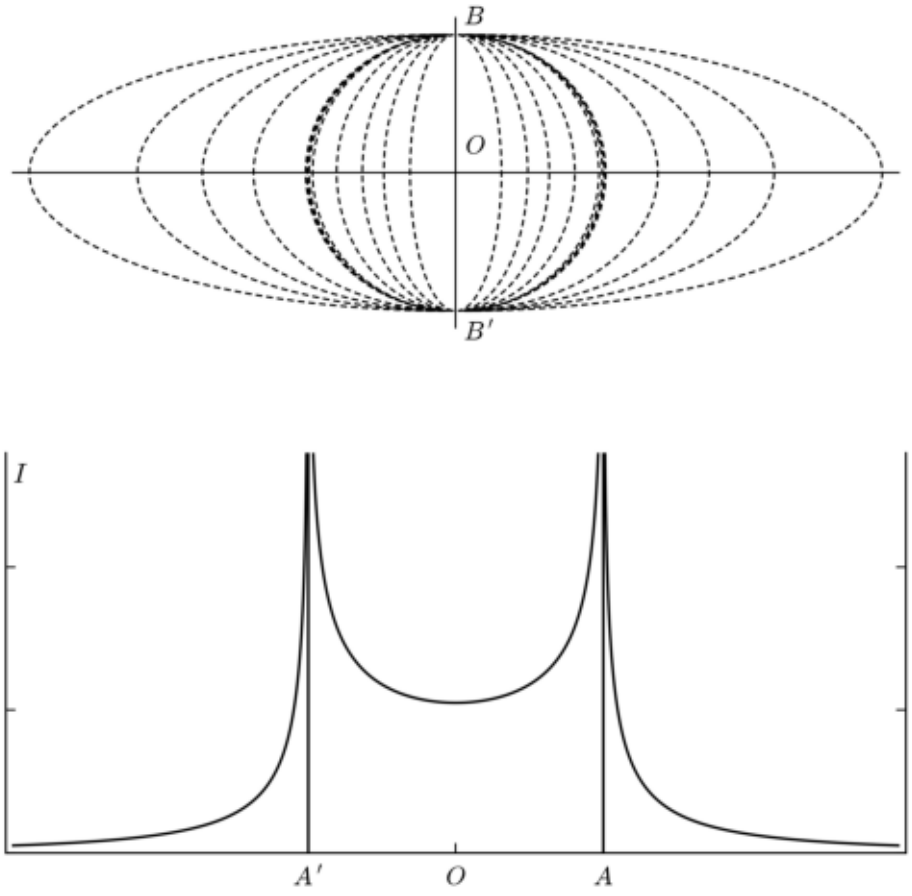
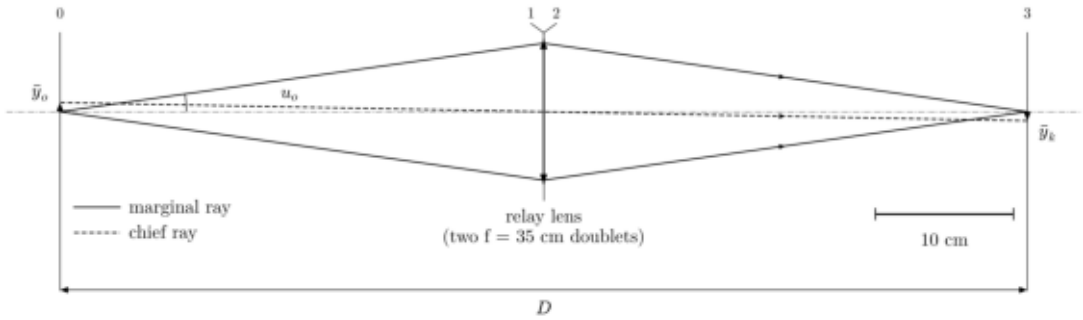


Figure 12.1 — DONE (layout)

1962 scan

(no scan crop)

rebuilt



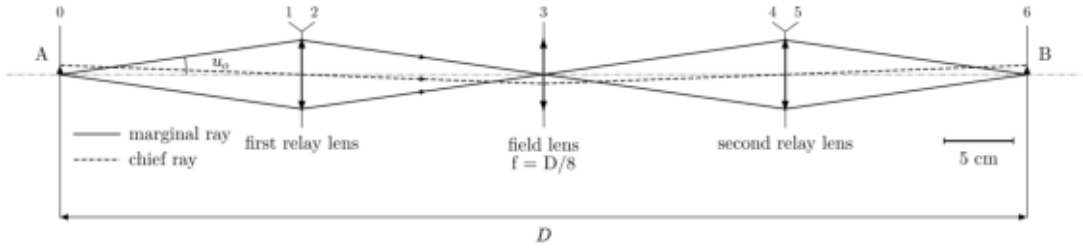
(printed caption not found on page)

Figure 12.2 — DONE (layout)

1962 scan

(no scan crop)

rebuilt



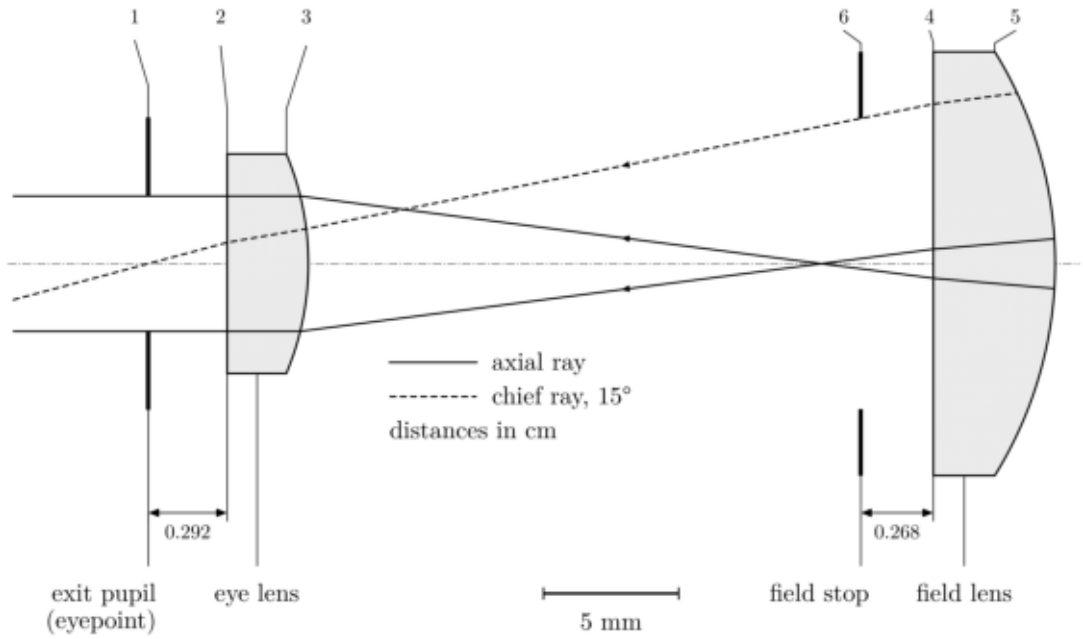
(printed caption not found on page)

Figure 14.1 — DONE (layout)

1962 scan

(no scan crop)

rebuilt



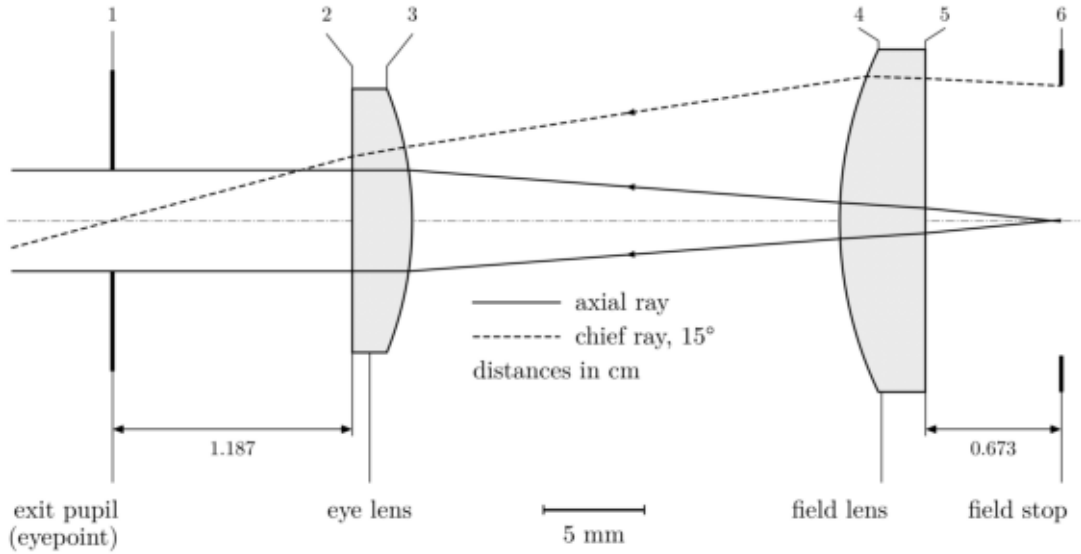
(printed caption not found on page)

Figure 14.4 — DONE (layout)

1962 scan

(no scan crop)

rebuilt



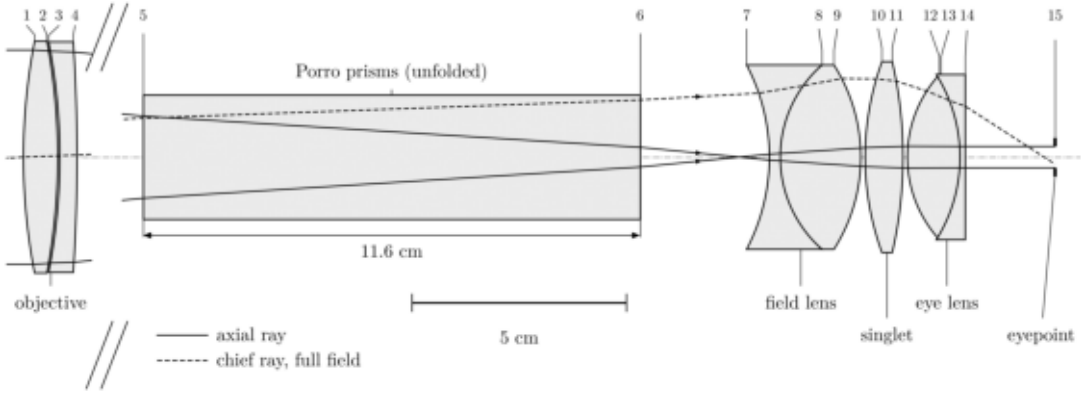
(printed caption not found on page)

Figure 15 (telescope) — DONE (layout)

1962 scan

(no scan crop)

rebuilt



(printed caption not found on page)

Figure 23.7 — DONE (layout)

1962 scan

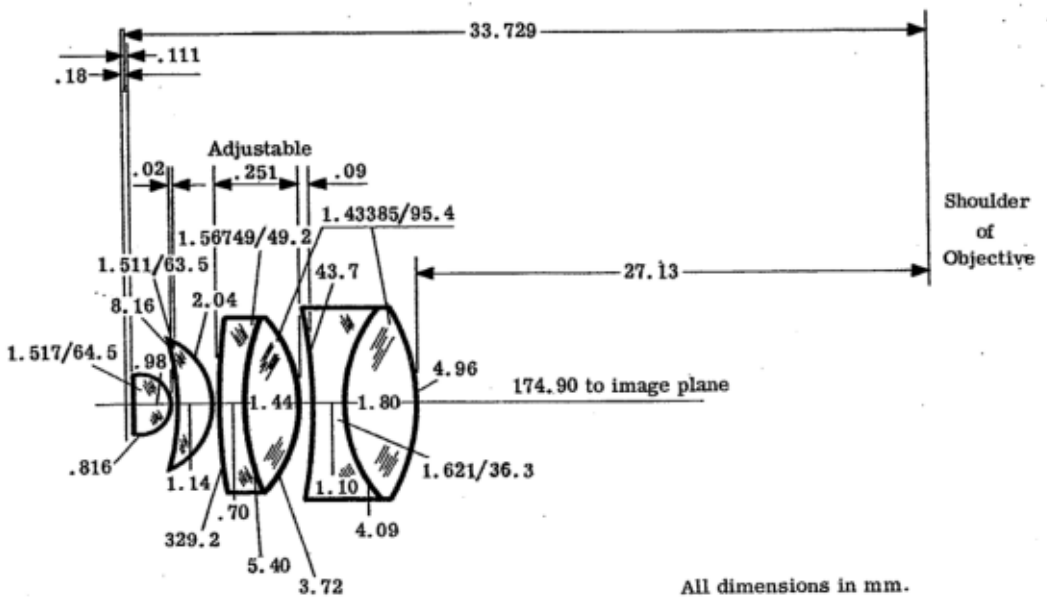


Figure 23.7- Optical layout of a 1.8mm flourite objective.

23-9

Figure 23. 7- Optical layout of a 1.8mm flourite objective.

rebuilt

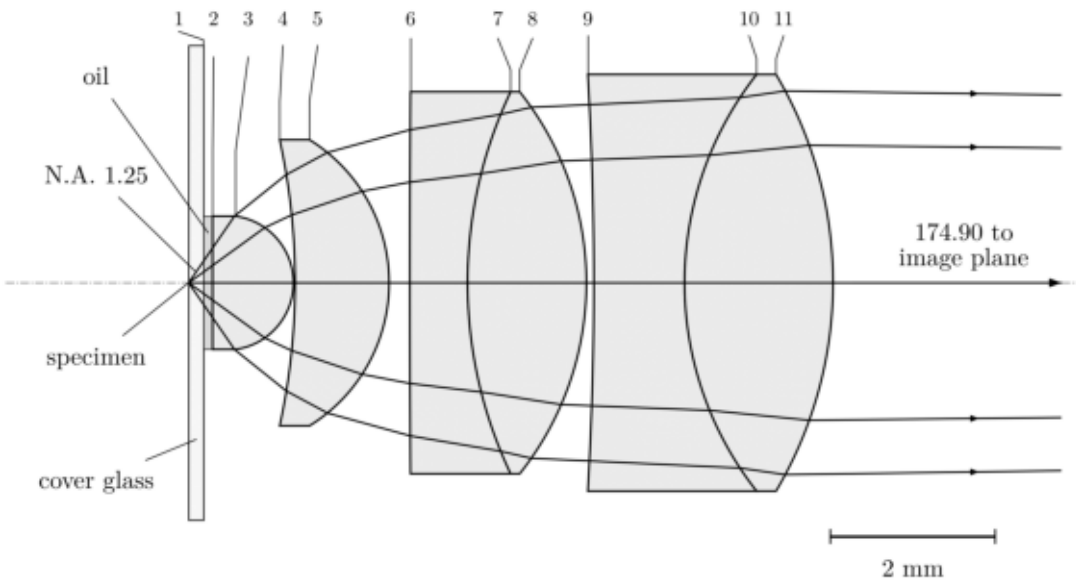
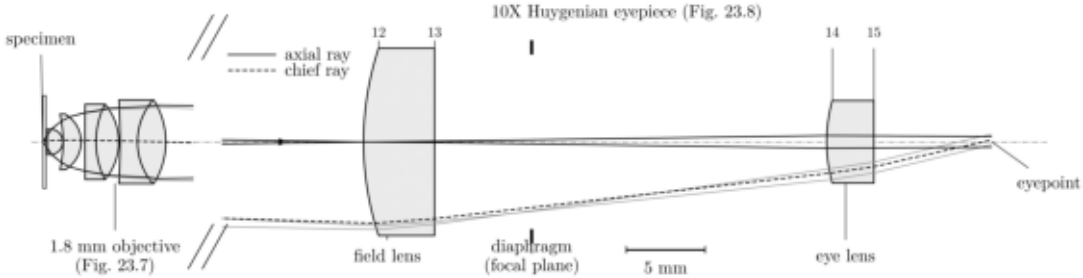


Figure 23 (microscope) — DONE (layout)

1962 scan

(no scan crop)

rebuilt



(printed caption not found on page)

Flagged - not built

spec under-determines the geometry; scan shown so you can judge

Figure 22.1 — FLAGGED

1962 scan

rebuilt

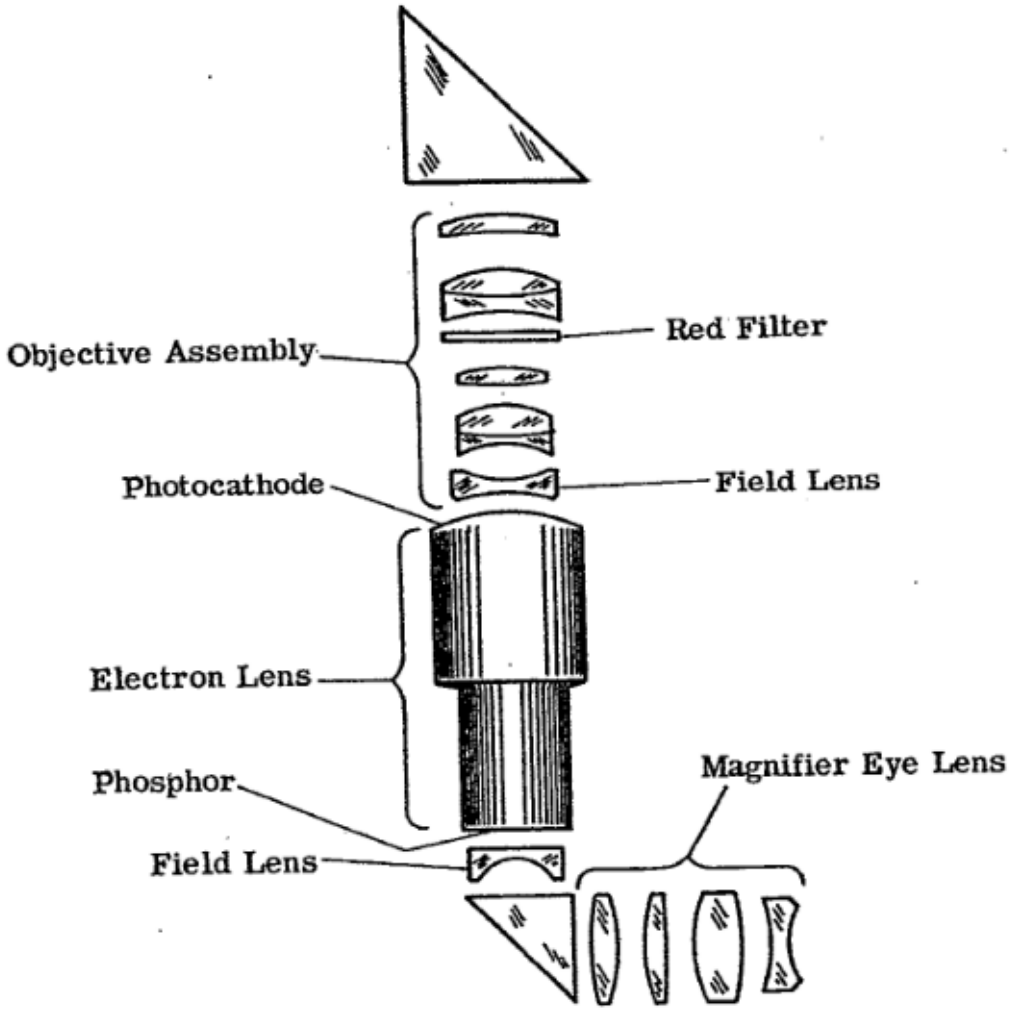


Figure 22.1- Optical schematic of an image converter system.

(not built)

Figure 22.1- Optical schematic of an image converter system.

FLAGGED: the spec names the objective assembly (4 elements + red filter), tube, field lens, two prisms and a 4-element magnifier but gives no radii, glasses, spacings or prism dimensions

Figure 22.2 — FLAGGED

1962 scan

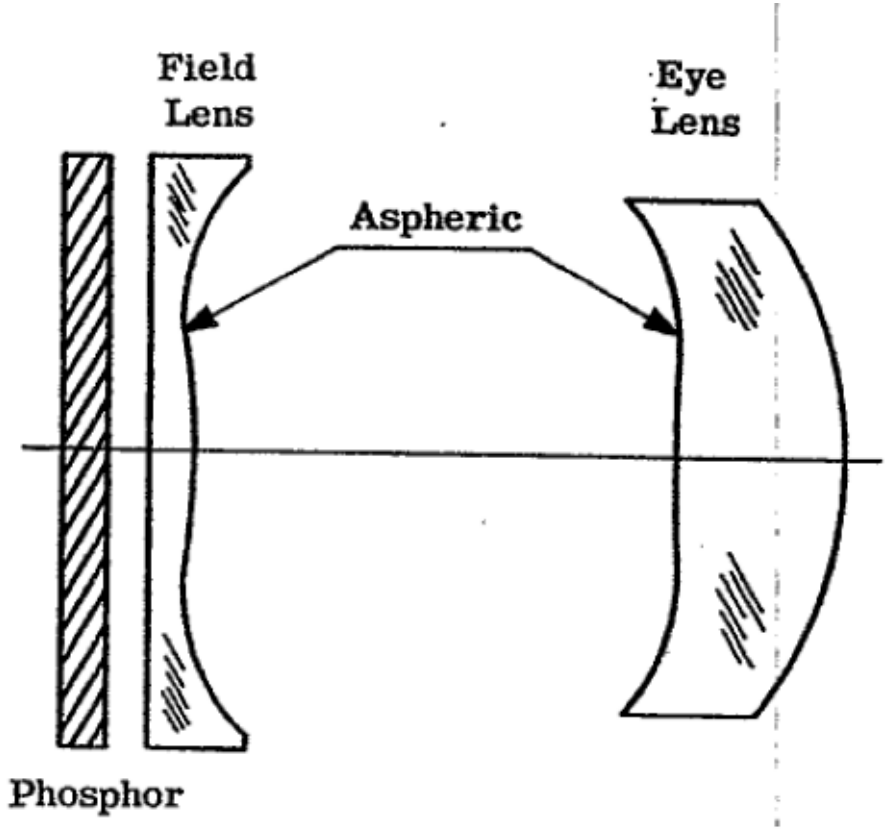


Figure 22. 2- Aspheric Magnifier.

rebuilt

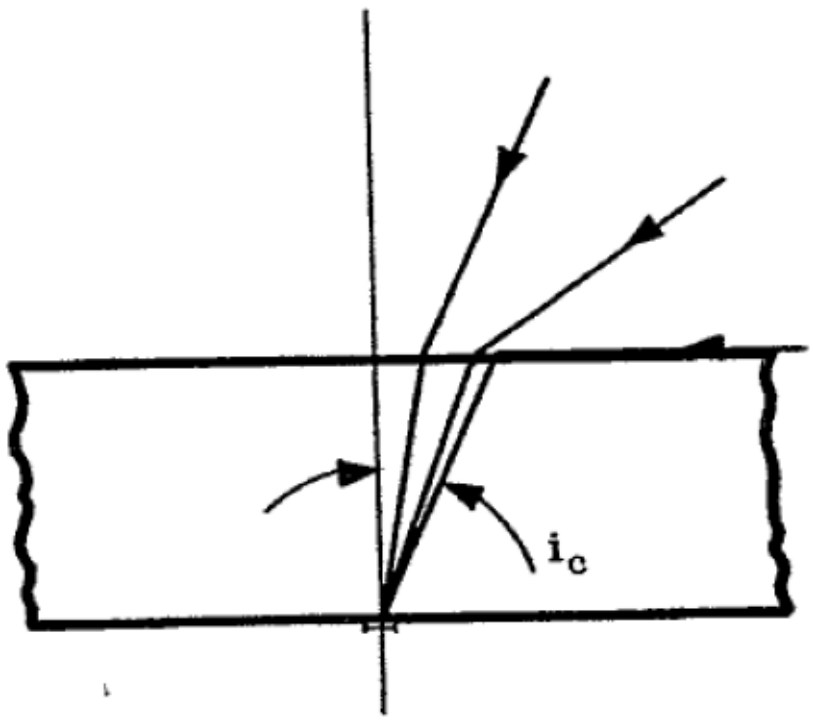
(not built)

Figure 22, 2- Aspheric Magnifier.

FLAGGED: field lens and eye lens are named with two aspheric surfaces, but no radii, thicknesses or aspheric coefficients are given

Figure 22.4 — FLAGGED

1962 scan



Rays beyond i_c are not usable.

Figure 22.4- Illustratio

rebuilt

(not built)

Figure 22. 4- Illustration of advantages of an immersed detector.

FLAGGED: the spec is ambiguous about the dashed "image in space" versus solid "immersed image" ray cones

Figure 23.2 — FLAGGED

rebuilt

1962 scan

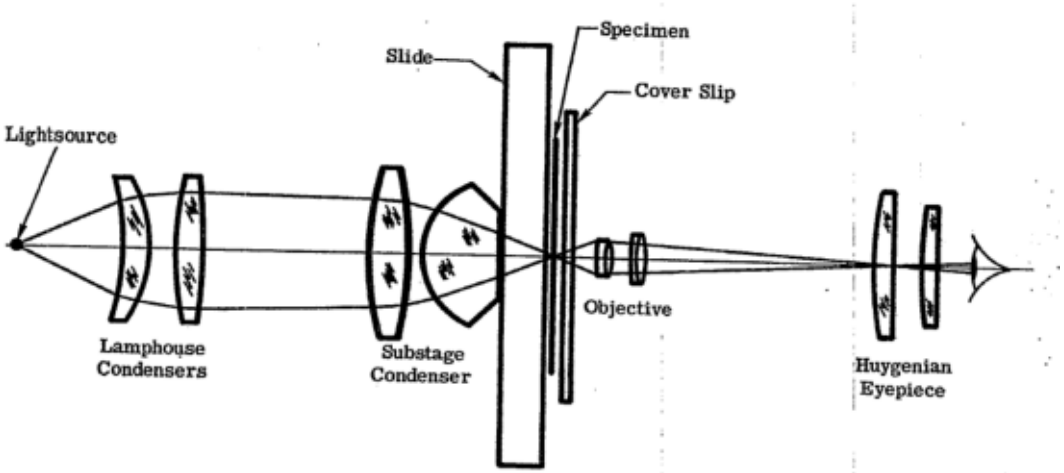


Figure 23. 2– Optical schematic of a compound microscope.

(printed caption not found on page)

(not built)

FLAGGED: the illumination train (lamp condensers, substage condenser) has no prescription

Figure 23.3 — FLAGGED

1962 scan

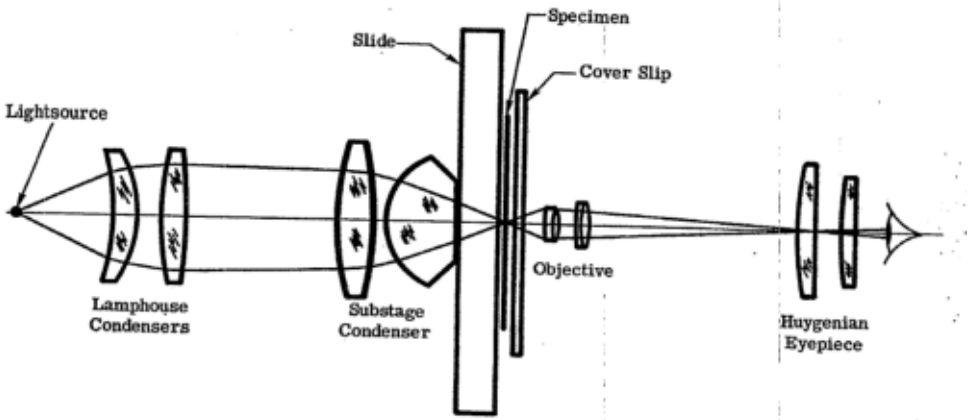


Figure 23. 2– Optical schematic of a compound microscope.

rebuilt

(not built)

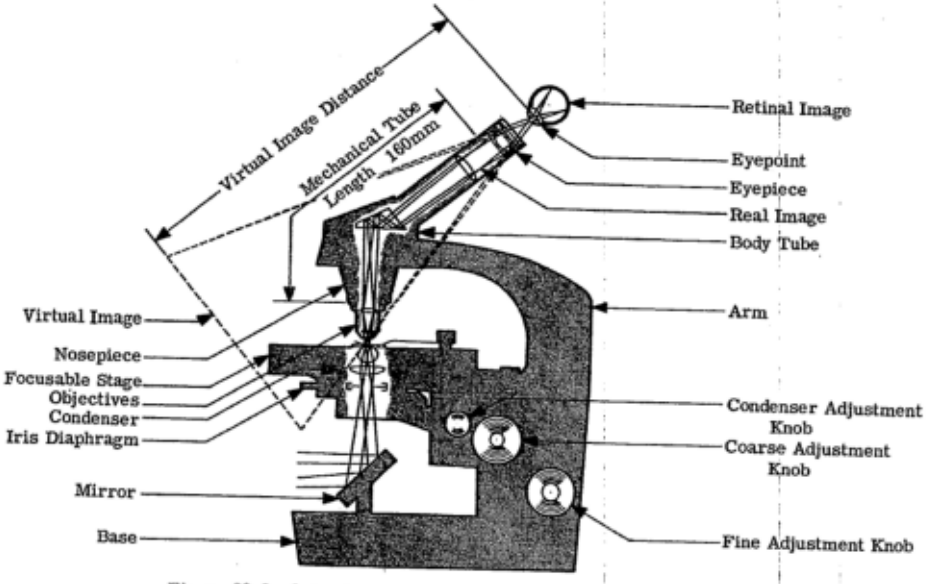


Figure 23. 3– Optical and mechanical features of the microscope.

Figure 23.3- Optical and mechanical features of the microscope.

FLAGGED: a mechanical stand cutaway with no dimensions beyond tube length 160 mm and virtual image distance ~250 mm

Figure 23.9 — FLAGGED

1962 scan

rebuilt

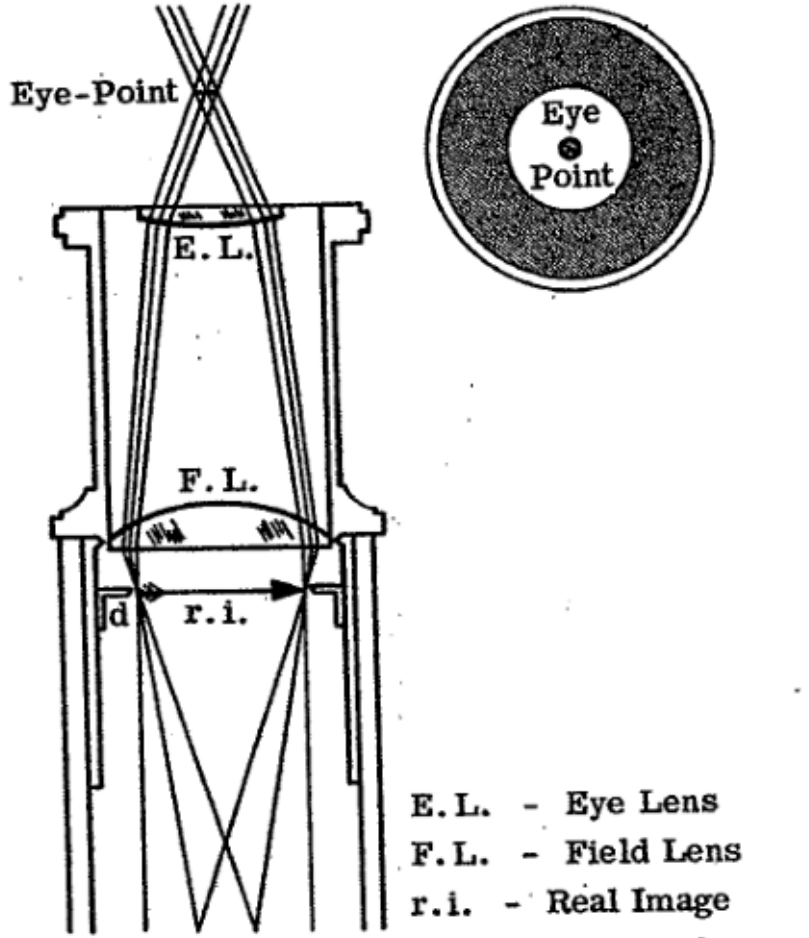


Figure 23. 9- Optical schematic of a Ramsden eyepiece.

(not built)

Figure 23. 9- Optical schematic of a Ramsden

FLAGGED: needs a mechanical barrel section with no dimensions

Figure 23.10 — FLAGGED

rebuilt

1962 scan

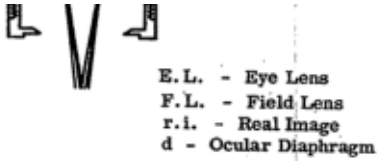


Figure 23.10- Optical schematic of a high eye-point compensating eyepiece.

Figure 23. 10- Optical schematic of a high eye-

(not built)

FLAGGED: field lens doublet and eye lens group are named but have no prescription

Figure 23.13 — FLAGGED

rebuilt

1962 scan

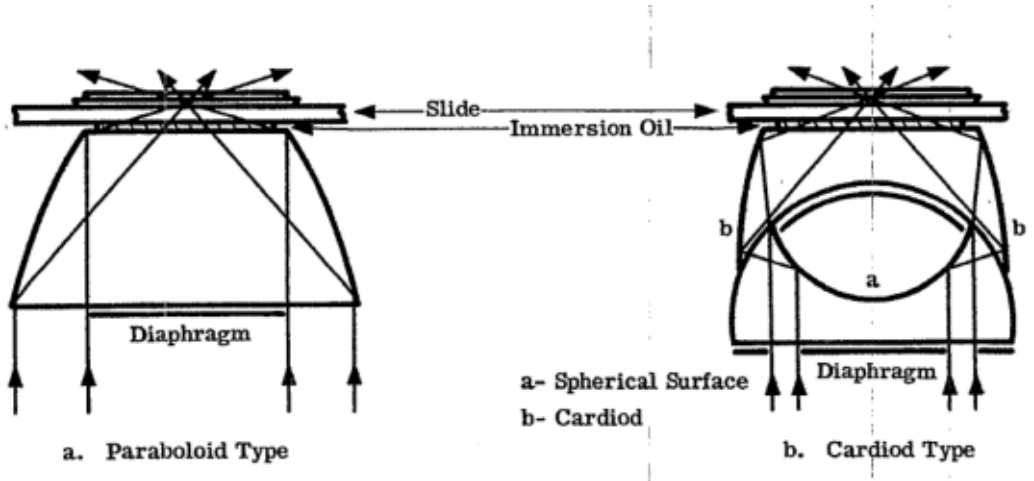


Figure 23. 13- Aspheric darkfield condensers.

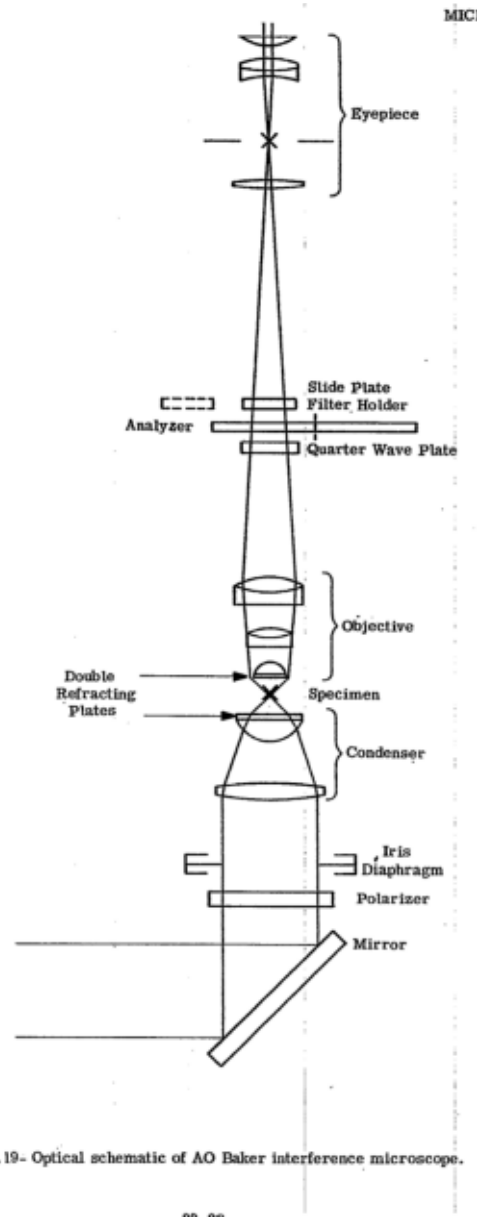
(not built)

Figure 23, 13- Aspheric darkfield condensers.

FLAGGED: paraboloid and cardioid condenser surfaces have no conic constants or dimensions in the spec

Figure 23.19 — FLAGGED

1962 scan



.19- Optical schematic of AO Baker interference microscope.

Figure 23. 19- Optical schematic of AO Baker interference microscope.

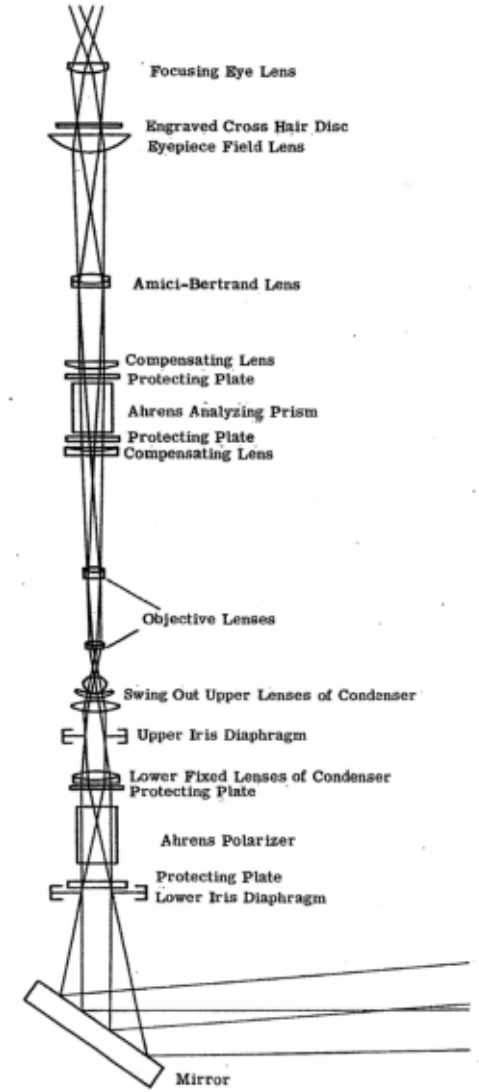
rebuilt

(not built)

FLAGGED: component prescriptions (condenser, objective, birefringent plates) are not given

Figure 23.22 — FLAGGED

1962 scan



23. 22- Optical system of a petrographic microscope.

Figure 23. 22- Optical system of a petrographic microscope.

rebuilt

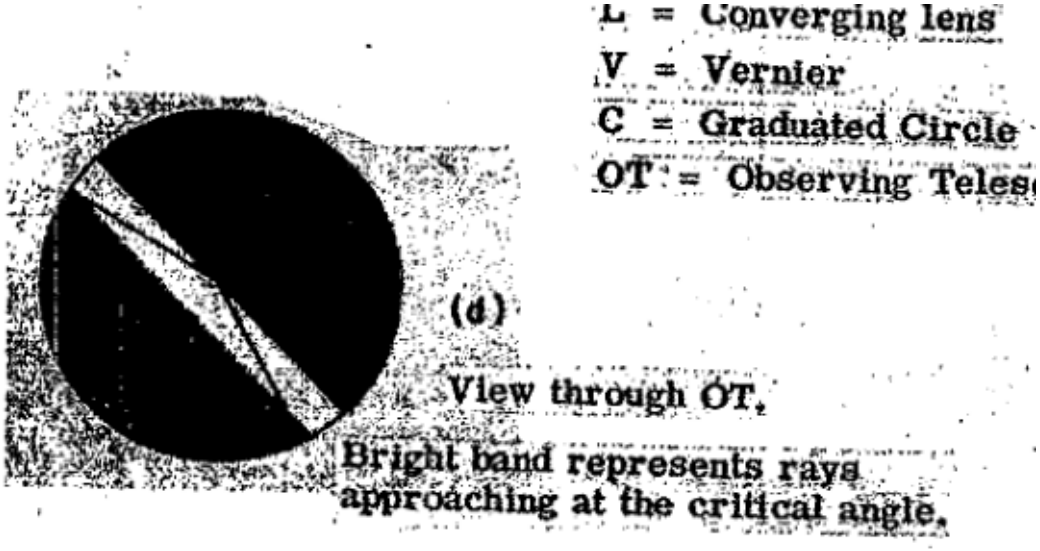
(not built)

FLAGGED: seventeen components are listed with no prescriptions or spacings

Figure 25.8 — FLAGGED

rebuilt

1962 scan



(not built)

Figure 25. 8 – Schematic of the I

5.8 OPTICAL DEVICES, TESTING SYSTEMS AND PROCE

Figure 25.8 - Schematic of the Pulfrich "prastomieter.

FLAGGED: sub-figure (a) is an instrument drawing with no dimensions

Figure 25.12 — FLAGGED

1962 scan

rebuilt

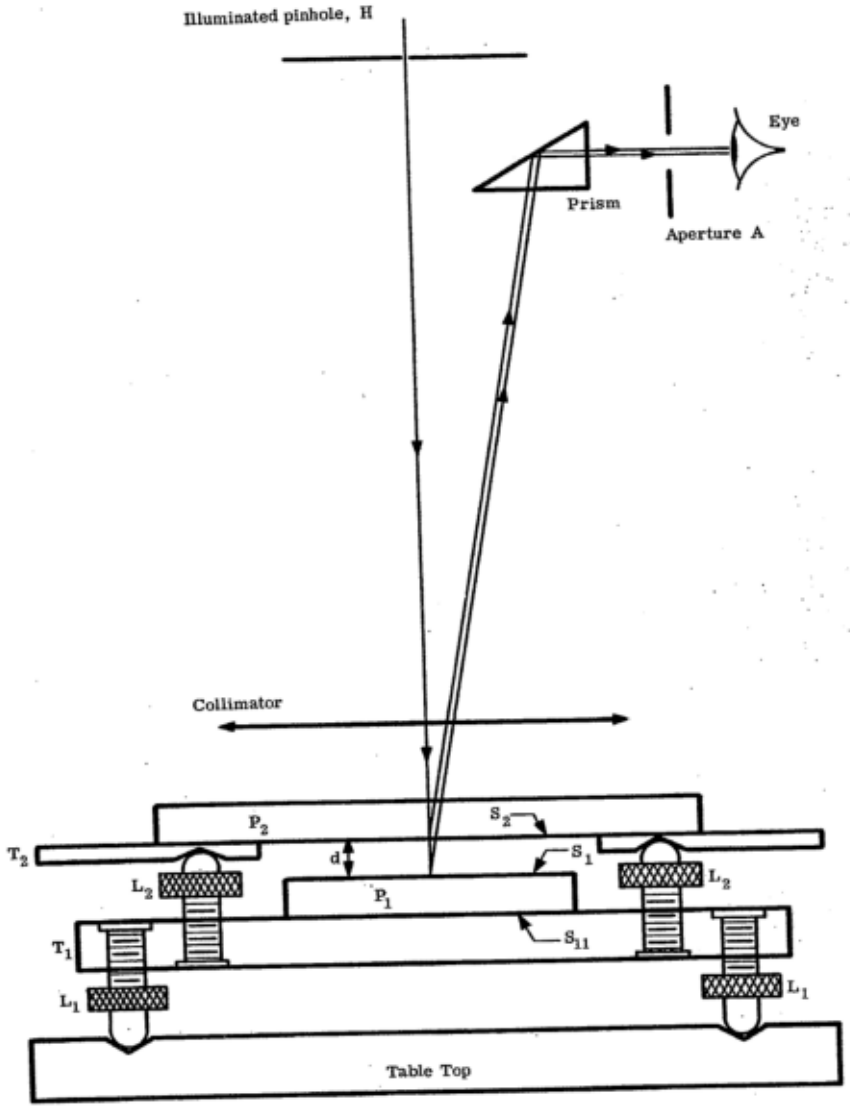


Figure 25.12 - A Fizeau interferoscope.

25-13

(not built)

Figure 25.12 - A Fizeau interferoscope.

FLAGGED: mechanical table, leveling-screw and V-groove detail with no dimensions

Figure 25.29 — FLAGGED

rebuilt

1962 scan

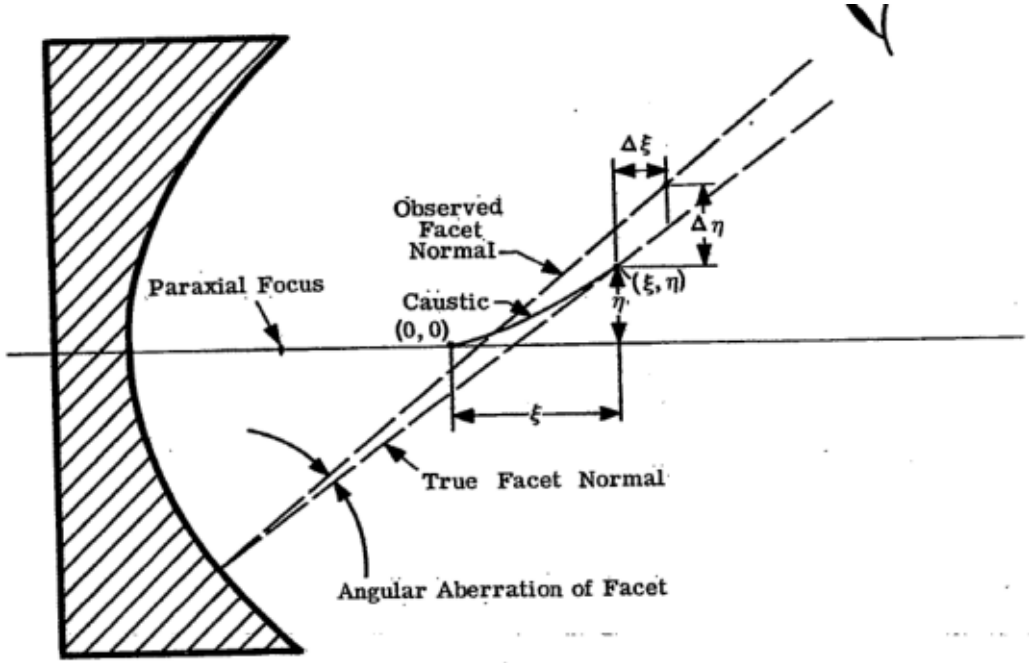


Figure 25.29–The Gaviola technique of Foucault Testing.

(After Strong's, Concepts of Classical Optics, W. H. Freeman and Co. 1958)

Figure 25, 29-The Gaviola technique of Foucault Testing.

(not built)

FLAGGED: the spec does not fix the geometry of the true/observed facet normals and the (xi, eta) caustic coordinates well enough to draw it without inventing it

HALFTONE PHOTO (scan only)

no rebuild - decide whether to crop the scan or regenerate

Figure 24.1 — HALFTONE PHOTO (scan only)

rebuilt

1962 scan

ture by control of aperture shape, variation of transmission, or illumination with radius has been known for some years. A few of the efforts in this direction are the work of Conder and Jacquinot [14] in spectroscopy, the work of Osterberg and Wilkins [15] with microscope objectives, and the work of Silver [16] on tapered illumination of microwave antennas.

20° Off-axis at Gaussian Focus

10° Off-axis, 0.3mm in from Gaussian focus.

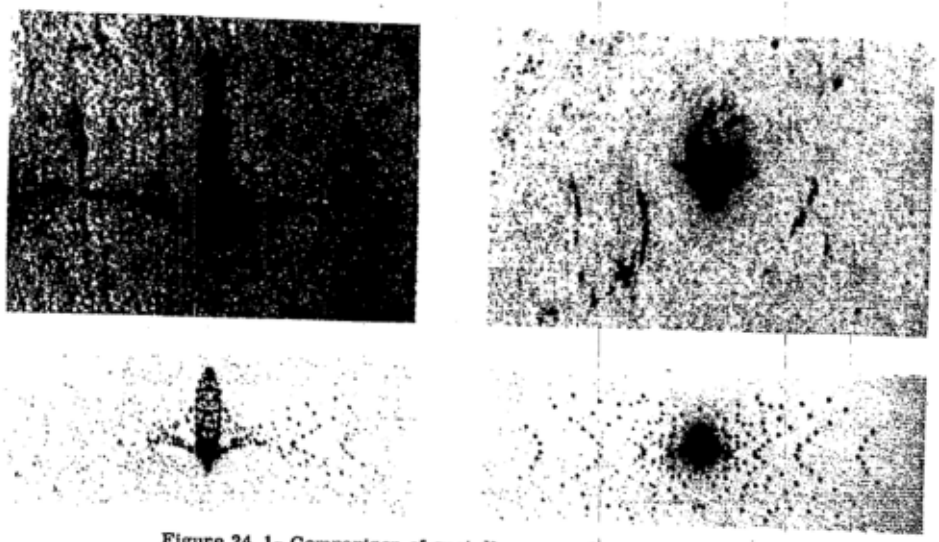


Figure 24. 1- Comparison of spot diagram and actual photograph.

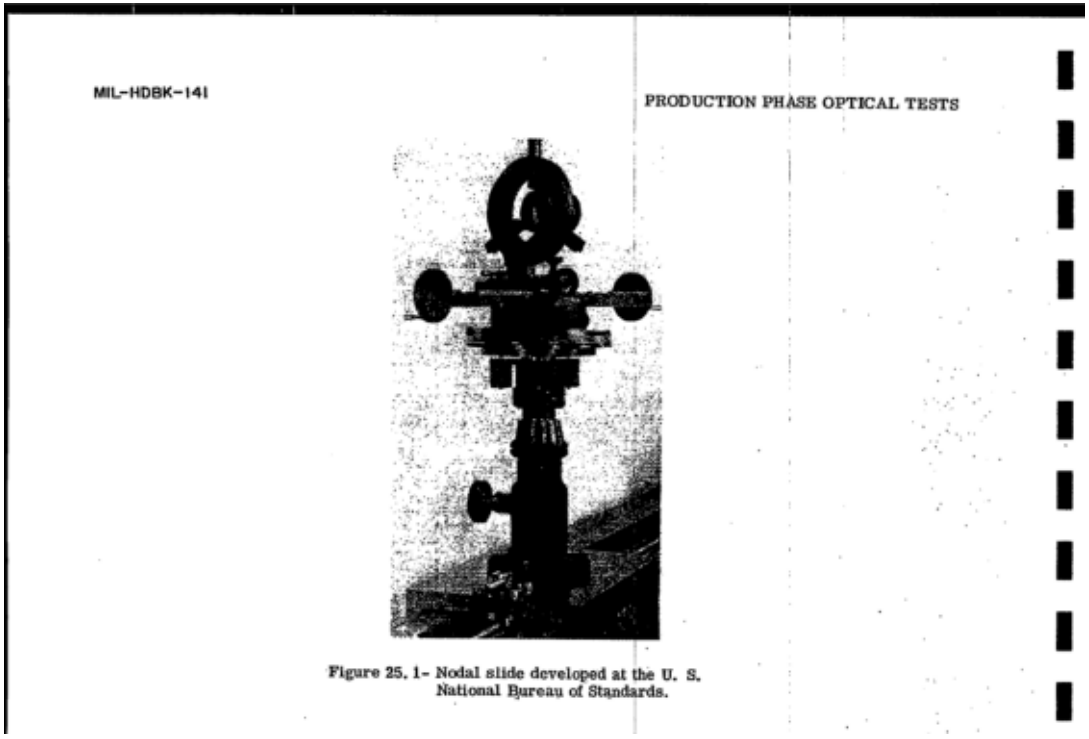
(not built)

Figure 24. 1- Comparison of Spot diagram and

Figure 25.1 — HALFTONE PHOTO (scan only)

rebuilt

1962 scan



(not built)

Figure 25, 1- Nodal slide developed at fhe U U.S.

Figure 25.6 — HALFTONE PHOTO (scan only)

1962 scan

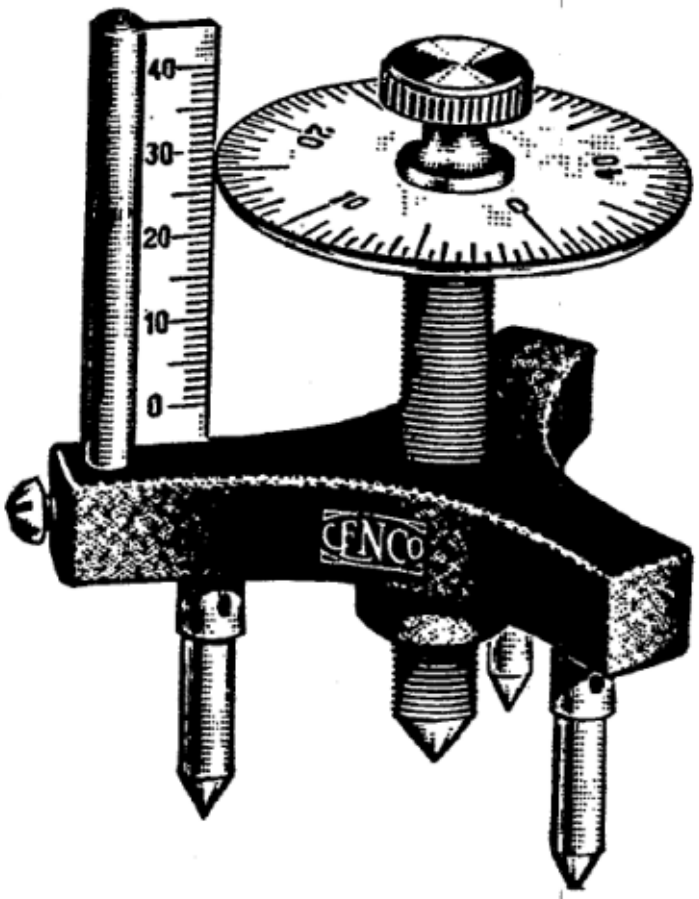


Figure 25.6- Elementary spherometer.

25-6

Figure 25.6- Elementary spherometer.

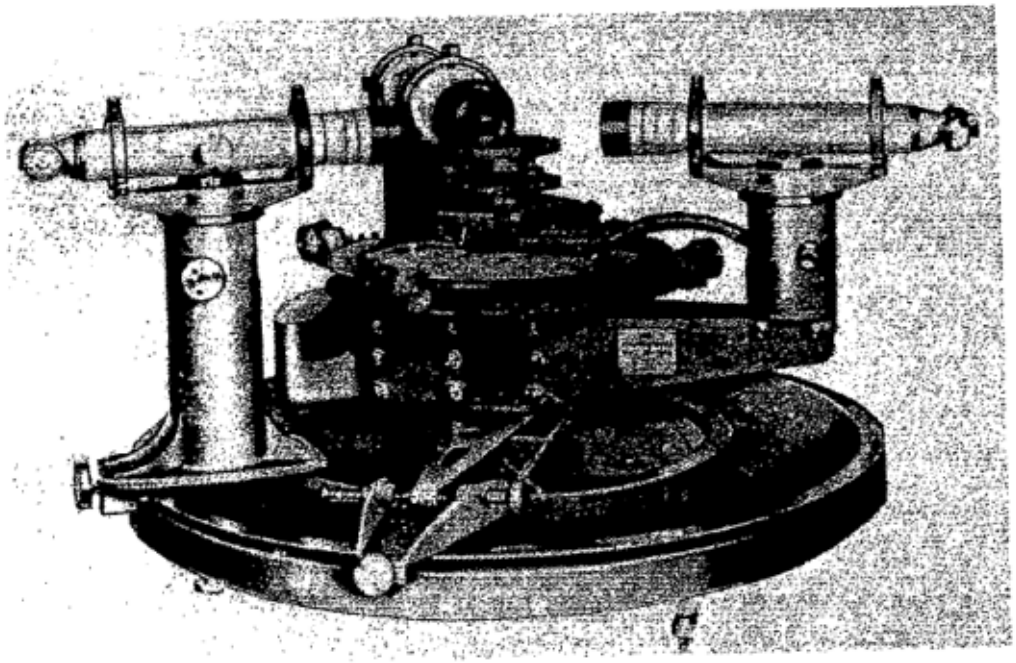
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(not built)

Figure 25.7 — HALFTONE PHOTO (scan only)

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1962 scan



(not built)

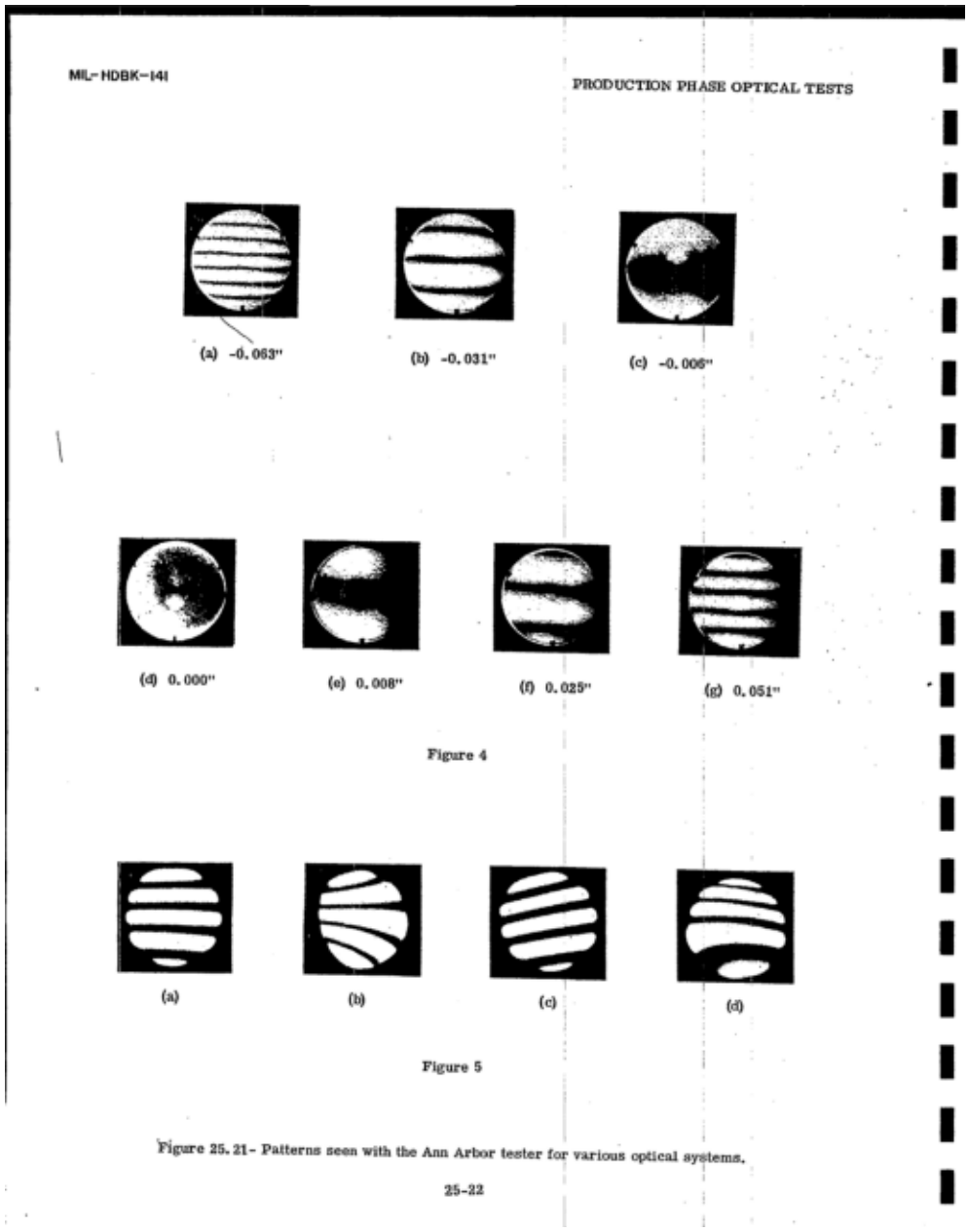
Figure 25.7 - The Guild-Watts precision spectrometer.

10), 549
xperimental Spectroscope, 55, Prentice Hall, (1944).

Figure 25.7- The Guild-Watts precision spectrometer.

Figure 25.21 — HALFTONE PHOTO (scan only)

1962 scan



rebuilt

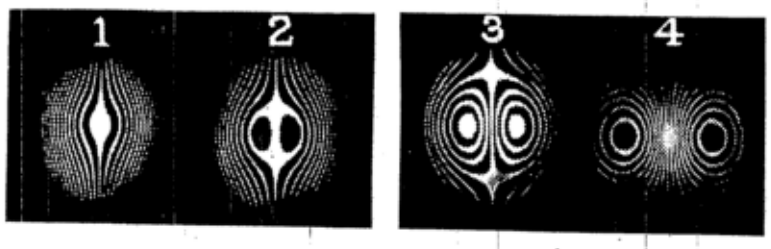
(not built)

Figure 25. 21- Patterns seen with the Ann Arbor tester for various optical systems,

Figure 25.23 — HALFTONE PHOTO (scan only)

1962 scan

rebuilt



1 and 2 are shadow fringes outside the caustic. 3 and 4 are shadow fringes inside the caustic.

The appearances follow in the progression 1, 2, 3, 4.

Figure 25.23-Jentsch's grid method.
(From Martin's, Tech. Optics, Vol. II, Pitman Pub. Co. 1950)

25.10 FOUCAULT TEST

(not built)

25.10.1 Introduction.

25.10.1.1 Having been given a lens or mirror surface prescription and having ground and polished the surfaces by hand or machine methods, the question arises as to what areas need to be "figured", i.e. be repolished to achieve perfectly the prescription. While helpful, the viewing of the image in toto is of less value than might be supposed, in that it represents the summation of the contributions from all parts of the surface. A technique which allows for inspection of the surfaces themselves is obviously required. One of the simplest and yet most delicate of all such techniques of surface testing was developed by Foucault (22) in 1859. The method requires, in its simplest form, merely a pinhole, a knife edge, the lens or mirror, and the eye of the observer. The system is shown diagrammatically in Figure 25.24 for a mirror.

25.10.1.2 In essence, the pinhole provides a small source of light which illuminates the surface of the mirror but which is so shielded that it sends no light directly into the eye. If we assume that the spherical surface is perfect and that the longitudinal aberration is negligible, then all rays striking the mirror will be focussed at some point, F. If the pinhole is located at the center of the curvature of the system, then F also will be at the center of curvature.

25.10.1.3 Assuming that the eye is placed close enough to the image so as to view the mirror in the Maxwellian sense (i.e. the eye receives all the rays coming to the focus), then the mirror surface will be evenly illuminated. (Actually even a perfect surface will show some deviations which will be discussed later.) If now a knife edge K is advanced in the direction indicated, the mirror surface will appear to go from completely bright to completely dark as the knife passes through F. Again for reasons to be discussed later this does not quite happen. If the knife edge is displaced toward the mirror from F then as it cuts into the beam the lower side of the mirror is darkened gradually and not until the beam is completely occulted will the eye see no light. If the knife edge is displaced from F away from the mirror, the reverse occurs. Clearly then the point where the intensity varies most rapidly with knife edge movement from bright to dark is the focus. If the knife edge always remains in the plane of the pinhole, then there will be only one place where the mirror darkens uniformly and rapidly with lateral displacement of the knife edge. This point is of course the center of curvature.

25.10.1.4 The extreme delicacy of this measurement will become more obvious if we study Figure 11.28 where a highly exaggerated error is present. Suppose there is an error of slope, angle, θ. The rays hitting

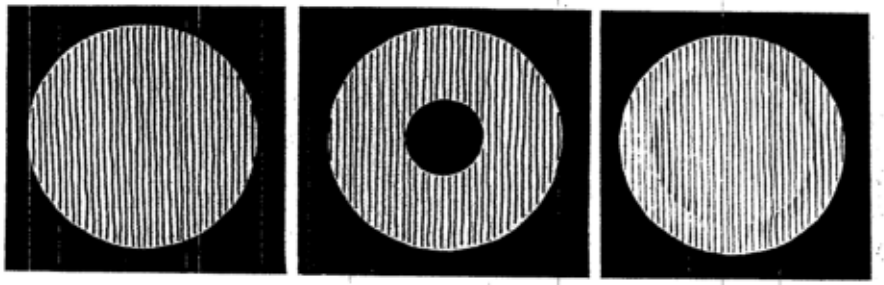
(22) Foucault, Ann de L'Obs. de Paris, V, 197 (1859).

(printed caption not found on page)

Figure 25.26 — HALFTONE PHOTO (scan only)

1962 scan

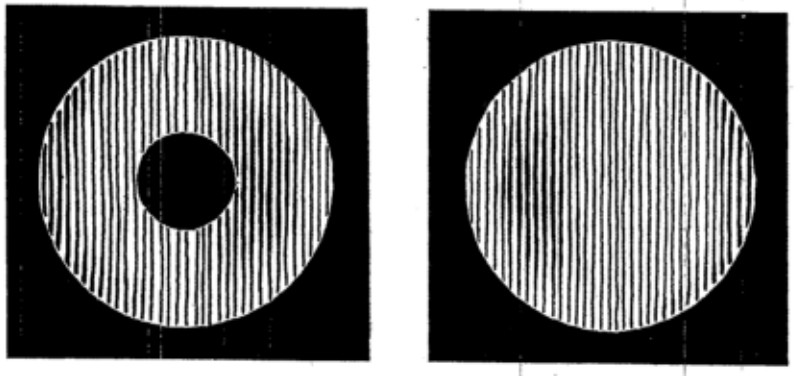
rebuilt



(a) Spherical mirror tested at the center of curvature.

(b) Parabolic mirror tested with a flat testing mirror.

(c) Spherical mirror with a raised annular ridge as tested at the center of curvature.



(d) Spherical mirror tested with a flat testing mirror.

(e) Parabolic mirror tested at the mean center of curvature.

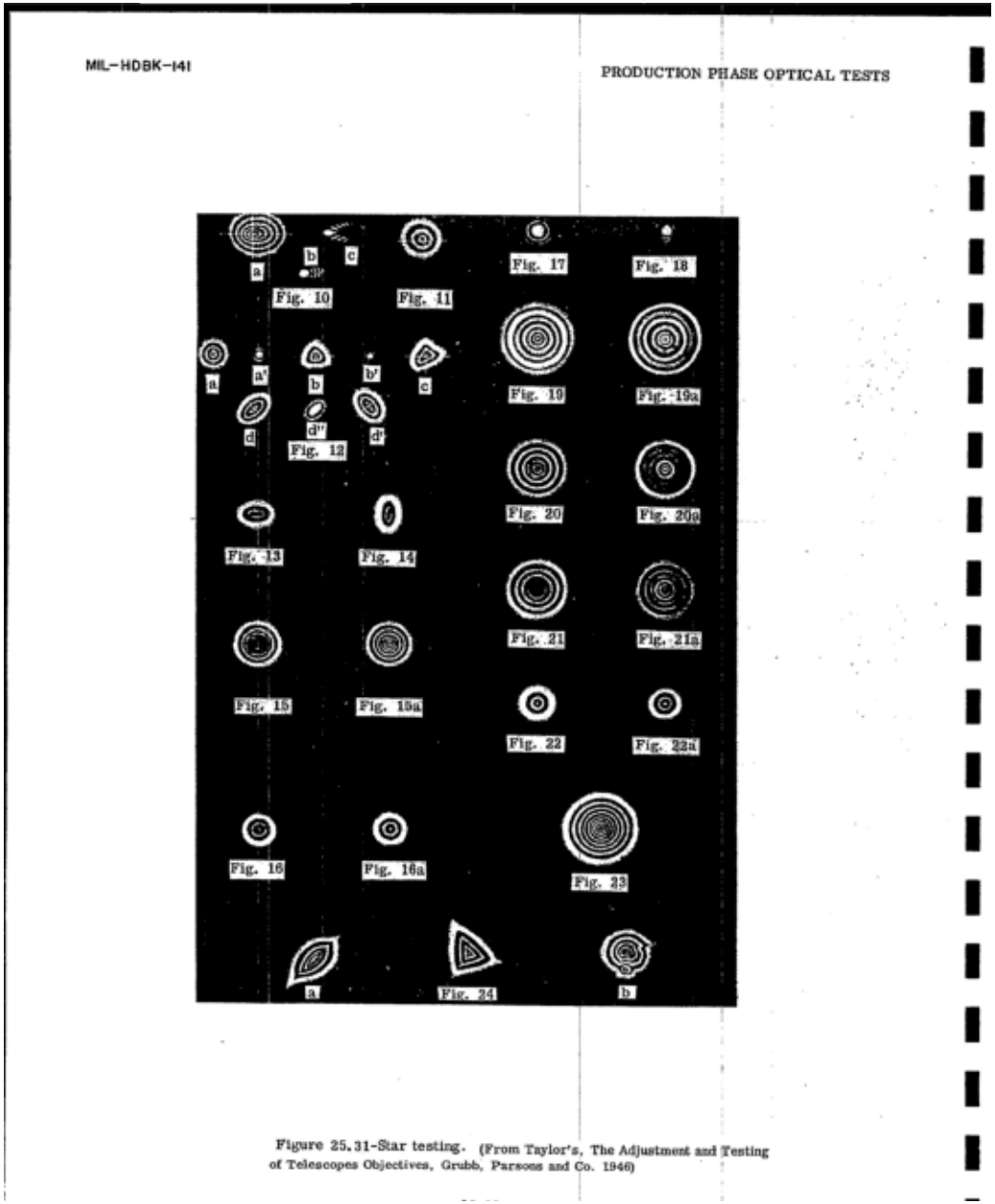
Figure 25.26-Foucault test appearances.
(From Strong's, Procedures in Experimental Physics, Prentice-Hall Inc., 1938)

(not built)

Figure 25. 26-Foucault test appearances.

Figure 25.31 — HALFTONE PHOTO (scan only)

1962 scan



rebuilt

(not built)

Figure 25,31-Star testing. (From Taylor's, The Adjustment and (esting

Figure 25.32 — HALFTONE PHOTO (scan only)

rebuilt

1962 scan

of focus. and viewed under a high power.
Fig. 23. – A section of the cone of rays yielded by a perfect objective, taken at about 1/4-inch on either side of focus, and viewed under a moderately high magnifying power.
Figs. 24 and 24a.– Examples of violent mechanical strain, due to imperfect mounting or bad annealing.
Fig. 24b.– Example of the effects due to the presence of veins in the material of the objective.

Index to Figure 25.31
(From Taylor's, The Adjustment and Testing of Teles. Objectives, Grubb, Parsons and Co., 1946)

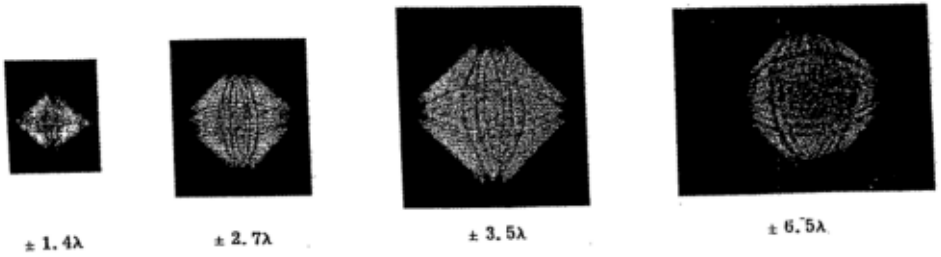


Figure 25.32-Star tests showing astigmatism of varying degrees.
(From Linfoot's, Recent Advances in Optics, Oxford Univ. Press, 1955)

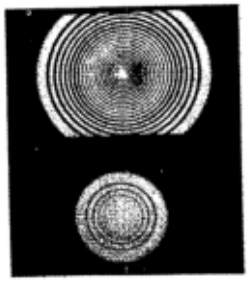
(not built)

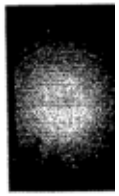
Figure 25, 32-Star tests showing astigmatism of varying degrees.

Figure 25.34 — HALFTONE PHOTO (scan only)

1962 scan

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17.5λ

(b) Images i
primary

(not built)



17.5λ

(c) Images i
primary

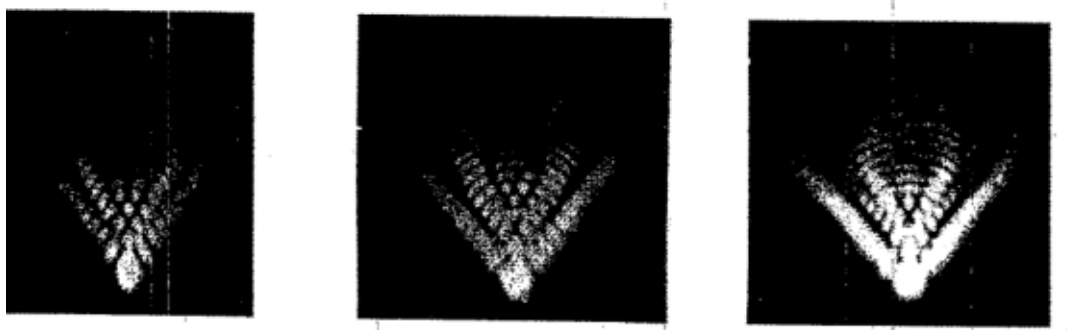
Figure 25.34-Star imag
(From Li

Figure 25, 34-Star images showing various amounts of primary spherical aberration.

Figure 25.35 — HALFTONE PHOTO (scan only)

rebuilt

1962 scan



(not built)

Primary coma - ϕ is $2\lambda (r^3 - 2/3\, r) \cos x$ at focal settings corresponding approximately to the given values of p.

Figure 25.35-Star images exhibiting coma.
(From Linfoot's, Recent Advances in Optics, Oxford Univ. Press, 1955)

Figure 25. 35-Star images exhibiting coma. :

3D-COORDINATE (scan only)

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Figure 16.1 — 3D-COORDINATE (scan only)

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1962 scan

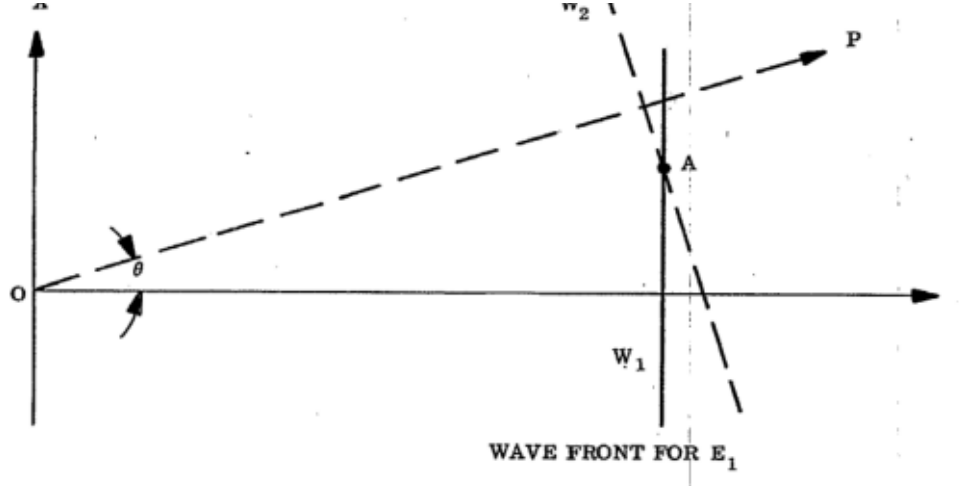


Figure 16. 1 – Interference between two plane wavefronts W_1 and W_2 that are propaga along different directions.

(not built)

Figure 16.1 - Interference between two plane wavefronts W , and We that are propagated

Figure 16.33 — 3D-COORDINATE (scan only)

rebuilt

1962 scan

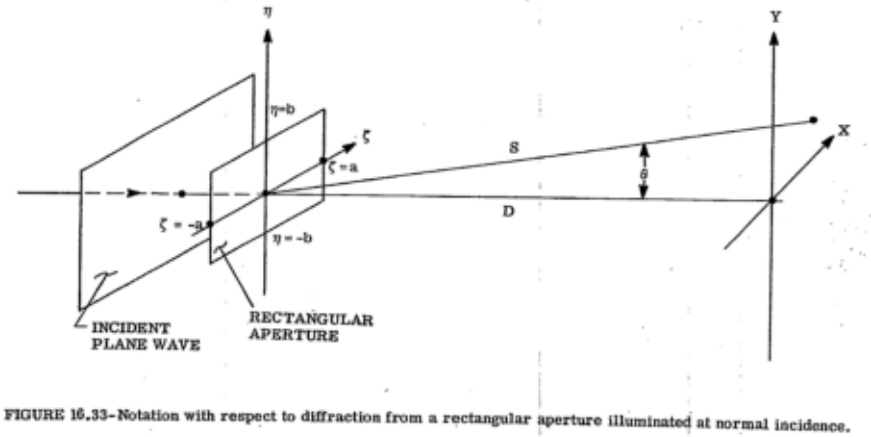


FIGURE 16.33-Notation with respect to diffraction from a rectangular aperture illuminated at normal incidence.

(not built)

FIGURE 16.33-Notation with respect to diffraction from a rectangular aperture illuminated at normal incidence.

Figure 16.34 — 3D-COORDINATE (scan only)

rebuilt

1962 scan

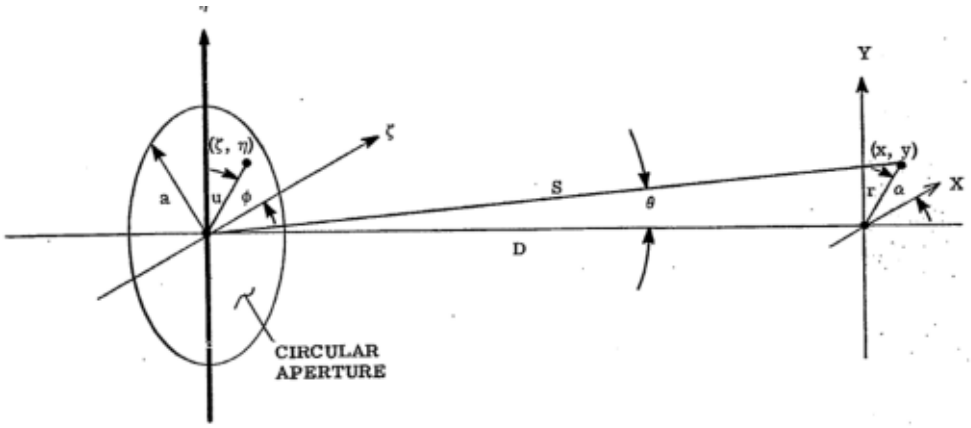


FIGURE 16. 34–Notation with respect to Fraunhofer Diffraction from circular apertures.

(not built)

FIGURE 16, 34-Notation with respect to Fraunhofer Diffraction from circular apertures.

Figure 25.19 — 3D-COORDINATE (scan only)

rebuilt

1962 scan

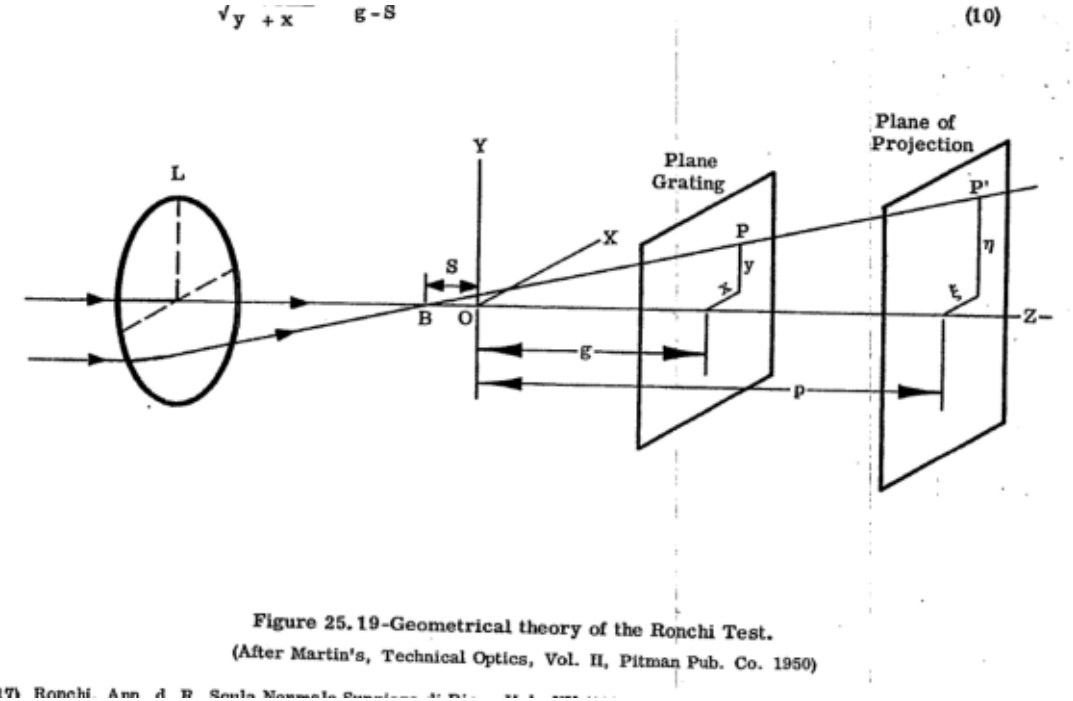


Figure 25. 19-Geometrical theory of the Ronchi Test.
(After Martin's, Technical Optics, Vol. II, Pitman Pub. Co. 1950)

(17) Ronchi, Ann. d. R. Scuola Normale Superiore di Pisa, Vol. [illegible]

Figure 25. 19-Geometrical theory of the Ronchi Test.

(not built)

Figure 25.22 — 3D-COORDINATE (scan only)

rebuilt

1962 scan

arrangement as shown in Figure 25.20.

The resulting patterns for the Tester in the correct and other positions with respect to the axis can be seen from the patterns pictured in Figure 25.21. Figure 5(a) in 25.21 shows the pattern obtained with the Tester on the optical axis of a 30" focal length parabolic mirror cut 7° off-axis. This position is illustrated above, with the Tester at A and the edge of the parabolic mirror closest to the optical axis at A'. Figure 5(b) in 25.21 shows the pattern when the Tester, parabolic mirror, and plane mirror are still in the correct plane, but the grating is located outside the optical axis (B in the Figure above).

(not built)

Figure 25.22- Determining the optical axis of off-axis parabolic mirrors with the Ann Arbor Tester.

25-23

Figure 25. 22- Detérmining the optical axis of off-axis parabolic mirrors with the Ann Arbor Tester.

Tables

transcribed into ocr_output/ch20.md - scan shown for checking

Figure Table 20.1 — TABLE

rebuilt

1962 scan

MATERIAL	REFRACTIVE INDEX	TRANSMISSION RANGE .	COMMENTS
Hot pressed Mg F_2 (Irtran 1)	1.37	200 mμ to 7.5 μ	
Calcium fluoride	1.42	180 mμ to 6 μ	
Barium fluoride	1.42	150 mμ to 13 μ	
Irtran 3	1.42	1000 mμ to 9 μ	
Fused quartz	1.46	180 mμ to 3.5 μ	1
Vycor (high-silica glass)	1.46	250 mμ to 3.5 μ	
Glass	1.51 to 1.70	320 mμ to 2.5 μ	2
Sapphire	1.70	<20 mμ to 7.5 μ	
Hot pressed ZnS (Irtran 2)	2.26	2000 mμ to 14 μ	
Arsenic trisulfide glass	2.40	800 mμ to 12 μ	
Irtran 4	2.40	1000 mμ to 20 μ	
Silicon	3.50	1100 mμ to 8 μ	
Germanium	4.05	1900 mμ to >22 μ	

NOTES:

1. The wavelength of the absorption edge in the ultraviolet depends upon the purity of the quartz.
2. In general, glasses with a lower refractive index are transparent to shorter wavelengths than glasses with a higher index. However, there are exceptions to this rule. Most glasses have a strong absorption band in the vicinity of 2.6 μ. Very thin glass plates (i.e. "cover slips") are transparent between 2.7 μ and 4.5μ.

Table 20.1 – Common substrate materials.

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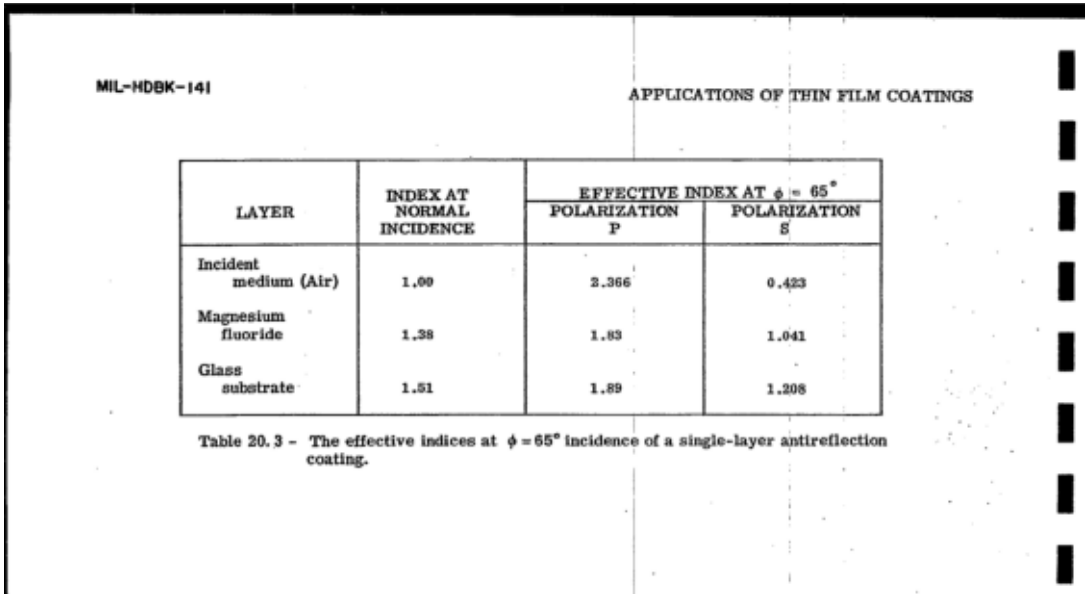
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Figure Table 20.3 — TABLE

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LAYER	INDEX AT NORMAL INCIDENCE	EFFECTIVE INDEX AT $\phi = 65^\circ$	
		POLARIZATION P	POLARIZATION S
Incident medium (Air)	1.00	2.366	0.423
Magnesium fluoride	1.38	1.83	1.041
Glass substrate	1.51	1.89	1.208

Table 20.3 – The effective indices at $\phi = 65^\circ$ incidence of a single-layer antireflection coating.

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16.11 20.2 20.98 21.2 25.13 25.15 25.25 Table 16.1 16.26 20.21 20.45 20.95 20.96 20.97 20.100 20.101